



# FOREWORD



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It is with great pleasure that I present the “ICAR Kharif Agro-Advisories for Farmers 2025”- a timely and comprehensive guide developed to empower India’s farming community with science-based, region-specific advisories for the upcoming Kharif season. This publication marks a significant step toward strengthening climate-resilient agriculture and enhancing farm productivity across diverse agro-climatic zones.

Agriculture in India stands at a critical juncture. While it continues to serve as the backbone of rural livelihoods, it faces unprecedented challenges due to climate variability, resource degradation, pest dynamics, and market volatility. In this context, the role of timely and location-specific agro-advisories has become more crucial than ever. These advisories act as a bridge between cutting-edge agricultural research and its practical application in farmers’ fields.

This publication offers practical guidance on region-based suitable varieties, nutrient management, pest and disease control, and harvesting practices tailored for major Kharif crops such as rice, maize, pulses, oilseeds, and cotton. It is the result of collaborative efforts by ICAR Institutes, Agricultural Universities, Krishi Vigyan Kendras (KVKs), and Subject Matter Experts working at the grassroots level.

At ICAR, we remain committed to the vision of a knowledge-driven, inclusive, and sustainable agricultural system. The advisories compiled herein reflect our continuous efforts to translate scientific knowledge from laboratories into actionable insights for every farmer. I urge extension officers, KVK teams, farmer producer organizations (FPOs), and progressive farmers to widely disseminate and implement the recommendations provided in this book.

I extend my heartfelt appreciation to all scientists, experts, and field professionals who have contributed to the preparation of this publication. I am confident that this agro-advisory compilation will serve as a valuable resource for farmers during upcoming Kharif season, enabling them to make informed decisions that lead to improved productivity, profitability, and sustainability.

Let us move forward together to build a resilient, prosperous, and empowered farming community #Viksit Bharat 2047.

Jai Kisan, Jai Vigyan!

(M.L. Jat)

# INTRODUCTION

Agriculture remains the cornerstone of India's economy, supporting nearly half of the population and providing employment, nutrition, and rural development. The Kharif season, spanning from June to October, is particularly vital due to its reliance on the south-west monsoon. Approximately 55% of India's net sown area is rainfed, making the timely and equitable distribution of monsoon rains crucial for agricultural success.

In 2025, the India Meteorological Department (IMD) forecasts an above-normal monsoon season, with cumulative rainfall expected at 105% of the Long Period Average (LPA) of 87 cm. The monsoon's onset is anticipated to be slightly delayed in southern India but should reach Maharashtra by May 27, facilitating timely sowing preparations. The IMD's forecast of an above-normal monsoon offers a promising outlook but still carries risks of intra-seasonal dry spells and localized flooding. The Indian government's initiatives, such as the "Panchayat Mausam Seva," aim to deliver hyper-local weather forecasts at the panchayat level, ensuring that farmers receive accurate and timely information.

The Kharif season witnesses the cultivation of a diverse array of crops across India's agro-climatic zones, including rice, maize, cotton, soybean, groundnut, bajra, jowar, tur, urad, and moong. Despite advancements in agricultural technology, the sector remains vulnerable to climatic fluctuations such as delayed rains, uneven distribution, and extreme weather events. These challenges can adversely affect crop yields, farm incomes, and rural livelihoods.

To mitigate these risks, it is imperative to provide farmers with location-specific, research-backed guidance. Agro-advisories play a pivotal role by offering timely, region-specific, science-based information on weather patterns, crop management, pest control, harvesting, and storage. These advisories integrate seasonal outlooks with crop calendars and state-wise sowing windows, enabling farmers to select appropriate crop varieties and sowing dates, optimize resource use, and prevent yield losses.

This advisory serves as a crucial tool for agricultural officers, Farmer Producer Organizations (FPOs), and NGOs involved in rural development. By bridging the information gap between research and practice, it empowers farmers with the knowledge needed to make informed decisions, thereby contributing to the vision of a self-reliant, prosperous, and inclusive India @ VIKSIT BHARAT 2047.

case of stormy weather conditions, while heaping of incompletely dried paddy sheaves, 25 kg of salt per acre of paddy sheaves may be sprinkled and heaped to avoid damage.

### **Maize**

In normal conditions, rainfed crop should be sown between June 15 and July 15. In the Kharif season, under rainfed conditions, maize should be sown only after a substantial rainfall (more than 50 mm). Seed rate is 17.5-20 kg/ha for normal hybrids, 7.5-10 kg/ha for sweetcorn, 12.5 kg/ha for popcorn and 25 kg/ha for baby corn. Within two to three days after sowing the crop, spray Atrazine herbicide @ 2 kg/ha in light soils and 3 kg/ha in heavy soils mixed with 500 liters of water when there is sufficient moisture which helps in controlling broadleaved and some grassy weeds for about 30 DAS. To control broadleaved and grassy weeds in normal and zero tillage maize crop at 15-18 DAS or 4 leaf weed stage, Tembotrione 34.4% SC @ 287.5 ml/ha + Atrazine 50% WP @ 1000 g/ha or Topramezone 33.6% SC 75 ml + Atrazine 50% WP @ 1000 g/ha should be sprayed. If the problem of *Cyperus spp.* is more, spray Halosulfuron methyl @ 100 g/ha mixed with 500 L of water.

*Fall army worm:* Summer ploughing, collection & destruction of egg masses, installation of pheromone traps (*S. frugiperda*) @ 4/acre sowing of border crop with grain sorghum & inter crop with cowpea (few rows) spraying of Azadirachtin 10000 ppm @ 2ml/l (10 to 15 DAS), EPN or Bt spray @ 2 ml/l (15 to 21 DAS). First insecticide spray - Emamectin Benzoate 5SG @ 0.4g/l (or) Spinosad 480SC @ 0.5 ml/l (21-28 DAS). Metarhizium anisopliae spray (1x107) @ 2 ml/l (30 -35DAS). Second Insecticide spray - Flubendiamide 480SC @ 0.3 ml/l (or) Chlorontriliniprole 18.5SC @ 0.3ml/l or Spinetoram 11.7 SC @ 0.3 ml/l (36 - 42 DAS). Poison Baiting - 45-65 DAS using Thiodicarb 75WP (250 g thiodicarb + 25kg rice bran + 5 kg jaggery / ha).

### **OILSEED CROPS**

#### **Groundnut**

Farmers are advised to take up summer ploughing and also suggested to select high yielding varieties like TCGS 1694, K-Chitravathi, K-Lepakshi. The suitable time for sowing of rainfed groundnut is July to first week of August. Farmers are advised to adopt intercropping with redgram at 7:1/11:1/15:1 ratio as per their convenience and availability of sowing equipment. Use fertilizer cum seed drill instead of manual labour for sowing behind the country plough. Formation of conservation furrows every 3.6m interval at sowing/20-30 days after sowing will help in soil moisture conservation.

Farmers are also suggested to follow seed treatment with imidacloprid 600 FS @ 1ml per kg seed and followed by tebuconazole 2 DS @ 1g or mancozeb @ 3 g/kg or *Trichoderma viride* @ 10g/kg seed. Take up sowing of bajra/jowar (4-8 rows) as border crops all around the field to avoid transmission of vectors like thrips which spread PSND/PBND diseases. Spray KNO<sub>3</sub> @ 0.5% or 19:19:19 immediately after dry spell. Provide supplemental irrigation 20mm at flowering and peg penetration stage. Install sticky traps (10/acre) and pheromone traps (4/acre) in field to monitor

enrich the soil. Tractor operated FYM Pulverizer cum spreader is useful for FYM application in the field without any labour. It covers up to 10 to 12 acres per day. Cost of the implement is Rs. 76000/-.

DSR with Seed cum fertilizer drill increases 14% higher yields compared to traditional transplanting. Crops matured 7-10 days earlier. Farmers saved on 12 men labour and 25 women labour per hectare leading to a 17 % reduction in cost of cultivation. Tractor drawn Ananta groundnut planter (8 rows) is used for groundnut sowing. The intercrops of redgram or castor can also be sown with this implement. It covers 6 to 7 ha per day. For small and marginal farmers, use Handpush seed drill in light soils for Maize, Castor, Redgram, Cowpea, field bean and other bold seeded crops and in heavy soils on ridges sowing should be done for better crop stand establishment. Motor-Operated Groundnut Sheller is capable of shelling 25 to 250 kg of pods per hour. The distance between the seed drum and the concave can be adjusted according to pod size for efficient shelling. Powered by a 1 or 2 HP motor/engine depending on size. Use cono weeders for Drum seeder/transplanted rice in wet conditions. Power weeders help in upland rice systems or DSR & Early-stage weeding (15 to 30 DAS) is crucial for effective control. Self-propelled Weeder in paddy fields is operated by a 3 to 5 HP diesel engine, it can cover 0.15-0.25 acres per hour at low speed. & with a 5 to 6.5 HP engine, it can cover up to 1 acre per hour. Cost of cultivation is Rs.52,000 to Rs.56,000/ha.

Rain pipe irrigation system is effortless to install, reallocate and reduce the consumption of energy and water by more than 50 percent. These pipes are light weight, portable and flexible. Works on low water pressure. Expected life for 3 to 5 Years. Solar Fence system is used to control Wild boars a major problem in groundnut, paddy, maize and other crops. The solar fence cost varies from Rs. 15,000 to 25,000 by different companies like Pavani Solar systems, Nellore (Mobile numbers: 9848582332 & 9440171938) and Krupa Solar fence, Hyderabad. This unit protects 5 acres field. Use Acoustics (Sound emitting machinery) to repel the Birds, Wild boars, Monkeys etc., in crops like Groundnut, Sunflower, Maize, jowar etc. Use Bird scarer to repel the birds in the fields. The implement will create sound by rotating fan when natural air occurs. The cost is Rs.800/-. Straw Baling Machine is used to bundle paddy straw into either round or square/rectangular shapes for easy handling and transport. ANGRAU blade guntaka (Groundnut Digger) separates groundnut pods from the soil efficiently without wasting time. Built to dig up to 30 cm depth and can harvest up to 1 acre per hour.

Groundnut Digger, shaker cum Windower digs plants up to about 25 cm depth, shakes off the soil, and arranges the plants into windrows behind the machine. Cost: Rs.1,68,000/-. Groundnut Stripper separates pods from the plant by using a 1 to 3 HP motor. Requires two labour. Capacity: 100-125 kg per hour. Tractor operated pit digger for sugarcane planting will dig pits 250 to 300 holes per hour with a diameter of 90 cm. Cost is Rs.1,20,000. Cutting sugarcane sett with single bud by Sugarcane sett cutter. Capacity about 1500 setts per hour and Cost of the implement is Rs.5000/-. Damage is less when compared to manual cutting. Sugarcane detrasher can be used for detrashing. Easy for handling and detrashing, reduces the labour

requirement and Cost of the implement is Rs.1200/-. Cotton Picker mechanically plucks open cotton bolls from the plant without damaging the fiber. Used in large-scale operations.

### **Fruit Crops**

Secateurs are used during June to August for pruning mango branches lesser than 1 inch thickness and for removing dry twigs. Cost of the falcon company implement is Rs. 450. Loppers are used during June to August for pruning mango branches and dry twigs less than 3 inches thickness. Cost of the falcon company implement is Rs. 1500. Mango harvester is used to harvest mangoes without damaging the fruits. Cost of the implement is Rs. 3150. Banana bunch cutter is used to harvest banana bunches. Coconut climbing cycles are used to easily climb coconut trees for harvesting coconuts and fronds and for pest and disease management. Cost of the implement is Rs. 3500/-. Coconut dehusker is used to remove the husk from coconut shells. Cost of the implement is Rs. 950. Oil palm harvester is used to harvest oil palm bunches, bring them down, and load them into a tractor.

### **Livestock**

A milking machine is beneficial for farmers with more than 4-5 dairy cows as it extracts milk 3-4 times faster than manual milking and reduces physical labour. Farmers rearing more than 30-50 backyard poultry birds can use a vaccinator to administer the specified dose of vaccine in a very short time. When raising backyard poultry birds in large numbers, de-beaking with the help of a debeaker is a good management practice to reduce pecking. With the help of a duplex poultry shed, 10 - 15 chickens can be kept in a protective shelter providing with integrated feed and protection, and manage very easily at very low cost.

### **Drone Technology**

Spraying of pesticides through Drones reduces time and cost of application and it is the Measure against labour scarcity. In Paddy, Maize, Jowar, Groundnut, Bengalgram, Chilli crops, this technology is already proved effective. An Acre can be sprayed with 10 litres capacity of Drone in 6-10 minutes.

### **ENTERPRISES**

Adopt secondary agriculture through value addition to primary agriculture for enhancing farm income and sustainable rural livelihoods. Agripreneurship promotion through farm mechanization, crop diversification, processing, value addition, organic farming. Take up Agro based enterprises such as commercial horticulture, cultivation of high value crops, animal husbandry, fisheries, poultry, and Mushroom cultivation for doubling farm income. Setting up of Agro produce processing units like Mini dhal mills, Rice Mills and decorticating mills for promotion of agripreneurship among rural youth & women. Value addition of agriculture producing through Agro produce manufacturing units such as preparation of millet value added products, bakery products, ready to cook products, ready to eat products and nutraceuticals. Entrepreneurship by agro- input manufacturing units such as production of organic manures, fertilizers, agricultural inputs, vermin compost units, Bio fertilizers and agricultural implements. Establishments and

## Maize

**Variety selection:** Recommended varieties are Megha Maize-1, Megha Maize-2, Ganga 5, Vivek Maize Hybrid 47, Vivek Maize Hybrid 53

**Sowing:** Seeds should be dibbled at a depth of 3-4 cm in rows to rows and plant to plant spacing of 65-75× 20-25 cm respectively. Depending upon the seed size, about 18-22.5 kg of seeds/ha (2.5-3 kg/ bigha) are required to ensure a high seed viability rate for optimal germination and establishment.



**Maize Variety Megha maize 1**

**Fertilizer Application and interculture operation:** Compost or FYM @ 10 t/ha should be applied. Fertilizer @ 60:40:40 kg N,  $P_2O_5$  &  $K_2O$  per hectare is recommended. Light hoeing and earthing up should be done as and when necessary. For further and adequate control of weed, atrazine should be applied @ 1.0 kg a.i/ha in 1000 liters of water as pre-emergence spray.

**Plant protection measures:** When infestation of stem borer is noticed, spraying of 5% Neem Seed Kernel Extract (NSKE) or Azadiractin 1500 ppm @ 5ml/litre of water may be done. In an area where Fall Army Worm (FAW) has been noticed, FAW pheromone traps @ 12 nos./ha may be set up.

## MILLETS

### Finger millet

**Variety selection:** Local varieties for organic condition are Kakhi, Hinnai, Chasa, Chalay and recommended high-yielding variety are VL-379, VL-380, VL-352.

**Sowing:** 4-5 kg/ha for transplanting and 10-15 kg/ha broadcasting. Treat the seed with Ridomil @ 2g/kg, carbendazim @ 2g/kg before sowing. Maintain a spacing of 25-30 cm (row to row), 8-10 cm (plant to plant). The seed should be planted 2-3 cm in depth.



**Finger millet cultivation in Arunachal Pradesh**

**Fertilizer Application and interculture operation:** Apply compost or farmyard manure @ 5-10 tonnes/ha about a month before sowing. Generally, fertilizers recommended to get a good crop are 40 kg Nitrogen, 20 kg of  $P_2O_5$  and 20 kg  $K_2O$  per ha. Apply, entire quantity of  $P_2O_5$  and half of Nitrogen at the time of sowing and remaining half of Nitrogen at 30 days after sowing. Two inter-cultural operations and one hand weeding in line sown crop is recommended. Intercultural operation using a tyne-harrow when crop is 30 days old is also recommended. In broadcast crop 1<sup>st</sup> weeding after 15-20 days of emergence of seedling and 2<sup>nd</sup> weeding 15-20 days after 1<sup>st</sup> weeding is recommended.

condition. Sprouted seeds should be sown in lines 20 cm apart with a seed rate of 75 kg/ha should be maintained. Need based fertilizer application is advocated. Pre-emergence herbicide pretilachlor @ 0.75 kg a.i./ha should be applied 2-3 days after sowing. Wherever, water management is possible, two irrigations at panicle initiation and flowering stages are recommended.

## **Maize**

**Selection of varieties:** Recommended hybrid variety are Ganga 5, Vivek Maize Hybrid 47, Vivek Maize Hybrid 53, Bio 9544 and composite variety are Dhawal, Navjot



**Maize Variety Bio 9544**

**Land preparation & sowing:** Any well drained soil is suitable for maize. Sandy and sandy loam fertile soils are preferable. It requires a deep fine and firm tilth. Depending upon the seed size, about 18-22.5 kg of seeds/ha (2.5-3 kg/bigha) are required. Seeds should be dressed with carboxin @ 2 g/kg of seed. Seeds should be dibbled at a depth of 3-4 cm in rows 65-75 cm apart and at a distance of 20-25 cm from seed to seed within the rows. Two seeds per hill are to be sown followed by thinning 10 days after germination retaining one seedling/hill.

**Nutrient management and intercultural operation:** A combination of organic and inorganic fertilizer gives better results than inorganic fertilizer alone. Compost or FYM @ 5 t/ha should be applied. In addition, 60 kg N/ha, 40 kg  $P_2O_5$ /ha and 40 kg  $K_2O$ /ha should be applied. For Hills Zone, a dose of 90 kg N/ha, 40 kg  $P_2O_5$ /ha and 40 kg  $K_2O$ /ha are recommended under rainfed condition. FYM or compost should be applied during land preparation. The entire quantity of phosphorus and potash and half of the total urea is to be applied in furrows (8-10 cm deep) and covered with 4-5 cm of soil. Sowing of seed should be done at least 2 days after fertilizer application. Light hoeing and earthing up should be done as and when necessary. For further adequate control of weed, simazine should be applied @ 0.5-1.0 kg a.i./ha in 1000 liters of water a pre-emergence spray.

**Plant protection:** When infestation of stem borer is noticed, fipronil 5 SC @ 1.0 kg/ha in 200-250 liters of water should be sprayed with power sprayer or in 700-800 liters of water with hand sprayer. To control the fall army worm apply emamectin benzoate 3% WG+ thiamethoxam 12% WG @ 0.1 % followed by application of chlorantraniliprole 18.5 % Sc @ 0.025% at 7 days interval.

## **OILSEED CROPS**

### **Sesamum**

**Selection of varieties:** Varieties like ST 1683, AST-1, SP 1181 (Madhavi), Gouri, Vinayak are suitable for Assam condition. Well drained sandy loam soils are preferable. Land should be prepared to a fine tilth by 3-4 ploughings followed by laddering. Sow the seeds from July to first fortnight of August @ 4 kg/ha for both



**Nursey raising for transplantation in paddy**

Effective management of yellow stem borer (*Tryporyza incertulus*) and leaf folder (*Cnaphalocrocis medinalis*) in paddy can be achieved by clipping terminal shoots prior to transplanting, followed by the application of Cartap Hydrochloride @2.0 g/L of water. A second application at the same concentration should be administered 20 days after the first treatment. To protect paddy nursery and other crops from grasshopper by spraying of cypermethrin 25% EC @1ml/L water or cypermethrin 10% EC @2.5 ml/L water or Deltamethrin 2.8% EC @1.5 ml/L of water is advised. After 2-3 days spray Neem oil @5ml/L of water.

### Maize

For successful maize cultivation, farmers are advised to begin land preparation in May and incorporate 3-4 quintals of lime per hectare. Sowing should be carried out in the first week of June or with the onset of the monsoon. During land preparation, the application of 100-150 quintals of well-decomposed cow dung compost is recommended to enrich soil fertility. Preferred maize varieties include PMH-3, Orange Flint, HQPM-1, and HQPM-5 short to medium duration hybrids: HQPM-1, Rajendra Shankar Makka-3, Vivek-27. Bio 9681, All Rounder and P-3522 are the Long-duration hybrids. A seed rate of 20 kg per hectare with a spacing of 60 cm × 20 cm is advised for optimal plant growth. Additionally, proper drainage systems must be established to prevent water stagnation in the field.

For effective management of Fall Armyworm (*Spodoptera frugiperda*) in maize, application of sand should be carried out at the whorl formation stage and upon the



**Management of fall army wall**

vaccination schedule should be followed before the onset of rainy season for prevention of Foot and Mouth Disease (FMD), Haemorrhagic Septicaemia (HS), Black Quarter (BQ) in cattle and buffaloes, and Peste des Petits Ruminants (PPR) in goats. Poultry must be protected from Ranikhet and fowl pox.

To increase the feed intake in high yielding dairy animal during heat stress, farmers should practice urea treatment of dry straw (4 kg urea in 40 litres water per 100 kg straw) to enhance its digestibility. Supplement animal diets with mineral mixture and common salt, and in lactating animals, give bypass protein and calcium supplements to maintain milk production. Ensure availability of green fodder during this season. Cultivate fodder crops like Maize, Sorghum, Cowpea and Napier to provide fresh and nutritious feed. In goats and sheep, provide bamboo slat flooring or dry raised platforms to prevent foot rot during wet months. Clean water should be provided to the animals three times a day. Farmers are advised to administer Electrol Powder 2g and Vimeral 5 ml per 5 litre water for adults. In case of disease outbreak in animals, wash the infected animal tongue, mouth, feet and nose by 1% red potash or 2% baking soda 2-3 times in a day. For adult birds, mix 2 g of Electrol powder and 5 ml of Vimeral in 5 litres of water. If the chick shows symptoms like drowsiness, cough and distorted feet appearing or change in colour of excreta then immediately consult the nearest Veterinarian.

### **Fisheries**

Fisherman are advised to prepare their own feed from locally available materials like oil cake, undecomposed grains, rice bran, mineral salt etc. (To prepare 100 kg fish feed there is need of 40 kg oil cake, 19 kg undecomposed grain (floor), 40 kg rice bran and 1 kg mineral mixture). In order to maintain the water colour light green, fill the pond with fresh water as and when needed. Also to maintain the proper water quality, apply lime @250 kg/ha and Potassium permagnate 500g/ha. Harvest only that much quantity which can be sold locally. Beware of gathering of fish while catching and selling the fish and at any cost maintain the social distancing.



**Diagnostic visit to pond**

should be applied within 3 days after sowing. Followed by post emergence application of Bispyribac sodium (10%) @ 250 ml/ha.

**Pest and disease management:** If symptoms of sheath blight disease are observed in paddy field, spray Hexaconazole @ 1000 ml/ha. Bacterial Leaf Blight in rice fields can be controlled by draining excess water and spraying a mixture of Validamycin @ 1250 ml, Streptomycin @ 30 g, and Potash @ 37.5 kg per ha. It is recommended not to apply/spray urea during this period. For the management of panicle mite, spray Diafethiuron 50% WP @ 1.2 g/L water along with Propiconazole @ 1ml/L at late tillering stage. If adult insects of Yellow Stem Borer are observed in the rice crop, destroy the egg masses immediately. Remove 'dead hearts' (dried central shoots) from the plants. Install pheromone traps @ 5 traps/ha to monitor stem borer activity in the rice field. For control, spray Fipronil 5% SC @ 1000 ml/ha. For control of leaf folder and army worm caterpillars, observe the population of larvae if it >1 per leaf, take up a spray of Fipronil 5% SC 1000 ml/ha.

### **Utera**

When the rice crop is approaching maturity and the field retains some moisture, farmers can opt for relay cropping (utera) with linseed or lathyrus. Before sowing Kharif crops, farmers should arrange for necessary agricultural tools and equipment, carry out any required repairs, and perform deep ploughing using a MB plough. Cover the fields with polythene can help destroy weeds, soil-borne diseases, and insect eggs. If green algae appear in paddy fields, drain the water. Place a pouch of copper sulfate (blue vitriol) where water enters the field.

### **Maize**

Use of high-yielding varieties such as Vivek Hy-17, Vivek Hy-43, Pusa Vivek QPM-9, VLQPM-45, Hybrid Maize-1, HQPM-1, Vivek Makka Hy-51, Pusa HM-4, DMRH-1308, IMH 230, and IMH 231. A germination test is essential to ensure an optimum number of healthy plants in the field. This should be followed by bio-priming or seed treatment using Carbendazim 50% WP @ 2 g/kg seed, Imidacloprid 600 FS @ 2 g/kg seed, *Trichoderma* @ 10 g/kg seed, and *Azotobacter* @ 10 g/kg seed to enhance productivity. The optimum sowing time is from June 15 to June 30, with a spacing of 60 x 25 cm (row x plant). Apply balanced fertilizers, including zinc-based fertilizers, for better crop growth.

For weed control, apply Tropicamezone 33.6% SC @ 30 ml per acre as a post-emergence spray at 20-25 days after sowing. The pupae of Fall Armyworm and stem borer, as well as the spores of *Rhizoctonia solani* (the causal agent of sheath blight), remain in the soil. Therefore, deep summer ploughing is crucial to reduce the inoculum potential. The infestation of Fall Armyworm can be significantly minimized by dry sowing or sowing at the onset of the monsoon, practicing crop rotation, and alternating maize with other crops that are not affected by Fall Armyworm. Additionally, apply a basal dose of nutrients and use the ridge and furrow method of bed preparation. These practices should be completed by the last week of May or the first week of June. Monitor regularly to prevent Fall Armyworm infestations. If necessary, spray Spinetoram 11.7 SC @ 350 ml/ha or Emamectin Benzoate 5 SG @ 200 g/ha. To control leaf blight disease, take up spray of Mancozeb 75 WP @ 1000 g/ha.

and crossbred animals are more susceptible to heat stress. Buffaloes are particularly vulnerable due to their dark skin and fewer sweat glands.

Animals experiencing heat stress or other health issues may exhibit rapid breathing, characterized by panting, fast and shallow breaths, and mouth breathing. Their skin often becomes dry and rough, while the eyes may appear red and watery. A noticeable reduction in feed and water intake is common, accompanied by digestive disturbances such as acidity, loose stools, and slowed gut movement. Urine output tends to decrease, indicating impaired hydration or kidney function. Affected animals may also display lethargy or weakness, often sitting or lying down frequently. Additionally, an increased heart rate is typically observed, often linked to heavy or labored breathing.

**Housing and Environment:** Keep animals in the shade and protect them from direct sunlight. Ensure adequate space in the shed, as overcrowding increases heat. Plant trees around the animal shelter to keep the surroundings cool. Construct well-ventilated sheds that allow hot air to exit and cool air to enter. Keep the shed roof high for better airflow. Paint the roof white to reduce heat absorption. Place mulch or wet jute sacks on the roof to lower the temperature.

**Cooling:** Install fans, foggers, and water sprinklers. Hang wet gunny sacks or soaked sheets around the animal shed. Ensure a constant supply of cool, clean water in cement tanks.

**Feed and Water Management:** Provide animals with continuous access to cool and clean drinking water. Feed them in the morning and evening to avoid generating excess body heat. Give enough salt or electrolytes to maintain the body's mineral balance.

**Vaccination and treatment:** If an animal shows signs of heat stress, consult a veterinarian immediately. Meanwhile, move the animal to a cool place, bathe it with cold water or wrap it in a wet cloth, and use a fan to cool it down. Some animal diseases can spread to humans (zoonotic diseases). Vaccination helps prevent the transmission of such diseases from animals to humans and from one animal to another. Never vaccinate animals that are sick, weak, or under stress. Treat them first and vaccinate only after recovery. Follow the prescribed vaccination schedule for livestock and get them vaccinated regularly through government or veterinary centers. Timely vaccinate to animals through FMD and H.S. Vaccination should be administered to prevent brucellosis (infectious abortion) in female calves aged 4 to 8 months.

Artificial Insemination (AI) should be continued in consultation with veterinary officer on door-to-door basis rather than direct visit to the dispensaries. To maintain productivity and improve immune status of animals, it is suggested for regular supplementation of mineral-60g/ day in adult dairy animals and 20-30g in small ruminants. Also, administer deworming medication. Offer clean and cool water three times a day. In case of injury in lactating cows, apply Himex ointment. In case of temporary shortage of availability of green fodder or concentrate feed farmers can prepare home-made feed by using grain like wheat, maize, wheat bran, oilcakes and salt. They can also prepare silage for their adult dairy animals. Due to increased maximum temperatures, goats should be kept in shaded areas and given clean, fresh water three times daily. Vaccinate goats against PPR (Peste des Petits Ruminants), Contagious Pneumonia, and ETV (Enterotoxaemia). Also, implement control measures for ticks and lice. To control ectoparasites, apply Butox or Cleaner @ 2 ml/L of water over the animal's body, ensuring the mouth is covered during application. Use smoke in animal shelters to keep away mosquitoes and other insects.

To ensure the well-being of livestock, provide adequate shade, cool water, and a mineral mix for proper nutrition. Regularly shower animals to help keep them cool. Vaccinate for Hemorrhagic Septicemia (HS) and Black Quarter (BQ), and deworm young animals to maintain their health. For tick control, use Deltamethrin. Additionally, use sanitizers in the sheds to reduce the presence of flies and mosquitoes, promoting a clean and hygienic environment for the animals.

## CENTRAL GUJARAT

This region of Gujarat includes Ahmedabad, Anand, Kheda, Vadodara, Panchmahal, Mahisagar, and Dahod. They receive 600-900 mm of rainfall annually. The region has shallow to medium-depth black and red soils, which are typically low in fertility and organic carbon.

### Maize

For sowing in Central Gujarat, use improved varieties like Narmda Moti, GM-2, GM-4, and GM-6, and hybrids such as GAYMH-1 and GAWMH-2. The ideal sowing time is from June 15<sup>th</sup> to July 15<sup>th</sup>. Prepare the land with 1-2 harrowings and 1 planking, applying manure before sowing. For spacing, use 60x20 cm for improved varieties and 75x20 cm for hybrids. The seed rate is 20-25 kg/ha for improved varieties and 25-30 kg/ha for hybrids. Fertilizer application should be 100:50:00 (N:P:K) kg/ha for improved varieties and 120:60:00 (N:P:K) kg/ha for hybrids. For plant protection against Fall Armyworm, use light traps, pheromone traps, and Neem seed powder/oil spray.

### Rice

For transplanted rice cultivation, use 20-25 days old seedlings, with 2-3 plants per hill. Treat seedlings with *Azotobacter* and PSB liquid before transplanting. Irrigate with 5-7 cm of stagnant water. For top dressing, apply 40-48 kg N/ha at the tillering stage. For plant protection, control stemborer with light traps and spray Flubendiamide or Indoxacarb. Manage Leaf Blight with Streptocycline + Copper Oxychloride, and control Blast with Tricyclazole spray.

### Soybean

**Use recommended varieties:** NRC-37, JS-335, KDS 726, Gujarat Soybean-2, and JS-9305. The ideal sowing time is from June to mid-July, with early sowing if conditions permit. Land preparation involves applying 10-15 tonnes of manure, followed by ploughing and harrowing. For spacing, use 45x7.5 cm or 30x5 cm, with a seed rate of 70-80 kg/ha. Fertilizer application should be 45:60:00 (N:P:K) kg/ha before sowing. Weeding can be done manually or with a Propaquizafop + Imazethapyr spray. For plant protection, control Bihar Hairy Caterpillar with Neem powder or pesticide spray, and manage Girdle Beetle with Novaluron or Quinalphos spray.

### Pigeon Pea

For pigeon pea cultivation, early varieties include GT-1 and Vaishali, while late varieties are BDN-2 and IPCL-87119. The ideal sowing time is June-July, after rainfall. Land preparation involves applying 8-10 tonnes of manure, followed by ploughing and harrowing. For spacing, use 45-60 x 15-20 cm for early varieties and 90-120 x 25-30 cm for late varieties. Fertilizer application should be 25:50:00 (N:P:K) kg/ha.

**Plant protection:** Application of Malathion 50 EC @ 0.75-1.25 ml/L of water effectively controls sucking pests like jassids, aphids, and whitefly.

**Disease management:** To manage bacterial blight effectively two sprays with Streptocycline @ 75g and copper oxychloride @ 500g in 500L water per ha at initiation stage of disease or at 55-60 days after sowing is beneficial.

### GREEN MANURING PRACTICES

The period between harvesting of wheat crop and transplanting of rice crop (May-June) can be utilized for growing of green manure crops. Dhaincha (*Sesbania sesban*) is very suitable cover crop which can be grown for green manuring. Sowing of dhaincha @ 30 kg seed/ ha is done in first week of May. The crop is ploughed up in 3<sup>rd</sup> week of June (45-50 DAS). Transplanting of rice is done after one week of ploughing of dhaincha crop. Apply 2/3<sup>rd</sup> dose of NPK fertilizers in fields where green manuring is done.

### WATER CONSERVATION PRACTICES

Transplanting of rice requires huge volume of water. It can be saved by bringing change in planting method of the crop. Direct seeding of rice crop has been found very successful technology and is helpful in conservation of water resources. Levelling of fields with laser Level Machine helps in water saving in transplanting conditions.

### LIVESTOCK

Temperatures are high during this month and some areas experience severe dust storms accompanied by thunder showers. Heat related diseases in animals that can be seen to affect animals during this time are fever, dehydration, decrease in body salts, loss of appetite and decrease in productivity. Animals should be protected from the heat and strong, hot, and dry summer afternoon winds. Adequate efforts should be made for fodder collection/purchase and storage for periods of shortage. To avoid loss of essential body salts in animals ensure that a salt mixture in appropriate quantity is mixed with the feed and water and given to animals. Depending upon the season, the content of the animal feed should be changed. Currently increase the quantity of wheat chaff and Jowar in the feed. Give dairy animals a balanced feed so that their milk production capacity is enhanced. Deworming of animals should be carried out. Maize, perennial grasses and other fodder species should be harvested now. Sheep should be sheared during this month.

## DEMONSTRATION OF METHOD OF SOIL SAMPLING IN FIELD

**Use of Compost/Farm Yard Manure (FYM):** Apply compost or Farm Yard Manure (FYM) at least 3 weeks before sowing to improve soil fertility and water retention. Enriching FYM with bio-agents like *Trichoderma* and *Pseudomonas* enhances its effectiveness.

**Post-Harvest Storage:** Ensure grains, pulses and millets are thoroughly dried before storage. Avoid reusing old gunny bags; instead, use treated bags soaked in 5 per cent neem solution or insecticide to prevent pest infestation.

**Pro Tray Seedling Production:** Adopt pro tray seedling technology using coco peat, perlite and vermiculite mixture for crops like tomato, brinjal, capsicum, and cauliflower. This method ensures healthy, uniform seedlings, reduces disease risk, supports timely transplanting, and is especially suitable for women farmers.

**Snails attack:** To prevent snail attacks, maintain clean and weed-free bunds. In case of infestation, apply metaldehyde baits as a control measure.

**Rainwater Harvesting:** Farmers are advised to adopt low-cost rainwater harvesting through silpaulin-lined ponds. These structures are easy to construct, cost-effective, and help store rainwater for irrigation during dry spells. Select a low-lying area, dig a pond, and line it with a durable silpaulin sheet to minimize seepage. This method enhances water availability during lean period and supports sustainable farming practices.



Demonstration on rainwater harvesting

## CEREAL CROPS

### Maize

In high hills, the best time for sowing of maize is from 15<sup>th</sup> April to 15<sup>th</sup> May using Girija variety for timely sowing and Bajaura Makka, Bajaura Popcorn, Bajaura sweet corn for late sowing condition. In mid and low hills, sowing operation should be done from 20<sup>th</sup> May to 15<sup>th</sup> June and 15<sup>th</sup> June to 30<sup>th</sup> June, respectively. University tested hybrids such as PSC-3322 Gold, PSC-4455, SMS-137, K-25 Super, Star-35, Vyas Gold, AHC-2337. HP-333 Gold, Sambha Gold can also be procured from the agriculture department sale centres at subsidized rates.

Optimum seed rate for maize is 20 kg per hectare. Sowing should be done at a depth of 3.5 cm, with a spacing of 60 cm between rows and 20 cm between plants to ensure optimal plant population (75,000-83,000 plants/ha). Kera method should be used for sowing of seed instead of broadcasting. Complete sowing 12-15 days before the monsoon (last week of June).



**Use of Kera method for seed sowing in maize**

Apply 10-15 t of FYM per hectare 2-3 weeks before sowing. In hybrid varieties, apply NPK @ 120:60:40 kg and 90:45:30 kg per hectare in high and low rainfall areas, respectively. In local varieties, apply NPK @ 80:40:30 and 60:30:20 kg per hectare in high and low rainfall areas respectively. Apply full dose of P & K at sowing and two split doses of N at first earthing up and one month thereafter. In case of Zn deficiency, apply 25 kg of zinc

sulphate heptahydrate (21%) or 16.25 kg zinc sulphate monohydrate (33%) per ha during sowing.

For weed control, use pre-emergence herbicides like Atrazine @ 1.5-2.0 kg a.i./ha for sole crops while alachlor @ 1.5 kg /ha for mixed cropping systems. Weeds can also be controlled post emergence by application of herbicide Tembotrione @120 g a.i./ha with surfactant 1 L/ha at 2-3 leaf stage or herbicide Topramezone 25.2 g a.i./ha + atrazine 750 g a.i./ha at 2-3 leaf stage.

In areas where cutworm and white grubs are serious pests, chlorpyrifos @ 2.0 litre per ha mixed in 60-70 kg sandy soil may be used for soil amendment before sowing or cypermethrin 10EC @ 1ml per litre may be used for foliar application. Monitor the crop regularly for fall armyworm incidence. If the leaves are damaged or insect faeces are seen, then apply flubendamide (Fame) or emamectin benzoate or chlorantraniliprol @ 0.5 ml/L and repeat application at 15 days interval if needed. Proper earthing up should be done to prevent lodging of kharif maize due to high wind, cyclone etc. If there is heavy rainfall, care must be taken for drainage of excessive water from the maize field.



**Demonstration of post-emergence herbicide in maize at farmers field**

## **Paddy**

Paddy nursery sowing @ 25 kg seed/ha after seed treatment with bavistin @ 2.5 g/kg for timely sown conditions may be completed from 20<sup>th</sup> May to first week of June using high yielding varieties HPR 1068, HPR 2143, RP 2421, HPR 2720, HPR 2880 and transplanting may be completed using 25-30 days old seedlings by end of June. In case of basmati rice, high yielding and disease resistant varieties Kasturi and HPR 2612 could be used. For late sowing conditions also, high yielding short duration variety HPR 2612 can be used. Varieties Bhurghu Dhan, Varun Dhan and Naggur Dhan could be used in Kullu valley and other colder areas. Red rice varieties HPR 2795 (upland areas) and HPR 2720 for irrigated transplanted area can be sown by mid-June. University tested hybrids Super-120, Star-795, AHR-8213 can be purchased from agriculture department.

Clean and cool drinking water must be available throughout the day. Ruminants should consume no less than 50-60 liters daily. In high-altitude areas, black-colored water containers help warm the water via passive solar energy. To mitigate heat stress, shade nets, white-washed roofs, and oral electrolyte supplementation are advisable. Migratory herds grazing in high-altitude pastures must be provided shaded rest areas using either natural tree cover or portable shelters.

Vector and parasite management involves the application of acaricides every 3-4 weeks, ideally alternating chemical classes to prevent resistance. Integration of neem-based herbal sprays adds a sustainable touch to ectoparasite control. Deworming should be carried out in May and again in August based on fecal sample results to rationalize the use of broad-spectrum anthelmintics. Disaster preparedness should include maintaining a district-level veterinary emergency contact list, strategic fodder stockpiling in vulnerable areas, and pre-positioning of veterinary first-aid kits in remote villages.

### **Cattle and Buffalo**

Vaccination against Hemorrhagic Septicemia and Black Quarter should be conducted during May-June, with biannual Foot and Mouth Disease vaccination scheduled before and after monsoon. Feeding must be based on balanced rations including green fodder like maize and cowpea, combined with dry roughage such as wheat or paddy straw. Exclusive feeding of fermented fodder must be avoided to prevent bloat and ruminal tympany. Lactating animals should be supplemented with 1.5-2.5 kg concentrate daily (depending on yield), along with 50 g mineral mixture and 30 g salt.

Weekly footbaths with 5% zinc or copper sulfate are recommended to reduce foot rot. Teat dipping with iodophor-based solutions after milking significantly lowers mastitis risk. Reproductive management should focus on insemination during early morning or evening hours in the June-August window to avoid heat stress-related infertility and ensure spring calving.

### **Sheep and Goats**

Sheep and goats must be vaccinated against PPR and enterotoxemia by June. Deworming of migratory flocks is essential before they ascend to summer pastures. To enhance conception rates, flushing rations composed of grains, legumes, and mineral mixtures should be introduced 3-4 weeks prior to breeding. During monsoon, diet supplementation with leaves from Bhimal (*Grewia optiva*), Timla (*Ficus palmata*), and Dhaman (*Cenchrus ciliaris*), along with dry hay, improves nutritional resilience.

Hoof trimming on a monthly basis and periodic footbaths with antiseptics help prevent foot rot. Pregnant does and ewes must not be exposed to wet grazing in the early morning to avoid enterotoxemia and associated complications.

### **POULTRY (BACKYARD AND SEMI-COMMERCIAL)**

A rigid vaccination protocol must be followed: Lasota for Ranikhet Disease on day 5 and every 2-3 months; Fowlpox at 6-8 weeks; IBD at 2-3 weeks; and Marek's Disease on day one. Thermal comfort must be maintained by keeping litter moisture under

25%, using lime beneath the bedding, and ensuring 14-16 hours of light daily in layers to sustain egg production. Vitamin C and electrolytes in water during heat peaks are essential. Biosecurity practices include use of footbaths with disinfectants; restricted access to poultry sheds by wild birds and visitors, and fogger or fan usage during the humid summer months.

### Swine

Swine units must ensure annual vaccination against Classical Swine Fever. Nutritional care of breeding sows includes daily supplementation with 60-80 mL of calcium and 50-60 g of mineral mixture. Pigs should have access to wallowing areas or showers for thermal relief. Deworming using combination drugs like Levamisole and Albendazole should be conducted quarterly.

### FODDER AND FEEDING MANAGEMENT

Fodder production and conservation must be tailored to the agro-ecological zone. In low hills, crops such as maize, bajra, cowpea, and Napier grass are ideal however in Mid-hill regions farmers may benefit from intercropping maize with cowpea and using Napier hybrids. In high hills, July-sown oats, Setaria grass, and perennial forages can be preferred. In cold dry zones, Lucerne, short-duration barley, and dried willow leaves offer high-value feed options.

African Tall maize should be sown during May-June in irrigated areas at a rate of 50 kg/ha. Silage should be prepared in July-August using chopped green fodder mixed with jaggery and salt, stored in drums or pits. Dry fodder should always be stored on raised platforms to prevent fungal spoilage. Alternate feed sources such as Azolla, hybrid Napier, and tree-based forages like *Grewia optiva*, *Bauhinia variegata*, and *Morus* spp. should be promoted. Pasture improvement through sowing *Cenchrus ciliaris* and *Stylosanthes* spp., along with rotational grazing and fencing, can prevent overgrazing and improve long-term forage productivity.

### RAINY SEASON DISEASE SURVEILLANCE

The monsoon period predisposes animals to a surge in diseases such as foot rot, mastitis, pneumonia, and enterotoxemia due to high humidity and wet housing. Livestock should be actively monitored for symptoms of Lumpy Skin Disease (LS), vector-borne conditions like theileriosis, babesiosis, and anaplasmosis. Prompt diagnosis and early reporting are crucial. In case of animal mortality, carcasses must be disposed through deep burial (at least 4 feet) with lime layering to prevent environmental contamination and disease transmission.

### Fisheries

Maintain DO >5 mg/L, pH 7.0-8.5; reinforce bunds and screen inlets to prevent escape. Apply lime @ 200-250 kg/ha; fertilize based on plankton levels. Use prophylactic  $\text{KMnO}_4$  (2-5 ppm) or salt (1-2%) baths biweekly. Feed 25-28% CP floating pellets @ 2-3% biomass/day. Stock only acclimatized fingerlings. Prevent livestock runoff in integrated systems. Maintain records for water quality, feeding, and growth.

# Jammu & Kashmir



## JAMMU

Jammu region has three agro-climatic zones; sub-tropical (upto 800 m above msl), sub-temperate (800-1500m above msl) and temperate (above 1500m above msl). Rice is the major crop in irrigated plains, while maize is generally grown in hills under rainfed ecosystem. After Rabi harvest, deep plough fields to utilize high temperatures for killing soil-borne pathogen spores, and take preventive steps to avoid post-harvest pest infestation. The crop-wise agro-advisory for kharif season is presented as below:

### CEREAL CROPS

#### Paddy

Soak paddy seed in carbendazim 10 g + streptocyclin 3 g/5 kg seed (in 8 L water) for 24 hrs before nursery sowing. In sub-tropical zone, use blast-resistant coarse/semi-fine varieties (SJR-5, Jaya, PR-113, PHB-71, Ratna) and superfine types (Jammu Basmati-129, Pusa Basmati 1121). For temperate/sub-temperate areas, prefer early, blast and cold-tolerant types like China-1039, Giza-14, K-84, K-343, K-39.



Blast disease



Stem Borer

Prepare nursery beds (dry or wet) with 15 kg FYM, 60 g urea, 50 g DAP per 10 m<sup>2</sup>. Use 16-18 kg/acre seed for fine/coarse varieties (2+ seedlings/hill), 6 kg/acre for hybrids; treat seed with copper oxychloride @ 3 g/kg.

Control weeds manually or use butachlor 5G (1.5 kg/500 m<sup>2</sup>) or pretilachlor (20 g/500 m<sup>2</sup>). Apply 30:12:8 kg N:P:K + 8 kg ZnSO<sub>4</sub>/acre; split N at puddling, tillering, and panicle stages. Apply full DAP, MOP, ZnSO<sub>4</sub>, and 1/3<sup>rd</sup> N at puddling; split remaining N at tillering and panicle initiation stages, draining water before top dressing.

For chemical weed control, use butachlor 12 kg/acre after transplanting or anilophos + ethoxy sulfuron (500 + 60 ml/acre) at 10 DAT. Manage weedy rice with stale seed-bed and glyphosate (0.6 kg/acre) or paraquat (0.32 kg/acre).

For management of blast, spray the crop with tricyclazole @ 0.06% or azoxystrobin + difenoconazole @ 0.1%. For stem borer, apply Cartap hydrochloride 4G (8 kg/acre) in 5-7.5 cm standing water. Do not drain or irrigate for 72 hours. Spray Cartap hydrochloride 50 SP (240 g/acre) when there are 5% dead hearts or one egg mass/sq.m.

### **Maize**

Grow hybrids (Ganga Safed-2, Vivek-25, HQPM-1) or composites (Mansar, C-6, C-2); use 8 kg/acre seed for line sowing, 12 kg for broadcast, and 14-16 kg in hills. Treat seed with Thiram/Captan (3 g/kg) or Carbendazim (2 g/kg) + Chlorpyrifos (5 ml/kg) in cutworm-prone areas. Apply 6 t FYM/acre 2-3 weeks before sowing; with FYM, reduce NPK by 25%; otherwise, use 24:16:8 kg NPK + 4 kg ZnSO<sub>4</sub>/acre.



**Blister Beetle**



**Fall army worm**

**Maintain spacing:** 75×20 cm (hybrids), 60×20 cm (composites); use Atrazine @ 0.4 kg/acre or weed manually at 15 & 30 DAS. Keep weed-free till 40 DAS; do earthing-up at 30 DAS for stability and moisture.

For pests, apply chlorpyrifos 1.5% D (10 kg/acre) in last ploughing, use light (2/acre) & pheromone traps (6/acre), spray methyl demeton, cypermethrin or cartap hydrochloride in whorls. For Blister Beetle, Cut Worm, Stem Borer & Army Worm in maize, use acetamiprid 20SP @ 0.2 g/L or chlorpyrifos 20EC @ 480 ml/acre, apply chlorpyrifos 1.5% D @ 10 kg/acre in soil for cut worm, cartap hydrochloride 4G @ 8 kg/acre in whorls for stem borer, and adopt trap cropping (okra/arhar) and manual beetle removal for blister beetle. For smuts, treat seed with carbendazim/carboxin @ 2 g/kg; for stalk rot, drench with copper

oxychloride 3 g/L or apply bleaching powder @ 10 kg/acre; for leaf blight, spray propiconazole @ 0.1% at disease onset.

### **FODDER CROPS**

Bajra and Jowar are important kharif fodder crops, with varieties like Giant Bajra, FBC-16, PCB 164 for Bajra, and MP Chari, Haryana Chari-260, Proagro Chari (SSG-998) for Jowar, suitable for both irrigated and rainfed conditions.

#### **Bajra (Pearl Millet)**

Use 2 kg of hybrid seed/acre, apply 40:24:10 kg NPK/acre, and manage pests like leaf caterpillars and weevils with chlorpyrifos 1.5% D @ 10 kg/acre.

#### **Hybrid Jowar (Sorghum)**

Sow 5 kg seed/acre with 20:12:6 kg NPK; for insect pests like shoot fly and stem borer, apply cartap hydrochloride 4G @ 8 kg/acre in central whorls at 10-20 DAS.

### **OILSEED CROPS**

#### **Sesame (Til)**

Use 800 g-1 kg seed/acre in July; for pests like hairy caterpillars and whitefly, spray chlorpyrifos 20EC @ 2 ml/litre or imidacloprid 17.8 SL @ 0.3 ml/litre.

### **PULSE CROPS**

#### **Mash (Black Gram)**

Use 6-8 kg seed/acre; varieties include Pant U-19, Uttara, Pant U-31, NDU 99-3, KUG 479; spray chlorpyrifos 20EC @ 2 ml/litre or imidacloprid 17.8 SL @ 0.3 ml/litre to manage hairy caterpillar and whitefly.

#### **Moong (Green Gram)**

Use 6-8 kg seed/acre; varieties include Pant Mung-6, Pusa Vishal, SML 668, Pusa 0672, Satya; manage YMV and sucking pests with resistant varieties and spray dimethoate 30 EC @ 1 ml/litre or thiamethoxam @ 0.03%.

#### **Rajmash (Bhaderwah Local/Chinta Selection)**

Intercropped with maize (row spacing 75 × 20 cm); sow 9.5 kg seed/acre; treat with copper oxychloride @ 3g/kg; control aphids and anthracnose with chlorothalonil @ 2 g/L spray.

### **COMMERCIAL CROPS**

#### **Sugarcane**

Grown in parts of Jammu and Kathua, high-yielding sugarcane varieties like COJ-64, COJ-81, and CO-1148 are preferred. Manage termites with chlorpyrifos, borers with cartap, and diseases like red rot and grassy shoot using resistant varieties and appropriate sprays.

#### **Saffron**

Saffron requires 20-24 q corms/acre, with fertigation using 8 kg of Urea, 24 kg of DAP, and 13 kg of MOP, and management involves deep ploughing, raised beds, disease-free corms, and rodent control.

**Bitter Gourd**

Varieties such as Puas Domausami, Punjab 14, Pusa Vishesh, Kalyanpur Baramasi can be used for summer and rainy season at spacing of 1-2 m × 75-90 cm with seed rate 2 kg/acre. Apply 87 kg/acre urea, 43.5 kg per/acre DAP and 33.2 kg per acre MOP. High nitrogen under High temperature conditions promotes staminate flowers.

**Pumpkin**

Pusa Vishwas, Azad Pumpkin1, Arka Suryamukhi, Arka Chandan, Punjab Samrat varieties can be raised at spacing of 2-2.5m x 75-90 cm with 2 kg seed per acre. Besides, FYM, apply 80 kg urea, 44 kg DAP, 17.2 kg per acre MOP per acre.

**Sponge Gourd**

Use Pusa Chikni & Pusa Supriya for the rainy season; sow 2 kg seed/acre at 1.5-2.5 m × 75-90 cm spacing; apply 34.8 kg urea, 26 kg DAP, and 20 kg MOP/acre.

**Cucumber**

Varieties such as Japanese, Long Green, Straight Eight, Poinsette and local kheera can be planted at the spacing of 1.0-1.5 m × 60-90 cm with seed rate of 0.8 to 1.2 kg per acre. Apply 34.8 kg urea, 22 kg DAP and 16.8 kg MOP per acre. Avoid excessive use of fertilizers and irrigation. Apply nitrogenous fertilizer at proper dose to avoid the staminate flowers. In cucurbits, control red pumpkin beetle with acetamiprid 20 WP @ 0.2 g/l; for fruit fly, install methyl eugenol traps (2-4/acre) or spray malathion 1 ml + 10 g gur/l in the evening. Similarly, manage powdery mildew with wettable sulphur @ 0.2% every 7-10 days. Also, prevent fruit rot by keeping fruits off the soil and harvesting ripe fruits promptly.



**Fruit fly and red pumpkin infestation in cucumber**

**Mushroom (Button/Oyster)**

It Can be processed into pickles, dried, or preserved using steeping solution.

**LIVESTOCK****Cattle**

Provide adequate shade, ventilation, and clean drinking water to protect animals from high humidity and heat. Ensure timely deworming and vaccination against prevalent diseases like Foot and Mouth Disease (FMD) and Hemorrhagic Septicemia (HS). Cultivate Kharif fodder crops like maize, sorghum, or bajra and ensure silage preparation for consistent feed supply.



**Foot and Mouth Disease (FMD)**

**Recommended fungicides sprays:** Ediphenphos 50 EC @ 100 ml/ 100 liter water; Tricyclazole 75 WP @ 60 g/ 100 liter of water; Isoprothiolane 48 EC @ 100 ml/ 100 liter water; Hexaconazole 5 EC @ 30 - 50 ml/ 100 liter water; Mancozeb 75 WP @ 200 g + Carbendazim 50 WP @ 50 g/ 100 liter water; Carbendazim 50 WP @ 50 - 100 g/ 100 liter water & Propiconazole 50 WP @ 100 ml/ 100 liter water. It is worthwhile to mention here that the spray of same fungicides during a single cropping season should not be repeated.

## **Maize**

**Nutrient Management:** Apply well-decomposed compost or FYM uniformly at 15-20 t/ha during land preparation. Vermicompost @ 2.5 t/ha can substitute 5 t FYM/ha and 25% of the recommended NPK.

**For Irrigated Maize:** Hybrid Varieties: Urea: 100 kg/ha, DAP: 160 kg/ha, MOP: 67 kg/ha & Zinc Sulphate: 20 kg/ha (basal dose). For Composite Varieties: Urea: 80 kg/ha, DAP: 130 kg/ha, MOP: 50 kg/ha & Zinc Sulphate: 15-20 kg/ha (basal dose).

**Rainfed Maize:** Hybrid Varieties: Urea: 60 kg/ha, DAP: 100 kg/ha, MOP: 34 kg/ha & Zinc Sulphate: 15 kg/ha (basal dose). For Composite Varieties: Urea: 28 kg/ha, DAP: 87 kg/ha, MOP: 33 kg/ha & Zinc Sulphate: 10 kg/ha (basal dose)

**Weed Control:** Apply Atrazine (Atratraf 50 WP or Gesaprim 500 FW) @ 1.0-1.5 kg a.i./ha in 600 litres of water 2-3 days after sowing.

**Earthing Up:** Perform at knee-high stage (35-40 DAS) in early-sown crops.

**Top Dressing of Urea:** Urea @ 80 kg/ ha is recommended for Irrigated Hybrid maize while the doses of urea for irrigated composite, rainfed hybrid and rainfed composite are 65, 50 and 40 kg/ ha, respectively. Apply during first weeding, hoeing, and earthing up at the knee-high stage. It is important to avoid moisture stress during the knee-high stage in early-sown crops.

## **PLANT PROTECTION**

**Cutworm Management:** Prefer early sowing (up to 3<sup>rd</sup> week of April). Apply Carbofuran 3% CG @ 32.5 kg/ha before sowing. Install light traps (5/ha) and pheromone traps (20/ha) for monitoring. Use pitfall traps to collect and destroy larvae.

**Armyworm Management:** Uproot stubbles of the previous crop. Spray Chlorpyrifos 20 EC @ 150 ml/100 litres of water.

**Disease Management:** Maintain field sanitation. Use disease-free seeds. Treat seeds with: Mancozeb 75 WP @ 3 g/kg seed, or Metalaxyl MZ 72 WP @ 2.5 g/kg seed. Foliar Spray for Disease Management in Maize: At the appearance of disease symptoms, apply a foliar spray of either: Mancozeb 75 WP @ 3 g/litre of water, or Hexaconazole 5 EC @ 0.05% concentration

## **PULSE CROPS**

### **Moong (Green Gram)**

Sowing should be done between 15<sup>th</sup> May and 30<sup>th</sup> May. Use a seed rate of 20-25 kg/ha. Maintain a spacing of 30 × 10 cm (row-to-row × plant-to-plant). Inoculate seeds with *Rhizobium* and *PSB* culture @ 5-10 g/kg of seed before sowing. Ensure the field is well pulverized and levelled before sowing. Ensure adequate soil moisture at sowing for better germination. Apply well-decomposed compost or FYM @ 10-15

**Important management tips for raising cucurbits:** Select a plot with an efficient drainage system. Plant the seed when the soil has attained the optimum temperature between 20- 25°C when the winter is over and follow space planting as recommended. Irrigate twice or weekly during hot summer. Apply fertilizers properly when the vine starts growing. Adopt timely plant protection measures against diseases and insect pests. Harvest the melons (except long melon) when they are mature whereas harvest the gourds when in immature state. Procure the seed of a recommended variety from a reliable source. Hand pollination may be adopted to get higher yields. Do not grow cucurbits under shade, if possible. Do not grow cucurbits under very humid conditions. Do not irrigate heavy soils after sowing. Do not allow tender fruits to go beyond marketable stage. Do not sow the seeds in nursery because cucurbits do not withstand transplanting shock. However, as recommended seedlings can be raised indoors in polythene bags.

### **Leguminous Vegetable**

#### **French Bean**

Bush type varieties like Bountiful, Master, Arka Komal, Contender and pole type varieties like Kentucky Wonder, Painted Lady and Asparagus beans can be grown successfully in Kashmir conditions with seed rate of 80-100 kg/ ha, 25-30 kg/ ha and 20-25 kg/ ha and spacing of 30 cm x 10 cm, 100 cm x 50 cm and 100 cm x 50 cm in Bush type, Pole type and Asparagus beans respectively. Sowing can be carried from April - June. Entire FYM, K and  $\frac{1}{2}$  N (recommended dose is of FYM is 20-25 t and 30:60:60 NPK at the time of sowing and other  $\frac{1}{2}$  N when true leaves emerge.

#### **Bhindi (Okra)**

Pusa Sawani (60-80 q/ ha) is Recommended variety of okra for Kashmir valley. Seeds pre-soaked in hot water at 50°C for 30 minutes and directly sown on the ridge tops which are 15 -20 cm high above ground and 45 cm apart can be sown in Late April-May with seed rate of 15-20 kg/ha. After the seeds have germinated and plant has attained height of about 7-10 cm, plants are thinned to a spacing of 30cm between plants. NPK @120:90:60 can be applied.

**Seed Production:** Besides all cultural practices required for commercial production of okra, retain first 10 fruits on main stem for obtaining higher seed yield. Foliar application of 100 ppm (1mg/L) NAA (Naphthalene Acetic Acid) after 30 days and 50 days of sowing gives higher yield.

### **LIVESTOCK**

#### **Cattle**

Maintain cleanliness in and around farms and ensure availability of clean water. Ensure that the sheds receive enough sunlight during the day to prevent dampness and to complement the cleaning process. Ensure 6-8 hours of daily grazing to animals if community pastures are available. Concentrate mixture should be fed as per the body weight, production and stage of pregnancy to ensure optimal productive performance. To mitigate scarcity of fodder during winter months, fodder crops such as maize, bajra, and sorghum should be sown, so that it can be preserved through ensiling.



**Management of paddy field**

per hectare within 1-2 days of sowing. Bispyribac 10 SC should be applied at 250 ml/hectare 20-25 days after sowing. In case of chlorosis (iron deficiency), spray 1% ferrous sulphate solution (2.5 kg in 250 litres of water) two to three times at weekly intervals, and ensure adequate irrigation. For Terai paddy, seedlings should be prepared as per medium land rice guidelines. Varieties like Rajendra Mahsuri, Rajshree, and BPT-5204 should be selected and purchased in advance, preferably in May.

### **Maize**

Farmers are advised to plough the land during May and apply 3-4 quintals of lime per hectare. Seeds should be sown in the first week of June or at the onset of the monsoon. To control termite populations and larvae, 100-150 quintals of well-decomposed farmyard manure (FYM) along with Methyl Parathion 2% dust at the rate of 25 kg per hectare should be incorporated into the soil during land preparation. Use 20 kg of seed per hectare of recommended maize varieties such as PMH-3, Orange Flint, HQPM-1, or HQPM-5. At the time of sowing, apply 86 kg of urea, 130 kg of DAP, and 68 kg of MOP per hectare. If the field remains dry, provide irrigation after 30 days. After proper weeding, apply an additional 65 kg of urea. Ensure the field has proper drainage to prevent water stagnation. Two weeks after germination, apply 10-15 granules of Fipronil 0.3G or Carbofuran 3G in each plant to manage early-stage pests. To control the incidence of Fall Armyworm in the maize crop, spray Emamectin Benzoate 5% SG at the rate of 1 gram per 3 litres of water.

### **PULSE CROPS**

Farmers are advised to grow pulses (especially pea, chickpea, and lentil) during Kharif season. The recommended varieties for upland field include: Pigeon pea- Narendra Arhar-1, Narendra Arhar-2, Rajeev Lochan, IPA 203, Malviya Arhar-Mal-13; Black gram- WBU-109, Birsa Urd -1, PU-31; Green gram- Virat, Shikha, HUM-12, IPM 2-14, IPM 2-3, HUM-16; Horse gram- Birsa Kulthi-1, Payur-2, Madhu, VLG-19 etc. Oil seed crops like Niger- Utkal Niger-150, DNS-4, BNS-3, Pooja, JNS-29, JNC- 6, BNS-1; Sesame-RT-351, Krishna, Sekhar, RT-346, GT-4, GT-6, Punjab-1; Groundnut- K1812, TLG-45, ICGV-00350, K-6, TG-37A, TG-38 Dharni, Grinar-3, TG-51, TCGS-1043, AK12-14 etc.

fungicide in 10 litres of water. This solution should be sprayed on the lac-coated branches using a sprayer (make sure to wear a mask while spraying). For Rangini-Baisakhi Beehan lac crops, cut the Ari Lac from the Plum trees and prune the entire Plum tree. Farmers who have not yet pruned the vacant Palas trees should complete this work as soon as possible.

### **LIVESTOCK**

Proper vaccination schedules should be followed before the onset of the rainy season to prevent FMD and Black Quarter disease in cattle and buffaloes. Vaccination of sheep and goats against PPR and Enterotoxemia is also recommended. For dairy units, farmers should prepare feed from locally available ingredients such as maize, wheat, and pulses. It is advised to provide 20-25 grams of mineral mixture as supplementary feed to the dairy animals. Feedings should be done during the cooler parts of the day, such as early morning or late evening. Incorporating green fodder and moist feed will help with hydration. Avoid spoiled feed to prevent digestive issues. Proper sanitation and hygiene are essential. The floor of the animal shed should be cleaned regularly to prevent disease infestation, and milk utensils and equipment should be frequently cleaned. Provide adequate space to the animals to prevent overcrowding.

#### **Piggery**

For pig units, farmers are advised to grow green fodder and add 40% green matter with 60% concentrated fodder, such as rice husk, maize, wheat husk, oil cake, mineral mixture, and salt. To prepare concentrated feed, mix 60% rice husk, 20% ground corn, 20% oil cake, and 20% rice bran. Also add 10% wheat bran, 8% salt, and 1.5% vitamin and mineral mixture. Providing extra immunity boosters like zinc and selenium in the feed is recommended. Ensure pigs are vaccinated against common monsoon-related diseases like Swine Fever and Foot and Mouth Disease (FMD).

#### **Poultry**

For poultry units, cleaning the poultry shed every other day is essential. Regularly sprinkle bleaching powder or lime to disinfect the area, and all equipment and sheds should be properly disinfected. In backyard poultry farming, birds should be allowed to roam in the open yard during the day. Deworm the birds using Piperazine Hydrochloride at 5 ml per 10 birds. Vaccinate the birds against Ranikhet and fowl pox. Provide clean and fresh water three times a day. For adult birds, mix 2 grams of Electrol powder and 5 ml of Vimeral in 5 litres of water. If any symptoms such as drowsiness, cough, cramps in legs, or changes in stool color are observed, contact the doctor immediately.



**Diagnostic visit to poultry farm**



In-situ moisture conservation measures covering both soil and land management practices to be practiced. They include field bunding, compartmental bunding, ridges and furrows, conservation furrows, broad bed and furrow systems, dead furrows at 30 DAS especially in red soils for groundnut, groundnut+redgram, finger millet + pigeonpea, pigeonpea + field bean and pigeonpea + cowpea intercropping. Frequent interculture required both in kharif and rabi for weed control and moisture conservation. Avoid dry sowing of crops such as cotton, groundnut especially in shallow soils. Sowing needs to be done only when sufficient moisture is available in soil root zone. Unless 50 - 60 mm rainfall is received over 2-3 days, sowing should not be carried out. Facility of crop insurance including weather-based insurance to be availed. Follow the weather advisories available for the region. Do not burn crop residue. Take up rotavator operation in harvested fields of Cotton, Wheat and Safflower for incorporation of stubbles and crop residue in the soil. Border crops (Jowar/Bajra/Maize) should be practiced in vegetable growing fields which will prevent the vector entry in the main field. Attend ploughing operation immediately after the harvest of rabi crops which reduces the pest & weed incidence in next season.

## CEREAL CROPS

### Paddy

Paddy is commonly cultivated using a seed rate of 62.5 to 75 kg per hectare, depending on the variety. Popular varieties include MO-4, Abhilasha, Intan, RNR 15048, Tunga, KHP-10, and BPT 5204. Sowing is generally done in June to August, with some varieties like RNR 15048 and BPT 5204 sown in the first fortnight of July and transplanted in August. The recommended fertilizer dose for most varieties is 75:75:90 kg of N:P:K per hectare, while BPT 5204 and RNR 15048 require a higher nitrogen input of 150 kg along with 75:75 kg of phosphorus and potassium. Seed treatment is a crucial practice and involves the use of Carbendazim 50 WP at 2 grams per kg of seed to prevent fungal infections. In some cases, seeds are also treated with

*Streptocycline* (0.1 g), copper sulfate (0.1 g), and biofertilizer *Azospirillum* at 400 g per 25 kg seed mixed with 50 g of jaggery.

Seedlings may also undergo root dipping in Carbendazim solution before transplanting. Transplanting is done either manually or with machines, maintaining a spacing of 20 by 10 cm or 8 by 4 inches depending on the practice. Weed management begins with pre-emergent application of Pendimethalin 30 EC at 1 L/acre or Pyrazosulfuron ethyl 10% WP at 200 g per hectare, followed by post-emergent spraying of Bispyribac sodium 10 SC at 100 ml per acre. Paddy crops can be rainfed, especially for varieties like MO-4 and Tunga, while others require irrigation during critical stages such as tillering, panicle initiation, flowering, and grain filling. Additional foliar nutrient application includes potassium nitrate at 10 grams per litre of water at 60 days after transplanting, along with nano urea, nano DAP, and micronutrient sprays such as Samriddhi. Zinc sulphate may also be sprayed at 10 g per litre.

Pest management involves several strategies. For stem borer control, clipping the terminal leaf during transplanting is recommended, along with spraying Chlorpyrifos 20 EC at 2 ml per litre or applying Fipronil 0.3G at 10 kg per acre and using pheromone traps at the rate of 7 per acre. Leaf folder can be managed with Flubendiamide 48 SC at 0.08 ml per litre, while gall midge is controlled with Chlorpyrifos sprays. For brown plant hopper and case worm, soil application of Carbofuran granules (10% at 5 kg or 3G at 8 kg per acre) is effective. In terms of disease management, blast is addressed through seed treatment with Carbendazim and spraying Tricyclazole 75 WP at 0.6 to 1 g per litre or Propiconazole 25 EC at 1 ml per litre. Brown leaf spot is managed by spraying Carbendazim at 1 g per litre. Bacterial blight is controlled using a combination of copper oxychloride (50 g) and *Streptocycline* (25 g) per acre. Udubatta disease can be prevented by treating seeds with Carbendazim 50 WP at 4 g/kg. To avoid grain discoloration during the harvesting stage, spraying Nativo (Trifloxystrobin + Tebuconazole) at 0.4 g / litre is recommended.

Other cultural practices include salt water treatment to eliminate chaffy seeds and incorporating green manure like sunhemp, particularly with the Intan variety. Ratoon cropping is practiced for varieties like Intan and IET 7191, but only if the main crop performs well.

### **Finger Millet**

Recommended variety is MR-6, with a seed rate of 12.5 kg/ha, sown during June-July. Fertilizer application is 100:50:50 kg NPK/ha. *Azospirillum* biofertilizer is applied. Line sowing is done at a spacing of 30 × 10 cm. The crop is grown under rainfed conditions with no intercrop. Hand weeding is the primary weed control method. Blast disease is managed by spraying Saaf at 2 g per litre of water. Stem borer is controlled using Coragen at 0.5 g per litre of water.

### **Maize**

Maize is cultivated during both Kharif and Rabi seasons using a variety of hybrids such as CP-818, Arjun, Dharma (G.H. 150125), Shruti (G.H. 0727), and private hybrids. The seed rate varies between 15 to 25 kg per hectare depending on the hybrid and season. Sowing is done from May to July for Kharif and from September to

October for Rabi crops. Recommended spacing is 60 cm × 20 cm or 60 cm × 30 cm, and sowing can be done by line sowing, seed-cum-fertilizer drill, or dibbling method. Fertilizer application varies with soil and irrigation conditions. For irrigated maize, a dose of 150:65:65 kg NPK per hectare along with 25 kg zinc sulphate is advised, while in dryland conditions, 100:50:25 kg NPK with 10 kg zinc sulphate is sufficient. Top dressing of urea at 45 days after sowing (DAS) improves yield. Application of micronutrients such as 19:19:19 at 5 to 10 g/litre of water and nano urea at 4 ml per litre is recommended during key stages like vegetative growth, tasseling, silking, and grain filling. Biofertilizers such as PSB, KSB at 500 g/ha, and Azospirillum at 500 g/ha for seed treatment further enhance nutrient uptake.

Intercropping systems such as maize with greengram, blackgram, cowpea, soybean, redgram (6:1), avare (8:1), and pigeonpea (4:2) are beneficial in increasing land-use efficiency. Irrigation should be provided every 8-10 days in red soils and every 10-12 days in black soils, with special attention during sowing, vegetative, and grain filling stages. Weed management includes one intercultivation at 30 DAS and hand weeding at 40 DAS. Atrazine 50% WP at 2-2.5 kg/ha is recommended as a pre-emergent herbicide, followed by post-emergent application of Tembotrione 42% SC at 287.5 ml/ha or Tambotrion 34.4% SC at 285 ml/ha. For disease management, Turicum leaf blight can be controlled with Mancozeb 75 WP at 2.5 g/litre, downy mildew with Metalaxyl + Mancozeb at 2 g/litre, and bacterial stalk rot with Copper hydroxide at 2 g/litre.

Seed treatment with Captan or Thiram at 4 g/kg of seed is also essential. Fall army worm is a major pest in maize and can be managed by seed treatment with Chlorantraniliprole at 0.4 ml/kg, spraying Emamectin Benzoate 5 SG at 0.2-0.3 g/litre, Chlorantraniliprole 18.5 SC at 0.2-0.3 ml/litre, and installing pheromone traps at 10 per hectare. Biological control with *Nomuraea rileyi* or *Beauveria bassiana* at 10 g/litre is also effective. Aphids can be managed with Azadirachtin 300 ppm at 5 ml/litre, while Spinetoram 11.7% SC at 0.5 ml/litre applied at 15-20 DAS helps manage early-stage armyworm attacks. In some cases, Diplodia ear rot has been observed at harvest.

## **OILSEED CROPS**

### **Sunflower**

Sunflower varieties suitable for cultivation include KBSH-44, KBSH-53, KBSH-78, KBSH-85, and KBSH-90. The recommended seed rate is 5 kg per hectare, and it can be sown during June-July, September-October, and December-January. Under irrigated conditions, apply fertilizers at the rate of 36:36:25 kg NPK/ha, and under rainfed conditions, apply 15:20:15 kg NPK/ha along with 4 kg of zinc and 6 kg of borax per hectare. Treat seeds with *Azotobacter* at 150 g/kg, Imidacloprid at 5 g/kg to control necrosis, and Metalaxyl at 6 g/kg for downy mildew. Maintain a spacing of 60 × 30 cm, and irrigate every 8-10 days in red soils and every 15-20 days in black soils. Apply a foliar spray of borax at 2 g/l during flower opening. For weed control, use Butachlor 50 EC or Alachlor 50 EC at 2.0 l/ha or Pendimethalin 38.7 EC at 1.8 l/ha. To manage diseases and insect pests, spray Imidacloprid 17.8 SL at 0.5 ml/l at 25 and 45 DAS for necrosis and sucking pests; Cypermethrin 35 EC at 1 ml/l for leaf-eating caterpillars; 5% NSKE or NPV (100 LE) for head borer. Practice thinning at 15-20 DAS and rotate with legumes for better productivity.

## **Chilli**

Chilli cultivation includes both irrigated and rainfed practices, with popular varieties being Byadagi Dabbi, Byadagi Kaddi, Hybrid 5531, Krishnaprabha Rudra, Noolvi Dabbi for dry chilli, and Arka Tanvi, Arka Yashasvi, and Nagavi Giddagayi for green chilli. Under irrigated conditions, the seed rate is about 250 to 400 grams per hectare, whereas for rainfed conditions, it ranges from 1 to 1.25 kilograms per hectare. The ideal sowing time is May-June for kharif and October or January-February for irrigated crops, with transplanting carried out from July to August. Fertilizer recommendations for irrigated chilli are 150:75:75 kg NPK/ha, while for rainfed conditions, it is 100:50:50 kg NPK/ha.

Seed treatment should be done using Imidacloprid 70 WS at 10 g/kg, Thiram 75 WP at 2 g/kg, *Trichoderma viride* at 4 g/kg, or *Azospirillum* at 200 g/kg, and seedlings can be dipped in *Azospirillum* solution (400 g per litre of water) during transplanting. Spacing should be 75 cm x 45 cm under irrigation and 90 cm x 90 cm under rainfed conditions. Both transplanting and direct sowing methods are practiced. Intercropping with marigold and crops like onion or Bt cotton is encouraged, along with border cropping using hybrid maize or sorghum. A foliar spray of NAA at 0.3 ml/L during flowering and Arka Vegetable Special at 3 g/litre enhances plant growth and yield. Irrigation should be scheduled once in 5-7 days in red soil and once in 15-17 days in black soil.

Weed management is achieved using Pendimethalin at 0.5 litres/ha or 4 ml/litre applied on the day of sowing or the next day. Major diseases like anthracnose, powdery mildew, *Alternaria* leaf spot, and murda complex can be managed by integrated practices. Specific foliar sprays include Carbendazim 50 WP at 1 g/litre or Fenconazole 10 EC at 1 ml/litre for powdery mildew and Mancozeb 75 WP at 2 g/litre or Copper Oxychloride at 3 g/litre for anthracnose. Pests such as thrips, mites, and whiteflies are controlled through IPM practices including seed treatment with Imidacloprid, foliar sprays of Difenthiuron 50 WP at 1 g/litre, and installation of 25 blue and yellow sticky traps per hectare.

## **Tomato**

Recommended varieties for tomato include Arka Rakshak, Arka Samrat, Arka Abhijith, Arka Vardhan, and Arka Vikas, with fruits weighing around 75-100 grams. Suitable sowing periods are June-July, October-November, and January-February. For nutrient management, hybrids require 250:250:250 kg/ha of NPK, while varieties require 115:100:60 kg/ha. Seeds are treated with *Trichoderma subtilis* @ 50 g/kg or Thiram @ 1.25 g/kg for protection against seed-borne diseases. The crop is planted on ridges and furrows (90 x 45 cm) or raised beds (75 x 45 cm) for rainy season crops. Foliar nutrition with Vegetable Special @ 3 g/l is applied fortnightly.

Irrigation is scheduled once in 5-7 days based on moisture and weather. Weed control is achieved using pre-emergent herbicides like Alachlor or Butachlor @ 1.5 litre/ha. Major diseases include damping off (managed by seed treatment or drenching with Metalaxyl or Mancozeb), Tomato Leaf Curl Virus and Spotted Wilt Virus (managed by vector control and removal of infected plants), early and late blight (controlled by sprays of Azoxystrobin, Copper Oxychloride, or Metiram), bacterial wilt (managed by soil drenching with Streptocycline and crop rotation),

powdery mildew (Myclobutanil spray), and nematodes (managed by trap cropping and soil drenching with Velum Prime or Carbofuron).

Pests such as thrips, whiteflies, and Tuta absoluta are managed by sprays of Imidacloprid 17.8 SL @ 0.5 ml/l + neem oil 2.5 ml/l, Fipronil 80 WG @ 40 g/acre, or combinations of Fipronil and Imidacloprid, Cyantraniliprole, Acetamiprid, Spirotetramat, and Pongamia oil. For bacterial wilt, soil drenching with Copper Oxychloride 50% WP @ 3 g/l is advised. Avoid over-irrigation and maintain optimal moisture. Intercropping with marigold (1 row every 16 tomato rows) and border cropping with maize or bajra helps reduce pest incidence.

## FRUIT CROPS

### Pomegranate

The preferred variety is Bhagwa, planted at a spacing of 3.2 x 2.6 meters with a planting density of 1200 plants per hectare between June to September. Propagation is done through air layering and tissue culture, with cuttings treated using 5000 ppm IBA. Intercropping with cowpea, cucumber, cabbage, onion, and pea is practiced during the first 5-6 years. Fertilizer dose is 197:99:99 NPK kg/ha. Borax is applied at 1 g/L of water. Drip irrigation is scheduled at 11 L/day during rainy season, 12 L/day in winter, and 22 L/day in summer. Herbicides such as Diuron (1 kg/ha in nursery) and Nephromide (2 kg/ha in the main orchard) are used for weed control. Pomegranate is prone to several diseases like bacterial blight, anthracnose, fruit rot, *Cercospora leaf spot*, *Alternaria fruit spot*, and *Fusarium wilt*. Management includes fungicide sprays (e.g., Dithane M-45, Captan, Bavistin), sanitation, and orchard hygiene. Pest issues include the anar butterfly (fruit borer), bark-eating caterpillars, whiteflies, aphids, mealy bugs, and fruit flies. Integrated Pest Management (IPM) practices include cultural sanitation, pruning, biological agents like *Trichogramma*, neem-based products, polythene bagging, and chemical treatments like malathion.

### Grape

Popular varieties include Thompson Seedless, Manik Chaman, and Sonaka, planted at a spacing of 2.4 x 1.8 meters with a planting density of 2314 plants per hectare from June to September. Propagation is done through hardwood cuttings, treated with IBA at 250 mg/L. Bitter gourd and ridge gourd are suitable intercrops. A fertilizer dose of 1000:500:1000 kg NPK per hectare is applied. Gibberellic acid (GA<sub>3</sub>) at 90 ppm is sprayed at different crop stages. Irrigation is managed through drip with 15 L/day after summer pruning and 10 L/day after winter pruning. Pre-emergent herbicide Diuron at 1 kg/ha is used. Major diseases include downy mildew (*Plasmopara viticola*), powdery mildew (*Uncinula necator*), black rot (*Diplocarpon vitis*), anthracnose (*Elsinoe ampelina*), *Botrytis* (gray mold), and Pierce's disease (*Xylella fastidiosa*). Pests such as leaf-eating caterpillars, mealybugs, mites, thrips, grape leaf folders, girdle beetles, flea beetles, and root-knot nematodes are managed through IPM practices, including sanitation, biological control using ladybugs and lacewings, and judicious use of insecticides and miticides.

### Coconut

The recommended variety is Tiptur Tall, with a spacing of 10 m × 10 m, planted during June-July with an average population of 100 palms per hectare. The fertilizer dose per

palm includes 500 g Nitrogen, 320 g Phosphorus, and 1200 g Potassium. Cocoa is a common intercrop. Micronutrients like boron and zinc are applied. Drip irrigation is used. Weed management is done by hand weeding. Stem bleeding is managed by root feeding with Hexaconazole. Black-headed caterpillar is managed through the release of *Goniozus nephantidis*, a biological control agent.

### **Banana**

Elakki Banana and G9 are the popular varieties grown in karnataka, apply NPK at 400:240:500 kg/ha along with 40 tons of FYM per hectare before planting. The recommended spacing for Elakki Banana is 1.5 x 1.5 meters in a triangular shape, accommodating 5,120 plants per hectare, while other banana varieties should be spaced at 2.1 x 2.1 meters, accommodating 2,270 plants per hectare. In high-density planting of G9 tissue culture banana, use a 6x6 feet spacing with 2 cm<sup>3</sup> pits, resulting in 7,500 plants per hectare and a potential yield of 135 tons per hectare. A spray of Arka Banana Special is recommended starting 5 months after planting and continuing for up to 10 months. Mix 75 grams of Banana Special with 15 liters of water, along with 1 lemon's juice and 1 sachet of shampoo, and spray every 20 days. Rhizome Weevil infestation can be prevented by treating rhizomes with 20% Neem Seed Kernel Extract. To control Leaf-Eating Caterpillars, apply 2.5 ml of Chlorpyrifos per liter of water. For Thrips, use 0.3 ml of Imidacloprid or 1.5 ml of Dimethoate per liter. To manage Leaf Spot Disease, spray 1 ml of Propiconazole per liter of water. Nematode infestation can be controlled by combined application of 12 grams of *Pseudomonas*, 12 grams of *Pachelomyces* and 500 grams of Neem Cake.

### **Cashew**

Cashew cultivation requires selecting recommended varieties such as Vengurla-4, Netra Jambu-1, Bhaskar, Ullal-1, Ullal-2, Ullal-3, Chintamani-1, and Chintamani-2. The ideal planting time for Vengurla types is between June and October, while Ullal and Chintamani types should be planted from May to June. For spacing, a 7 m x 7 m arrangement supports about 200 plants per hectare, while an 8 m x 8 m arrangement accommodates approximately 156 plants per hectare. The soil should be red sandy loam with good drainage. Nutrient management includes applying 60:60:60 g of NPK per plant in the first year, doubling the dose for the second to fourth year, and applying 500:250:250 g of NPK per plant from the fifth year onward. Foliar nutrition with 19:19:19 at 2 g/litre should be sprayed once every 15 days during summer. Suitable intercrops for cashew cultivation include turmeric, ginger, and field crops, excluding sunflower and maize.

Irrigation should be provided at 40-50 litres per plant, three times per month. Weed management can be done through hand weeding and light intercultivation as needed. Pest and disease management involves spraying Lambda Cyhalothrin @ 0.5 ml/litre or Quinolphos @ 1 ml/litre for tea mosquito bug, spraying COC 50 WP @ 3 g/litre for leaf spot, and swabbing with a 2% COC solution and 2 ml Chlorpyrifos 20 EC for stem borers.

### **Arecanut**

Recommended varieties of arecanut include Mangala, Mohitnagar, Sumangala, Swarna Mangala, Theerthalli Local, Channagiri Local, and South Canara Local. Planting is preferably done in May-June or September-October using pit planting (pit size: 90 cm x 90 cm x 90 cm) at a spacing of 2.7 m x 2.7 m or 3.3 m x 3.3 m, with a density of 1372 plants/ha. Wider spacing is preferred when intercropped with high-value crops such

as banana, cocoa, nutmeg, black pepper, or ginger. Fertilizer recommendation is 100:40:140 g NPK per plant annually, applied in two equal splits - first in September-October along with 12 kg organic manure and second in February (for irrigated) or June (for rainfed). Additionally, zinc sulfate and borax are applied at 5 g per plant during October-December. Irrigation is usually done through the basin method at 175-200 litres per palm per irrigation, adjusted as per season: every 7-8 days in November-December, every 6 days in January-February, and every 4-5 days from March to May. Drip irrigation is also practiced. Common diseases include bud rot, leaf spot, collar rot, crown rot, foot rot, and yellow leaf disease. Bud rot can be managed with sprays of 1% Bordeaux mixture, 0.3% Copper Oxychloride (COC), or 0.2% Metalaxyl + Mancozeb 72WP, applied before the onset of monsoon and repeated after 30-45 days. Pests such as spindle bug, Pentatomid bug, red palm weevil, mites, root grubs, inflorescence caterpillar, and ambrosia beetle affect the crop. Red mite infestations are managed by spraying wettable sulphur (80%) at 0.4%.

## LIVESTOCK

To ensure the health and productivity of livestock during the kharif season, it is essential to implement appropriate management practices. Cattle, buffalo, sheep, and goats should be vaccinated against Hemorrhagic Septicemia, while sheep and goats also require vaccination against Enterotoxaemia. Deworming of all animals using Piperazine, Albendazole, or Fenbendazole is recommended to control internal parasites during the rainy season. Along with green fodder, daily feeding of dry fodder to cattle, buffalo, and sheep is necessary for better digestion and nutrient absorption. Grazing of maize and jowar seedlings should be strictly avoided, as these can cause cyanide poisoning, potentially leading to animal death.

Pesticides, insecticides, and fertilizers must be stored safely away from animals to prevent accidental toxicity. Farmers can prepare animal feed using locally available ingredients, such as cereals (maize, jowar, wheat) 35-40 kg, bran or polish of grains 30-35 kg, oil cakes (groundnut, safflower, cottonseed) 25-30 kg, mineral mixture 2 kg, and salt 0.5-1 kg to make 100 kg of cattle feed. To prevent the spread of Lumpy Skin Disease (LSD), which is a vector-borne illness, livestock should be protected from mosquitoes, house flies, and ticks by using mosquito nets, repellents, and washing animals with ectoparasitocidal drugs. Additionally, burning dry neem leaves in the morning and evening can help reduce mosquito bites. Farmers are also encouraged to cultivate improved fodder and crops during the Kharif season, utilizing the monsoon rains. Furthermore, all backyard poultry birds should be compulsorily vaccinated against Ranikhet Disease.

## Fisheries

Maintain 6 feet water in the fish pond for proper growth and development of fishes. Remove unwanted plants and animals prior to release of fishes in the pond. As a supplementary food, floating pallets or groundnut cake (25%), paddy husk (40 %), sweet potato powder (10%), fish meal (24 %), vitamin and mineral mixture (1%) should be mixed and feed scientifically. Continue to feed fishes 3-5 % of body weight on daily basis. The suitable breeds for fresh water aquaculture are catla, rohu, mrigal, amur common carps, jayanti rohu, punggiasius, GIFT tilapia, grass carp, silver carp, murrel. Prior to release of fingerlings (15 days) mix cow dung (100 kg/100 m<sup>2</sup>) in the pond for plankton production.

## LIVESTOCK

### Dairy Cattle

Depending upon the altitude it is advised to use indigenous Ladakhi Cattle in altitude higher than 12000 ft ASL for milk production and Cross Bred Jersey fits best at altitude below 12000ft ASL.

Keep a routine watch for any heat signs in cows and immediately contact Animal Husbandry Department for AI if heat is detected in cows. Vaccinate your animal against FMD and HS before the start of season. Practice regular deworming with broad spectrum anthelmintics, after every 3-4 months or as per the local Veterinarians advice. Farmers are advised to clean the floor of animal housing regularly with phenol@1 ml per litre of water and white washing of the feeding and watering trough to prevent infection. Disinfect the farm premises with suitable chemicals such as 1% hypochloride+ bleaching powder (7gm in 1 litre of water) or 1% sodium hypochlorite or 0.5% hydrogen peroxide or 2% Sodium hydroxide etc. with proper care to prevent FMD outbreak.

Supplementation of mineral mixture @ 50-60 g/day in adult dairy animals is recommended for optimum production. If there is temporary shortage of availability of green fodder or concentrate feed, farmer can also prepare home-made feed by using grains like wheat, maize wheat bran, oil cakes and salt. They can also prepare silage for their adult dairy animals. For clean milk production adopt general hygiene practices while milking and handling milk and milk products. Washing hands before and after feeding and milking animals with soap and water or using alcohol-based sanitizer or 1% potassium permanganate solution.

Ensure animals are free from thirst, hunger and malnutrition and protect the animals against extreme cold climatic conditions along with providing a safe environment. Provide a comfortable and ventilated housing especially during winter. Don't allow animals to graze on alfa-alfa grass during the month of May-June, as this grass has high content of Hydrogen cyanide (HCN) during early growth stage, which causes acute bloat in ruminants resulting in instant death. In case of non-disposal of milk, farmers can process their milk to ghee, butter and other value-added products like curd or paneer that can be sold locally.

Farmers are advised to prepare hay, alfa-alfa and silage as scarcity fodder for their animals and enough quantity should be stored for long winter feeding. Farmers are advised to be fed edible tree fodders as well as chopped alfa-alfa fodder regularly to sustain the production. Regular supplementation of area specific mineral mixture @50-60g/day adult cows and calcium to the lactating and pregnant cows@ 60-80 ml/day /cow is very much essential for milking cows for optimum production and productivity. Provide 1 kg concentrate for every 2.5 kg milk for cows along with 5-6 kg of dry alfa- alfa fodder and hay. Feed colostrum to new born calf within twelve hours calving @1/10<sup>th</sup> of body weight. Provide tender leafy fodder to the calves for early development of rumen.

### Sheep & Goat

Changthangi sheep and goat herders are advised follow late breeding practices in late autumn preferably October so that the kids are borne in March which reduces kid

mortality. During breeding season, one male against every 50 females should be used for breeding to avoid inbreeding. Ensure deworming of sheep and goats every 6 months using a broad spectrum anthelmintic. Vaccinate your animal against FMD, PPR, Goat Pox and Sheep Pox. Give enterotoxaemia vaccination @ 2 ml s/c to pregnant Ewe/Doe before one month of expected date of lambing/kidding. Provide 25-50 g molasses per day to each pregnant Ewe/ Doe to prevent pregnancy toxemia. Follow rotational grazing, with a grazing period of 8 hrs during winter and 10 hours in summer. Give amprolium as a drench in kids to prevent coccidiosis in young ones. Herders are advised to take care of nutrition of pregnant and lactating ewes and does especially during winter and supplementary feeding with local barley or concentrate feed should be given @ 200-300 gms/day for adult and 150-200gms/day for young ones.

Treatment with Ivermectin for ticks and mites in sheep and goats should be done during winters and Dipping with Butox or N-cidol during summer months soon after combing/shearing. Stock enough dry fodder/ concentrate feed/ barley for the herds to feed then during heavy snow days. Practice Drum silage making using locally available ingredients (developed by KVK- Leh) to meet the nutrient requirement and to meet scarcity of fodder

### **Back Yard Poultry**

The poultry farmers are recommended, disinfect the premises of poultry houses with 1% sodium hypochlorite and inhibit the entry of outsiders to the poultry houses and premises. The litter materials in poultry shed need to be changed to prevent coccidiosis infestation. If red diarrhoea symptoms are noticed in the birds, immediately administer anti- coccidial drugs (in the drinking water for 3 -5 days.)

Poultry birds may be fed with green leaves along with grinded maize/rice. The saw dust should be dried completely in sun before using as bedding material. Apply hydrated lime powder @ 1-2 kg per 10 sq.ft area in the shed before birds are placed in the shed. Add 8% sugar solution containing 1% sodium chloride during first 15 hours after placement. Maintain proper shed temperature (35 C) during the first week of brooding. Provide good ventilation in the shed during the first three weeks of brooding. Provide minimum 23 hours light in the brooder continuously. The adult birds should be allowed to move out in the open yards during daytime. Vaccinate the adult birds against Ranikhet disease @ 0.5ml intramuscular after 2 months of age. Provide Layer mesh @ 20gms/bird in addition to free range feeding to the layers for good egg production. Deworm the birds using piperazine hydrochloride @ 5ml per 10 birds. Provide a minimum light of 16 hours for adult birds. Use anti-stress vitamins such as vimeral @ 10ml per 10 birds for 3 days in a week.

Additionally, Install pheromone traps @ 5 traps/ha to monitor and manage stem borer populations in the rice field. Regularly monitor rice fields for brown plant hopper infestations. If the pest population exceeds the economic threshold level, spray Imidacloprid 17.8% SC at 250 ml/ha. In case of severe infestations, spray Dinotefuran 10% + Pymetrozine 40% WG @ 250 g/ha. To control blast disease in paddy, treat the seeds with Tricyclazole @ 2 g/kg and spray Propiconazole 25 EC @ 500 ml/ha. If large, muddy-colored spots with purple margins appear on rice sheaths above the water surface, indicating sheath blight, spray Trifloxystrobin 25% + Tebuconazole 50% @ 0.5 gm/L or Carbendazim 50% WP @ 2 gm/L on the affected areas.

Repeat the spray after 12-15 days, if necessary. To conserve rainwater in fields, construct bunds that are high and wide to maximize water retention.

### **Maize**

Use of high-yielding varieties such as Vivek Hy-17, Vivek Hy-43, Pusa Vivek QPM-9, VLQPM-45, Hybrid Maize-1, HQPM-1, Vivek Makka Hy-51, Pusa HM-4, DMRH-1308, IMH 230, and IMH 231. A germination test is essential to ensure an optimum number of healthy plants in the field. This should be followed by bio-priming or seed treatment using Carbendazim 50% WP @ 2 g/kg seed, Imidacloprid 600 FS @ 2 g/kg seed, *Trichoderma* @ 10 g/kg seed, and *Azotobacter* @ 10 g/kg seed to enhance productivity. The optimum sowing time is from June 15<sup>th</sup> to June 30<sup>th</sup>, with a spacing of 60 x 25 cm (row x plant). Apply balanced fertilizers, including zinc-based fertilizers, for better crop growth.

For weed control, apply Tropicamezone 33.6% SC @ 30 ml per acre as a post-emergence spray at 20-25 days after sowing. The pupae of Fall Armyworm and stem borer, as well as the spores of *Rhizoctonia solani* (the causal agent of sheath blight), remain in the soil. Therefore, deep summer ploughing is crucial to reduce the inoculum potential. The infestation of Fall Armyworm can be significantly minimized by dry sowing or



**Integrated Weed Management**

sowing at the onset of the monsoon, practicing crop rotation, and alternating maize with other crops that are not affected by Fall Armyworm. Additionally, apply a basal dose of nutrients and use the ridge and furrow method of bed preparation. These practices should be completed by the last week of May or the first week of June. Monitor regularly to prevent Fall Armyworm infestations. If necessary, spray Spinetoram 11.7 SC @ 350 ml/ha or Emamectin Benzoate 5 SG @ 200 g/ha. To control leaf blight disease, take up spray of Mancozeb 75 WP @ 1000 g/ha.

### **PULSE CROPS**

#### **Green gram**

Use of high-yielding varieties such as IPM 410-3 (Shikha), IPM 205-7 (Virat). Before sowing, mix 2.5 kg *Trichoderma* with 1 quintal of FYM and incorporate it into the soil, to control soil borne diseases. Use of 20-25 kg/ha of improved green gram seed for sowing. To control yellow mosaic virus, treat the seeds with Imidacloprid 48% FS at 2 ml/kg of seed.

Animals experiencing heat stress or other health issues may exhibit rapid breathing, characterized by panting, fast and shallow breaths, and mouth breathing. Their skin often becomes dry and rough, while the eyes may appear red and watery. A noticeable reduction in feed and water intake is common, accompanied by digestive disturbances such as acidity, loose stools, and slowed gut movement. Urine output tends to decrease, indicating impaired hydration or kidney function. Affected animals may also display lethargy or weakness, often sitting or lying down frequently. Additionally, an increased heart rate is typically observed, often linked to heavy or labored breathing.

**Housing and Environment:** Keep animals in the shade and protect them from direct sunlight. Ensure adequate space in the shed, as overcrowding increases heat. Plant trees around the animal shelter to keep the surroundings cool. Construct well-ventilated sheds that allow hot air to exit and cool air to enter. Keep the shed roof high for better airflow. Paint the roof white to reduce heat absorption. Place mulch or wet jute sacks on the roof to lower the temperature.

**Cooling:** Install fans, foggers, and water sprinklers. Hang wet gunny sacks or soaked sheets around the animal shed. Ensure a constant supply of cool, clean water in cement tanks.

**Feed and Water Management:** Provide animals with continuous access to cool and clean drinking water. Feed them in the morning and evening to avoid generating excess body heat. Give enough salt or electrolytes to maintain the body's mineral balance.

**Vaccination and treatment:** If an animal shows signs of heat stress, consult a veterinarian immediately. Meanwhile, move the animal to a cool place, bathe it with cold water or wrap it in a wet cloth, and use a fan to cool it down. Some animal diseases can spread to humans (zoonotic diseases). Vaccination helps prevent the transmission of such diseases from animals to humans and from one animal to another. Never vaccinate animals that are sick, weak, or under stress. Treat them first and vaccinate only after recovery. Follow the prescribed vaccination schedule for livestock and get them vaccinated regularly through government or veterinary centers. Timely vaccinate to animals through FMD and H.S. Vaccination should be administered to prevent brucellosis (infectious abortion) in female calves aged 4 to 8 months.

Artificial Insemination (AI) should be continued in consultation with veterinary officer on door-to-door basis rather than direct visit to the dispensaries. To maintain productivity and improve immune status of animals, it is suggested for regular supplementation of mineral- 60g/ day in adult dairy animals and 20-30g in small ruminants. Also, administer deworming medication. Offer clean and cool water three times a day. In case of injury in lactating cows, apply Himex ointment. In case of temporary shortage of availability of green fodder or concentrate feed farmers can prepare home-made feed by using grain like wheat, maize, wheat bran, oilcakes and salt. They can also prepare silage for their adult dairy animals. Due to increased maximum temperatures, goats should be kept in shaded areas and given clean, fresh water three times daily. Vaccinate goats against PPR (Pestedes Petits Ruminants), Contagious Pneumonia, and ETV (Enterotoxaemia). Also, implement control measures for ticks and

## CEREALS CROPS

### Paddy

For optimal crop establishment, use certified seeds or ensure farm-saved seeds have over 70% germination. Perform salt treatment (300g salt/10L water), discard floaters, wash, dry, and treat with fungicide and *Trichoderma* (4-5g/kg seed). Sow from the 2nd week of June to 10<sup>th</sup> July (direct-seeded) or transplant from 2<sup>nd</sup> to 20<sup>th</sup> July. Deep plough with a sub-soiler, align with the slope, dig farm ponds, and prepare for drill sowing. Apply Pendimethalin 30% EC (50ml/10L water) after sowing and Bispyribac sodium 10% SC (6-7ml/10L water) 10-15 days later. Use 100:50:50 kg NPK/ha-full P & K with half N at planting; split the rest at tillering and panicle initiation. Begin seed preparation with a salt treatment (300 g salt per 10 L water); discard floaters, then wash, dry, and treat the seeds with fungicide and *Trichoderma* at 4-5 g/kg of seed. For optimal growth, direct seeding is recommended from the second week of June to July 10, while transplanting should be done between July 2<sup>nd</sup> to 20<sup>th</sup> July. Land preparation includes deep ploughing using a sub-soiler, ensuring cultivation is aligned parallel to the slope. Farm ponds should be dug for water harvesting, and the land should be prepared for drill sowing. Weed management involves applying Pendimethalin 30% EC (50 ml/10 L water) as a pre-emergence herbicide after sowing, followed by Bispyribac sodium 10% SC (6-7 ml/10 L water) as a post-emergence treatment 10-15 days later. Nutrient management consists of applying 100:50:50 kg NPK per hectare, with full doses of phosphorus and potassium and half of the nitrogen at planting. The remaining nitrogen is split and applied during the tillering and panicle initiation stages.



**Paddy after Transplanting**

### Maize



**Fall armyworm attack in maize**

Sowing of maize is recommended from mid-June to the first week of July, using the PDKV Arambha variety. Apply 20-25 cartloads of FYM per hectare before sowing. Treat seeds with biofertilizer at 25 g/kg. The recommended fertilizer dose is 120:60:60 (N:P:K), applied in three splits- 1/3 at sowing, 1/3 at 30 days after sowing (DAS), and 1/3 at 50 DAS. Maintain clean cultivation with deep ploughing, trap crops like Napier grass, bird perches, and pheromone traps. For pest management, spray 5% NSKE and release *Trichogramma* at 50,000 per acre weekly. For Fall Armyworm control, treat seeds with Cyantraniliprole + Thiamethoxam and apply Emamectin or NPV sprays if necessary.

Plant rhizomes at appropriate spacing (typically 20-30 cm between plants and 45-60 cm between rows). Ensure adequate moisture at the time of sowing. Depending on rainfall, you might need to provide initial irrigation.

**Pest, Nematode, soil born pathogen:**

Biological control - At the time of planting, apply 25 g powdered neem cake and mix well with the soil in each pit taken at a spacing of 20-25 cm within and between rows or application of neem cake @2 t/acre is also desirable. Plan for timely weed control through manual weeding or appropriate herbicides as per recommendations.



**FLD on Turmeric**

**LIVESTOCK MANAGEMENT**

In Livestock shed use of 0.01 % Deltamethrin spraying for control of ecto and endo parasites 15 days intervals. Provide ample amount of clean and cold water to livestock to avoid Colitis disease. Use of white paint lime to drinking water tank. Vaccination of livestock should be done twice in a year against FMD. Livestock deworming should be done once in a six month. In case of buffalo wallowing should be done in summer season for enhancing the milk yield and to maintain the body temperature. Grooming of milch animal to avoiding the falling of loose hairs in the milk and to get the clean milk production and enhancing the blood circulation. Washing of animal specially in summer season remove the dirt and dust and maintain the body temperature and to avoid the heat strock. As per thumb rule provide animal feed should be 60 % green fodder and 40% dry fodder. In case of milch animal give 30 gm mineral mixture in case of cattle and 60 to 80 g in case of Buffalo should be given in the diet.

**ENGINEERING**

Use sub-soiler for deep ploughing. Cultivate parallel to slope. Tools used are Paddy transplanter, power weeder, Cono weeder. Size of farm pond: 20x20x3 m per 2 ha.

**MARATHWADA REGION**

Marathwada region of Maharashtra is comprised of Aurangabad, Jalna, Beed, Latur, Osmanabad, Hingoli, Parbhani, Nanded. The region has moderate rainfall (600-800 mm) and has medium to deep black soils. This region possesses state agricultural university viz., Dr. Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani.

**CEREALS CROPS**

**Maize**

Maize should be sown at 75 x 20 cm or 60 x20 cm as per soil type by dibbling method with first onset of monsoon after receiving 75-100 mm rainfall. Improved and high yielding notified varieties or research hybrids should be used for sowing for achieving higher yield. Recommended dose of 100:75:75 kg NPK should be applied to maize out of which 50 % N and full P and K should be given at the time of

**Maize observation in field**

sowing while remaining 50 % N should be applied after 30 DAS. For the control of Fall Army Worm, treat the seed with 3 g/ Thiram per kg of seed, spray 5 % NSKE withing 30 days of sowing or at 5 % infestation. In addition, spray Emamactin Benzoate 5 % @ 4 g/ 10 L of water at 10-20 % infested plants. Earthing up should be done during dry spell for in situ moisture conservation.

## **COMMERCIAL CROPS**

### **Cotton**

Cotton should be sown after onset of monsoons or after 1<sup>st</sup> June only for effective management of pink bollworm. Well decomposed FYM should be applied to cotton @ 10 ton/ha before sowing. Recommended fertilizer dose of 120:60:60 kg NPK/ha should be applied to rainfed cotton out of which 40% N & complete P & K should be applied as a basal dose at the time of sowing. Remaining 30 % N should be applied after 30 DAS and 30 % after 60 DAS. Use of high yielding, short duration cotton variety like CICR-14, CICR-19, PKV-081, Suraj BT, Rajat BT, NH-1901,1902,1903,1904 etc. For irrigated cotton apply 150:75:75 kg NPK/ha out of which 20% N & complete P & K should be applied as a basal dose at the time of sowing. Remaining 40 % N should be after 40 DAS and 30 % after 60 DAS. Closer spacing and High-Density Cotton Planting (HDPS) should be adopted on medium and light soils respectively for higher yields by keeping plating distance of 90 x 30 cm for medium soils and 90 x 15 cm for light soils which will accommodate approximately 14000 and 29000 plants per acre respectively. Canopy management should be done by cutting monopodia at 45-50 DAS and nipping the tip at 90 DAS supported with one or two sprays of growth retardants if required under expert guidance. Earthing up should be done in cotton during dry spell in Kharif season to reduce evaporation losses of soil moisture. Follow IPM package for the control of pests and diseases in cotton as given below. Select short duration and sucking pest tolerant variety for sowing

**Whitefly on cotton**

- a. Follow crop rotation avoid cotton sowing followed by tomato, chickpea, etc.
- b. Sowing of intercrop or mixed crop like maize, cowpea, Black gram, green gram and use of trap crop like marigold, castor, etc.
- c. Sowing of okra as a trap crop in cotton for the control of Pink Bollworm.
- d. Avoid plant growth promoters and pesticides during first three months and use botanical pesticides or biopesticides
- e. Spray of 5 % NSKE at 60 DAS
- f. Install 40 pheromone traps per ha for mass trapping of pink bollworm.

## **FODDER MANAGEMENT**

Use Maize African Tall and new varieties of hybrid napier like Phule Gunwant, DHN-6, BNH-10 and Super Napier for round the year availability of green fodder. Excess green fodder should be converted into silage.

## **MILCH ANIMALS**

Provide clean, hygienic and plenty amount of drinking water to milch animals. To reduce the stress of heat in farm animals, provide roughages by mixing with solution of 1% jaggery and 0.5% salt separately. Milch animals should be offered 50 gm of mineral mixture daily in two divided doses to maintain milk yield and to overcome the problem of anoestrous cycle. Cow teats should be dipped in 2 % solution of Potassium Permanganate after every milking to avoid Mastitis in milch animals. Regular deworming should be done after every two months for one year for good health and weight gain of calves.

### **Buffalo**

Allow buffalo for wallowing in water as they have fewer sweat glands compared to cattle to maintain their body's thermo-equilibrium. Water-soaked gunny bag or other cloth should kept on a body to avoid occurrence of heat stroke.

### **Goat**

Animals should be vaccinated before monsoon season for Enterotoxaemia and Peste des petes (PPR) ruminants' diseases to avoid great losses. Improved breeds like Osmanabadi should be used for higher income. Regular deworming should be done after every month to goat kids for six months for good health and weight gain.

### **Poultry**

In the summer the poultry shed should be kept cool. Plants more trees around the shed. The roof of the shed should be covered with grass, so that the temperature remains under control. Exhaust fan should be kept in the shed, so that the hot air inside shed is thrown out. Use foggers, so the shade will cool down. Improved backyard poultry breeds like Grampriya, Shrinidhi, Kaveri, Giriraja and Vanaraja should be used to boost up the income of small and marginal farmers.

## **WESTERN MAHARASHTRA**

Western Maharashtra is comprised of Ahmednagar, Pune, Satara, Sangli, Kolhapur, Solapur. The region has moderate rainfall and Black to medium black, well-drained soils. This region has state agricultural university viz., Mahatma Phule Krishi Vidyapeeth, Rahuri, Ahmednagar.

## **CEREAL CROPS**

### **Rice**

In Western Maharashtra, nursery sowing for Kharif rice generally begins in late May to mid-June, aligning with the onset of monsoon. Recommended varieties include Phule Radha, Indrayani, Sahyadri, and Kolhapur-1, chosen for local adaptability and consumer preference. About 40-50 kg of seed per hectare of main field is required for a nursery area of 800-1000 m<sup>2</sup>. Seeds are soaked in water for 24 hours, then incubated for another 24-36 hours to promote uniform germination. Treat seeds

## Maize

In the Kharif season, the ideal sowing time is from June 15 to July 15, depending on the onset of monsoon rains. Apply 120 kg Nitrogen (N), 60 kg Phosphorus ( $P_2O_5$ ), and 40 kg Potassium ( $K_2O$ ) per hectare. At the time of sowing, apply half the nitrogen and the full dose of phosphorus and potassium. Apply the remaining nitrogen in two equal splits: one at 30 days and the other at 50 days after sowing. In case of micronutrient deficiencies, apply 5 kg/ha of Zinc Sulphate ( $ZnSO_4$ ) and 5 kg/ha of Ferrous Sulphate ( $FeSO_4$ ) at sowing and again 30 days after sowing. The first weeding should be done 20 to 25 days after sowing, and the second weeding around 45 days after sowing. To control weeds chemically, apply Atrazine 50% WP @ 1 kg/ha as a pre-emergence herbicide within 2-3 days of sowing. It should be applied uniformly on moist soil using a knapsack sprayer.

## COMMERCIAL CROPS

### Sugarcane

Well-drained deep loam with good water holding; avoid saline/alkaline soils; requires warm humid climate. Deep ploughing (1-2 times) + 2 cross harrowing; remove stubbles & weeds; level field with proper drainage. Varieties like CO-86032, CO-8014, CO-94012, VSI-434, CO-M-0265 (early maturing recommended). Use 9-10-month-old healthy canes; take top 2/3rd portion; avoid flowering canes. Recommended seed rate is 45,000-50,000 three-budded setts/ha (~6-8 tonnes/ha). Treat with 0.25% Bavistin or Carbendazim + 2.5% urea solution for 10-15 minutes before planting. Avoid the use of infested sets/seedlings. Paired Row planting system provides easy access for plant protection application. Judicious use of nitrogenous fertilizers to prevent infestation of sucking pests like Woolly aphids, Pyrilla, Scale insect, whitefly, etc.

**Plant Protection:** Remove and destroy the lower 4-5 leaves of cane to prevent infestation of Pyrilla and Scale insect. Use pheromone traps for the monitoring of Early Shoot Borer, Internode Borer and Top Borer and fortnight release of *Trichogramma* parasitoid insect (Trichocard 1.50 eggs/ha) to check the all-stem borer infestation. Preventive management of White Grub should be done by installing light traps in the field (1 trap/acre), collection and destroying beetles from bund trees during evening time. To control the initial infestation of white grub, apply *Metarhizium anisopliae* 15 kg or *Heterorhabditis indica* 7 kg per hectare as a biological insecticide in the month of June and repeat the application one month after first when incidence level is high. If the above treatments found unable to control follow the application of a mixed insecticide of Fipronil 40% + Imidacloprid 40% e at the rate of 15 ml per 10 litres of water. Also, if there is a high incidence of this pest, spray with Chlortraniliprole 18.50 SC @ 3-4 ml per 10 L of water. If symptoms of Rust disease observed, spray crop with Mancozeb 0.3% (30 g in 10 L of water) or Tebuconazole 0.1% (30 g in 10 litres of water). Regular monitoring of woolly aphids should be done, if colonies are found, they should be removed and destroyed, and predatory insects like *Conobathra*, *Micromus*, *Depha aphidivora* etc. should be conserved. For the Management of Smut disease spray the crop with Carbendazim 50 WP @ 10 g in 10 litre water. For the management of

soil. Clean the fields by destroying leftover nursery and removing weeds. Maintain bunds and level the fields to prevent water wastage and improve crop growth. Release *Trichogramma japonicum* and *T. chilonis* at 1 lakh/ha upon appearance of yellow stem borer and leaf folder egg masses/moths. Conserve natural biocontrol agents like spiders, dragon flies, and parasitoid wasps. Install light and pheromone traps to monitor pest populations. Regularly collect and destroy pest-infested plant parts. For gall midge, apply Fipronil 5% SC (2 ml/L) in the nursery. For stem borer, apply Quinalphos 25 EC or Profenophos 50 EC (1000 ml/ha). For leaf folder, use Fenitrothion 50 EC (1000 ml/ha) or Quinalphos 25 EC (1000 ml/ha) based on economic thresholds. Apply Phorate 10G or Carbofuran 3G (1.5 kg a.i./ha) for gall midge control within 10 days of sowing. Ensure that fields remain weed-free up to 50 days after sowing to minimize competition for nutrients and water. Follow integrated pest management practices for effective weed and pest control.

### Maize

Use high-yielding composite maize varieties like Vijaya Composite, RCM-75, RCM-76, and RC Manichujak-1, and hybrids such as All Rounder, Vivek QPM, HQPM-1, and HQPM-5 for better grain yield. Maintain a spacing of 75 cm x 18 cm. To manage stem borer and other pests, remove dead hearts and manually destroy white grubs and chaffer beetles during their adult emergence. Apply Trichocards @ 7-10 cards/ha or release



Varietal demonstration in maize

*Trichogramma chilonis* @ 1,60,000/ha at 7 and 15 days after sowing; repeat if needed. Conserve natural enemies like *Trichogramma chilonis*, *Cotesia flavipes*, *Chrysoperla*, spiders, and wasps by minimizing chemical pesticide use. When pest pressure is high, apply recommended insecticides such as Quinalphos 25 EC @ 1000 ml/ha, Phosphamidon 40 SL @ 600 ml/ha, or Profenophos 50 EC @ 1000 ml/ha.

### Millets

Grow millets like Jowar, Ragi, Bajra, and Sorghum as Kharif crops between June and November, especially as catch crops in areas affected by erratic weather. Use recommended high yielding ragi varieties such as VL Madua 376, 379, and 380; sow seeds at 3-4 cm depth using 6-8 kg/ha for line sowing, or transplant 3-4-week-old seedlings at 25 cm x 8 cm spacing using 4 kg seed for 1 ha. Begin ploughing at monsoon onset to achieve fine tilth for good germination and moisture retention. For line sowing, use 6-8 kg seed/ha at a depth of 3-4 cm; for transplanting, use 4 kg seed in nursery (May-July) to cover 1 ha. Maintain 20-25 cm row-to-row and 8-10 cm plant-to-plant spacing in direct sowing. For transplanted crops, place 3-4-week-old seedlings (2 per hill) at 25 cm x 8 cm spacing, 2-3 cm deep. Treat seeds with Thiram, Carbendazim, or Mancozeb @ 2-3 g/kg to control seed-borne diseases. For specific disease prevention: use Carbendazim for Grain Smut and Ridomil MZ for Green Ear. Apply FYM @ 5-10 t/ha one month before sowing. Follow up with a fertilizer dose of



## CEREAL CROPS

### Paddy

Farmers are advised to prepare a nursery area of about 500-1000 sq. meters for raising seedlings of recommended kharif paddy varieties such as Megha SA-2, Lungnilaphou, CAU-R1, CR Dhan-310, and CR Dhan-311 to cover one hectare area. Use 40 kg of high-yielding variety (HYV) seeds per hectare. Incorporate FYM and vermicompost into the nursery beds to improve soil health and seedling vigor. For disease prevention, treat seeds with



Varietal demonstration in paddy

*Trichoderma harzianum* @ 5 g/kg of seeds and drench the nursery with *Trichoderma harzianum* @ 5 g/L of water prior to sowing. Additionally, mix *Trichoderma harzianum* @ 2.5 kg with 50 kg of FYM and apply it to the nursery soil 10-15 days before sowing. Prepare the main field with 2-3 ploughings using a power tiller, and mix FYM @ 5 t/ha during the last ploughing. Transplant 20-25 day-old seedlings at a spacing of 20 cm x 15 cm for optimal plant population. For pest management, apply neem oil spray @ 3 ml/L of water at 10 days after transplanting (DAT). Release *Trichogramma japonicum* @ 50,000 eggs/ha at 35 DAT to control stem borers biologically. At the boot leaf stage, spray *Beauveria bassiana* @ 7 g/L of water to manage leaf folder and other pests. For direct seeding, use 80-100 kg of seeds/ha, while for transplanting, use 35-40 kg of seeds/ha. Apply FYM @ 10-20 t/ha and follow a fertilizer schedule of N:P:K @ 60:60:40 kg/ha.

### Maize

Farmers are advised to complete sowing of maize by the last week of May, using high-yielding varieties such as Megha Maize 1, Megha Maize 2, and RCM 1-3. Maintain a

spacing of 60 cm x 30 cm. Intercropping with Dwarf French bean (var. Pant Anupam) or soybean is recommended for better resource utilization and additional income. Treat maize seeds before sowing with *Trichoderma harzianum* @ 5 g/kg seed or *Azospirillum/ Phosphobacteria* @ 20 g/kg seed to improve germination and disease resistance. Carry out manual weeding and earthing up in early-sown maize to prevent lodging due to heavy winds. If any Fall Army Worm (FAW) infestation is observed, apply dry soil in the whorl of infected plants or spray neem oil (1500 ppm) as a first-line defense. For biological control, release *Trichogramma pretiosum* @ 50,000 eggs/ha and spray *Metarhizium anisopliae* @ 5 g/l of water into the leaf whorls 15-25 days after planting for effective FAW management.

## MILLETS

Finger millet is a major dual-purpose crop grown during the kharif season for both grain and fodder. It is highly drought-tolerant and is usually sown with the onset of the monsoon. Farmers are advised to select suitable drought-tolerant varieties such as VL 376 (105 days), VL 379 (early maturity in 100 days), and VL 380 (mid-late maturity in 120 days). Transplanting of seedlings should be done 21 to 30 days after sowing in the nursery. Delayed transplanting beyond this period may significantly reduce tillering, resulting in lower yields. To ensure an optimum plant population, use 4-5 kg of seeds per hectare.

## VEGETABLE CROPS

Farmers are advised to prepare the vegetable nursery for crops such as cucumber, bitter gourd, ridge gourd, bottle gourd, chilli, brinjal, and okra on raised bamboo structures, covering them with polythene to protect seedlings from heavy rain and hailstorms. Planting should be done in ridges and furrows with proper drainage channels to prevent waterlogging. It is essential to add farmyard manure (FYM) or vermi-



**In-situ moisture conservation in cabbage**

compost in the nursery for healthy seedling growth. To conserve soil moisture, prevent soil erosion, and control weeds, mulching with locally available materials like paddy straw is recommended. Farmers can also make their own compost using the 18-day compost method, pit method, or by producing vermicompost to meet their manure requirements. For pest management, it is advised to make biopesticides using locally available materials such as leaves, garlic, and tobacco. For crops under protected cultivation, including high-value crops, frequent irrigation should be provided at 10-15 day intervals, based on the weather conditions.

## SPICE CROPS

### Ginger and Turmeric

Farmers are advised to complete the sowing of ginger (variety Nadia) and turmeric (varieties Megha Turmeric 1 or Lakadong) without delay. Use a seed rate of 15-20

quintals per hectare and maintain a spacing of 30 cm × 30 cm. Before sowing, treat rhizome seeds by dipping them in a solution of *Trichoderma viride* @ 4 g/litre of water for 5 minutes or *Trichoderma harzianum* @ 10 ml/litre to prevent rhizome rot. For already sown crops, carry out proper weeding and earthing up to promote healthy growth. Ensure planting in ridges and furrows with good drainage. Farmers may also opt for intercropping with crops like soybean, rice bean, French bean, finger millet, or maize for better resource use and income diversification. To conserve soil moisture and control weeds, mulch the beds using dry paddy straw or green leaves. For effective disease management, especially soft rot, spray *Trichoderma viride* or *Trichoderma harzianum* @ 10 ml/litre or neem oil (10,000 ppm) @ 2 ml/litre as a preventive measure.

## **LIVESTOCK**

### **Piggery**

Farmers are recommended to select pig breeds like Lumsniang or Hampshire cross; supplement mineral mixture (50-60g/day) and calcium for lactating/pregnant sows; use local feed alternatives like banana pseudo stem, colocasia, and sweet potatoes. Ensure adequate clean drinking water and proper feed. Deworm pigs with Albendazole/Febendazole (7.5mg/kg body weight every 3 months); vaccinate against Swine Fever; regularly provide 20g of mineral mixture per pig with feed; and isolate sick pigs in a quarantine shed until fully recovered. Maintain dry, clean pig sheds with proper roofing and walls; avoid waterlogging; disinfect farm entrances and exits with 1% formaldehyde or 2% NaOH; and ensure footwear and equipment are disinfected. Quarantine newly purchased pigs for 4 weeks before introduction. Implement strict measures for African Swine Fever (ASF) by quarantining infected farms, prohibiting the movement of animals and personnel between farms, and ensuring the proper disposal of carcasses (deep burial with lime/bleaching powder). Avoid swill feeding unless properly boiled; maintain proper records of animal health and visitors; and prohibit animal movement between farms to reduce disease risk. Report suspected ASF cases to veterinary authorities immediately.

### **Poultry**

Farmers should maintain clean, ventilated poultry sheds with proper space (1 ½ sqft/bird); avoid overcrowding and heat stress in birds. Regularly change litter materials to prevent coccidiosis and maintain hygiene. Provide clean drinking water 3-4 times a day, occasionally adding electrolytes. Ensure the water is free from contamination and provide proper feed to maintain bird health. Backyard poultry



**Vaccination in poultry**

breeds like Vanaraja, Shrinidhi, and Kuroiler can be reared in low-cost housing systems. Ensure the houses are clean, dry, and properly ventilated to prevent infections and diseases. Administer anticoccidial drugs in drinking water for 3-5 days



## CEREAL CROPS

### Maize

**Drainage Management:** Farmers are advised to create drainage channels around fields, especially in low-lying areas, to prevent waterlogging.

**Varietal Selection:** Suitable composite varieties include RCM-75, RCM-1-1, RCM 1-76, RCM 1-3, and Vivek Sankul. Recommended hybrid varieties are Vivek QPM-9 and HQPM-1. Local varieties such as Mimban and Mimpui are also good choices. However, tall varieties like MZM 11, Ganga Hybrid, and local Popcorn should be avoided in areas prone to heavy rainfall and strong winds, as they are more susceptible to lodging and crop damage.

**Land Preparation and Sowing:** Prepare the field with at least 3 deep ploughings before sowing. Sow maize seeds at the rate of 20-22 kg/ha, maintaining a spacing of 60 cm × 20 cm.

### **Pest Management (Fall Armyworm):**

Install pheromone traps at the rate of 12 traps/ha during the 3rd to 4th week of May. For control, spray Emamectin Benzoate @ 0.4 ml/l, Spinosad @ 3 ml/l, or use other chemical options such as Spinetoram @ 0.5 ml/l, Lambda-Cyhalothrin + Thiamethoxam @ 0.25 ml/l, or Chlorantraniliprole 18.5% SC @ 0.4 ml/. Alternatively, apply ash or spray soap water into the whorls of affected plants. Avoid chemical sprays once tasselling begins.



Management of Fall Armyworm – FAW in Maize

local short duration var. Idaw, tai, Buhsakei are recommended for upland/Jhum cultivation with rich alluvial soil

**Pest & Disease Management:** To effectively control Gundhi bugs, deploy 100 rotten crab funnel traps per hectare during the critical milking stage of the crop. For organic pest management, utilize neem-based biopesticides, release *Trichogramma* egg parasitoids, and install yellow sticky traps to monitor and reduce pest populations. Additionally, conserve natural predators such as spiders, dragonflies, and ladybugs by minimizing chemical pesticide use and maintaining ecological balance in the field. Adopt SRI method in small areas for enhanced yields, conduct regular field monitoring for pests and nutrient deficiencies and maintain proper water management and spacing as per cultivation method.

## **OILSEED CROPS**

### **Soyabean**

**Varietal Selection:** RCS1-1, RCS1-9, RCS1-10, JS335, MAUS 71 is suitable option for Kharif cultivation in Mizoram.

**Irrigation:** Despite it is a rainfed crop in kharif, Supplemental irrigation is required depending on rainfall patterns. Flowering and pod filling stages are the critical stages of irrigation.

**Relay cropping:** Soybean can be grown as a relay crop with maize, planting between the rows of the maize crop is recommended.

**Integrated nutrient management:** Soybean being a leguminous crop does not require high dose of nitrogen, however 20 kg/ha is sufficient for healthy crop stand. A dose of 60 kg  $P_2O_5$  (SSP) and 30kg  $K_2O$  (MOP) are recommended. FYM @ 5t/ha should also supplement the chemical fertilizers.

**Integrated disease management:** Seed treatment with Thiram at the rate of 4.5g per kg of seed will control fungal diseases like Leaf Blight, Leaf Spot, Seedling Rot, Frog eye spot and Rust.

**Integrated pest management:** Spray application of Nuvacron 1.25ml/L or Dimethoate (0.04%) can be used for effective control of leaf folder, stem fly and semilooper like pests. Seed treatment with Imidachloprid @7g/kg of seed reduces the problem of Leaf folder

### **Groundnut**

For optimal groundnut production, farmers should complete sowing between 15th May and first week of June, using high-yielding varieties like ICGS 76, ICGV 86590, Girnar 4, TS 67 Before sowing, treat seeds with Bavistin (5g/kg seed) for disease prevention and apply Rhizobium + PSB culture for better plant growth. Prepare fields by incorporating lime to address soil acidity and use Butachlor (10ml/litre water) as a pre-emergence weedicide. Apply balanced fertilization at 20:60:40 kg NPK/ha, split into basal and top-dressing applications. To protect crops, spray Dichlorvos (2ml/L) against Blister beetles during flowering and implement monthly preventive sprays of Dichlorvos (2ml/litre) + Bavistin (2g/L) from 30 days after sowing. Ensure light

the flowering and maturity stages to maintain varietal purity and reduce disease spread.

**Pest and Disease Management of Paddy:** Install yellow sticky traps, light traps, and pheromone traps in the field. For organic pest control, apply biopesticides like *Trichogramma*, conserve biocontrol agents (e.g., coccinellids, spiders, damselflies), and use *Trichoderma harzianum* with FYM before sowing. Spray neem oil at 10 DAT and *Beauveria bassiana* at boot leaf stage. For stem borer control carry out deep summer ploughing to expose the pupae to predators and disrupt the insect life cycle. Release *Trichogramma* spp. @ 50,000/ha one month after transplanting, followed by 3-4 releases at fortnightly intervals to parasitize stem borer egg masses. To control Gundhi bug, monitor and manually collect nymphs at the tillering stage using sweep nets. Install improved Indigenous Technical Knowledge (ITK)-based crab traps during the milky stage of paddy to reduce unfilled grains. For blast disease management, ensure proper field sanitation and treat seeds with *Pseudomonas fluorescens* and *Trichoderma viride* @ 10 g/kg of seed. Upon disease onset, apply copper oxychloride 0.25% @ 2.5 g/L of water, repeating at 7-10 day intervals until the disease subsides.

### Maize

Farmers are advised for 2-3 deep ploughings followed by sun drying of the field before the onset of rains to prepare for kharif maize cultivation. Use high-yielding composite varieties like Vijaya Composite, Vivek Sankul, RCM-1-1, RCM 1-3, RCM 75, RCM 76, and DMRH 1301, or hybrid varieties like All Rounder, Vivek QPM, VLQPM 45, VLQPM 59, HQPM-1, HQPM-5, and HQPM-7. For short to medium maturity duration (85-110 days), especially suitable for re-sowing after hail damage, choose varieties like HQPM-1, HQPM-5, RCM-76, and Vivek 27. Sow maize @ 20-22 kg/ha with a spacing of 60 cm × 20 cm to achieve an ideal plant population of 60,000-65,000 plants/ha. Apply FYM or compost @ 5 t/ha and NPK @ 120:60:40 kg/ha. Apply nitrogen in three splits: at sowing, knee-high stage, and tasselling stage. Perform green manuring using Daincha, Sunhemp, or cowpea to improve soil fertility. Carry out earthing up twice (at 35-40 DAS and 60-65 DAS) to prevent lodging, and maintain a weed-free crop for the first 30-45 days. Ensure proper drainage during heavy rains to avoid waterlogging. Under drought conditions, irrigate at the knee-high, flowering, and grain-filling stages



Varietal demonstration in maize



Fall Armyworm in maize

for optimum crop performance. Harvest mature cobs promptly to prevent spoilage. For enhanced yield and soil health, adopt intercropping with soybean, kholar, or other legumes.

**Management of Fall Armyworm:** Sowing maize after the onset of rains reduces insect infestation. Regularly hand-pick and destroy egg masses and larvae by crushing or immersing in kerosene water. Apply sand or ash into the whorls of affected plants. Use pheromone traps @ 15/acre to trap male moths, and spray neem-based formulations (NSKE 5% or neem @ 4 ml/L) at early infestation stages. Apply a whorl spray of *Metarhizium anisopliae* @ 5 g/L at 15-25 DAS, followed by 1-2 sprays at 10-day intervals. For effective control of late instar larvae, use *Bacillus thuringiensis var. kurstaki* @ 2 g/L, Spinosad @ 0.3 ml/L, or prepare poison bait using rice bran, jaggery, and Thiodicarb.

## **MILLET**

Prepare the field with 2-3 harrowings followed by leveling. Sow the crop in March to utilize early soil moisture or in August to benefit from residual moisture after the kharif season or late monsoon rains. Use a seed rate of 8-10 kg/ha, sown at a depth not exceeding 3 cm, and maintain a line spacing of 25 × 8 cm for optimal plant population and intercultural operations. Farmers are suggested to select local varieties such as Shongtak, Langjai, Buyan, Tansho, Senyak, Anong, Yempang, Shonye, and Phuknyak, or recommended improved varieties like SiA 3085 and SiA 3156 for higher productivity and adaptability. Apply 10-15 tonnes of well-decomposed FYM per hectare. Keep the crop free from weeds through one or two manual weedings during early growth stages. Treat seeds with Bavistin or Captan @ 2-3 g/kg to control seed-borne diseases. Rogue out off-type and diseased plants at flowering and maturity stages to maintain genetic purity and reduce disease spread. For neck and finger blast, spray *Pseudomonas fluorescens* twice-once at flowering and again 10 days later. Foxtail millet generally does not face major insect issues; however, field sanitation and timely removal of diseased plants will help maintain crop health.

## **OILSEED CROPS**

### **Soybean**

Select high-yielding soybean varieties such as JS-95-60, JS-97-52, JS-2093, RVSM-1135, and MACS-1460 for better yield potential. Sowing should be completed between mid-June and the first fortnight of July in well-drained loamy or sandy loam soils, free from water stagnation. Treat seeds with *Rhizobium* and *Phosphotika* @ 20 g each per kg of seed to improve nodulation and nutrient uptake. Use line sowing with a spacing of 45 cm between rows and 5 cm between plants for optimal plant population and crop management. Maintain a weed-free crop up to 60 days after sowing (DAS). Conduct at least two intercultural operations-first at 25 DAS and the second at 45 DAS-to support healthy crop growth and suppress weed competition.



**Varietal demonstration in soyabean**



Punjab's agriculture is characterized by the cultivation of wheat and rice, which together dominate over 80% of the state's total cropped area. Other significant crops include sugarcane, maize, oilseeds, cotton, and various fruits and vegetables. The state has a high crop intensity, with many fields being sown multiple times annually, contributing to its status as a major food producer in India. Punjab's agriculture faces several challenges, including monocropping, depletion of water resources, excessive use of pesticides and fertilizers, etc. These issues contribute to environmental degradation, economic stress for farmers, and ultimately impact crop production. There is utmost need to educate the farmers with agro-advisories regarding advanced techniques in cultivation of field crops, vegetable & fruit crops, agroforestry and livestock management in order to promote the scientific cultivation along with conservation of natural resources. The following agro-advisory highlights the important points to be followed by the farmers during kharif season to sustain the productivity of field crops, horticulture crops, agroforestry and livestock.

## **SOIL TESTING**

Soil testing is the best tool to ensure optimum and balanced use of nutrients. Improper use of fertilizers leads not only to imbalanced nutrition but also deterioration of the environment. For collection of soil sample scrap away surface litter and make a V-shaped cut with a spade or a khurpa to a depth of 6 inches. Remove about 1" thick uniform slice of soil from one side of the cut. Similarly, collect samples from 7 to 8 places in the field of uniform texture and uniform fertility. Put the samples in a clean bucket, tray or cloth and mix it thoroughly. Take approximately half kg of soil in a cloth bag and label it with information such as field number, name of the farmer, address, date of sampling etc. If the samples are wet, dry them in shade before putting into the cloth bag. Soil and water samples are tested by Soil Testing Laboratory, Deptt. of Soil Science, Punjab Agricultural University, Ludhiana and Soil Testing Laboratory of all KVK's.

sowing (2-4 weed leaf stage). For swank, chinigha, broadleaf weeds, paddy mothas, gandiwalamotha, Novlect 12 EC (fenoxaprop-p-ethyl) @ 500 ml per acre should be sprayed at 20-25 days after sowing (2-4 weed leaf stage). For madhana, chinigha, chirigha, takrigha, Ricestar 6.7 EC (fenoxaprop-p-ethyl) @ 400 ml per acre should be sprayed at 20-25 days after sowing (2-4 weed leaf stage). For broadleaf weeds, paddy mothas, gandiwalamotha, Almix 20 WP (chlorimuron ethyl 10% + metsulfuron methyl 10%) @ 8 g per acre should be sprayed at 20-25 days after sowing.

**Fertilizers management:** Apply 130 kg urea per acre in three equal splits at 4, 6 and 9 weeks of sowing and for basmati rice, apply 54 kg urea per acre in three equal splits at 3, 6 and 9 weeks after sowing.

### **Irrigation**

**Direct seeding (flat/raised beds) in tar-wattar fields:** Apply first irrigation at around 21 days after sowing. After that, apply irrigations at 5-7 days interval depending on soil type.

**Direct seeding in dry fields:** Apply first irrigation immediately after sowing and second irrigation at 4-5 days after sowing. Subsequent irrigations should be applied at 5-7 days interval depending on soil type.

### **Maize**

**Improved varieties:** PMH 17, DKC 9144, Bioseed 9788, PMH 14, PMH 13, ADV 9293, JC 12, PMH 11, PMH 1, JC 4, PMH 2, JC 12 & JC 4 variety is recommended for kandi areas of the state.

**Time of Sowing:** Last week of May to end of June. In fields, which are prone to damage through water stagnation, sow the crop in end of May to early June, so that the crop gets firmly established before the rains.

**Seed rate and Method of Sowing:** Use 10 kg seed per acre for other varieties. Sow the seed 3-5 cm deep in lines with a maize planter or seed-cum-fertilizer drill, provided with a planting attachment. The row to row spacing should be kept at 60 cm whereas plant to plant spacing to be maintained at 20 cm. A four-row pneumatic planter with provision of bed making can be used for maize sowing.

**Weed management:** Give two hoeings about 15 to 30 days after sowing with khurpa/ kasaula/ wheel-hoe/ triphali/ tractor-drawn cultivator. Spray 800 g per acre Atrataf/ Atragold/ Masstaf/ Atari/Traxx 50 WP (atrazine) on medium to heavy textured soils and 500 g per acre in light soils within ten days of sowing, using 200 litres of water. Alternatively, spray 105 ml per acre Laudis 420 SC (tembotrione) in 150 litres of water at 20 days after sowing provides effective control of mixed weed flora. For the control of dila/ motha, apply 400 mL per acre 2,4-D amine salt 58 SL as post emergence 20-25 days after sowing in 150 litres of water.

**Fertilizer management:** Apply 110 kg urea/acre, 55 kg DAP/150 kg SSP per acre and 20 kg MOP/acre. If maize follows wheat, which had received the recommended dose of phosphorus, omit its application to maize. When 27 kg of DAP is used, reduce the urea dose by 10 kg and when 55 kg of DAP is used, reduce the dose of urea by 20 kg. To all recommended varieties, drill one-third of nitrogen and the entire quantity of phosphorus and potassium at the time of sowing. Top dress one-

third of nitrogen at the knee-high stage and the remaining one-third at the pre-tasseling stage.

**Insect pest management:** Regularly monitor the crop for the attack of fall armyworm. Use recommended insecticides for control of maize borer within 2-3 weeks of sowing. Control maize borer using Trichogramma cards on 10 and 17 days old crop.

## MANAGEMENT OF FODDER CROPS

Follow recommended time of sowing, seed rate, treat the seed before sowing and Use fertilizers in balanced amount. To get better quality of fodder, cut the fodder crop at proper stage. In summer months, Irrigate the fodder crop one week before cutting especially to sorghum and bajra to minimize anti-quality factors. Avoid use of un-recommended agrochemicals on fodder crops as this may be harmful to dairy animals. Plants attacked by any insect-pests should be uprooted and destroyed. Do not grow fodder crops on soils high in selenium. It is advisable to grow fodders, whenever possible, as mixtures rather than as monocultures. Crop mixtures, which combine a non-legume, such as maize, jowar, and bajra with a legume, such as cowpea and guara provide a balanced diet for animals because legumes are important source of proteins and non-legumes are rich in energy. Harvest the fodder mixture when maize is in the milk-ripe to dough stages, when sorghum has one half to one-third heads out and when bajra shows the emergence of ears from the flag leaves. An adequate supply of quality fodders round the year can be ensured by preserving the fodder as silage and hay.

### Silage making

Non legume kharif fodders, such as maize, sorghum, bajra, which are rich in soluble sugars and low in protein possess excellent qualities for conserving as silage. Harvest the crop for silage making when nutrient contents are at their peak stage and it has enough dry matter. A crop with 30-35 per cent dry matter conserves into a high-quality silage. Chop the harvested crop to the length of 2 to 3 cm and press the chopped crop in the trench thoroughly with a tractor and raise it to 1 metre above the ground level, to create proper anaerobic conditions to make quality silage. About 6-7 quintals of green fodder should be pressed in 1 m<sup>3</sup> area. Cover the fodder with polythen sheet and put the mud on it and finally mud-plaster. See that the silo-trench is completely air-tight. Silage will be ready within 45 days.

### Hay making

The fodder crops such as berseem, lucerne and cowpea having soft stems are suitable for hay-making. Harvest the fodder crops at pre-flowering stage. Chop the fodder to a length of 5 to 8 cm and spread it in a 10-15 cm thick layer on a hard-surface to dry it in the sun. The usual threshing floor can also be used for this purpose. To speed up the drying process, stir the fodder with a rake after every 2 to 3 hours during the day. When thoroughly dried (usually 2-3 days depending on the frequency of stirring), collect the dried material for storage. A kilogram of dried hay containing 90 per cent dry matter is equivalent to about 6 kilograms of green fodder containing 15 per cent dry matter.

PPH-1, and PPH-2 for pumpkin; Punjab Bahar, Punjab Barkat, and Punjab Komal for Bottle Gourd; Punjab Karela-15 and Punjab Jhaar Karela-1 for Bitter Gourd; Punjab Nikhar and PSG-9 for Sponge Gourd; PAG-3 for Ash Gourd; and Punjab Tinda-1 for Round Gourd. Irrigate every 6-7 days in summer, totalling 8-10 irrigations, but reduce frequency before harvest to prevent fruit rot. For fruit fly management, install 16 traps per acre in the 3<sup>rd</sup>-4<sup>th</sup> week of April for spring crops and the 4<sup>th</sup> week of June for rainy season crops, or use a bait spray of Malathion 50 EC at 20 ml plus 200 g Gur in 20 L water at 15-day intervals on the lower side of maize plants grown as a trap crop in rows 8-10 m apart. Manage fungal diseases with a mixture of Bavistin 1 g and Mancozeb 2 g per litre of water, and control sucking pests with Imidacloprid at 0.5 ml per litre. Additionally, control whitefly using 1200 ml of PAU Neem extract in 100-125 litres of water per acre, as recommended by Punjab Agricultural University.

### **Okra (Bhindi)**

Okra should be sown in the North Indian plains as a rainy season crop in June-July, using varieties like Punjab Suhawani, Punjab-8, Pusa Kranti, Varsha Uphar, or A-4 at a seed rate of 4-6 kg/acre with a spacing of 45 × 15 cm. To control jassid and whitefly, apply 1-2 sprays of Ecotin (Azadirachtin 5%) at 80 ml or PAU Neem extract at 2000 ml in 100-125 litres of water per acre. During the rainy season, when jassid attacks are severe, use chemical control with 40 g Actara 25 WG, 560 ml Malathion 50 EC, or 40 ml Confidor 17.8 SL in 100-125 litres of water per acre. Manage spotted bollworms by spraying Coragen 18.5 SC (Chlorantraniliprole) or Sumipleo 10 EC (Pyridalyl) at 50 ml in 100-125 litres of water per acre. Red spider mites, a serious issue during dry, hot summer weather, can be controlled with PAU Dharek extract at 1600 ml in 100 litres of water per acre or with 150 ml Oberon 22.9 SC in 100 litres of water per acre. Monitor for Yellow Mosaic Virus (YMV), rogue out affected plants, and spray Thiomethoxam 25WG at 60 g/acre to control its vector, whitefly.

### **Cauliflower**

For cauliflower, sow the nursery in May for the Kharif season and transplant by mid-June using early group cultivars like Pusa Deepali, Pusa Kartiki, and Pant Gobhi-3 with a spacing of 45 × 30 cm and only use early group cultivars avoid physiological disorders for Kharif season to like buttoning. Apply 10-15 tonnes of FYM, 50 kg N (110 kg Urea), 25 kg P<sub>2</sub>O<sub>5</sub> (155 kg SSP), and 25 kg K<sub>2</sub>O (40 kg MOP) per acre, with full P<sub>2</sub>O<sub>5</sub>, K<sub>2</sub>O, and 1/3 N at sowing, and the remaining N as top-dressing four weeks after transplanting. Irrigate every 6-7 days, totalling 9-11 irrigations, while avoiding over-flooding. Add Borax at 8 kg/acre to manage browning in cauliflower. To manage leaf spot, blight, and downy mildew, spray Mancozeb at 0.2% or Copper Oxychloride at 0.3%. Control stem borer and fruit borer with Spinosad 45 SC at 80 ml/acre in 200 litres of water, ensuring a waiting period of 5-7 days before harvesting.

### **Cowpea**

Sow cowpea as a rainy season crop in June-July using the variety Cowpea 263 with a spacing of 45 × 15 cm, applying 45 kg Urea, 100 kg SSP, and 16 kg MOP per acre at sowing, along with seed inoculation using Bavistin and Rhizobium culture. Control sucking pests like aphids and whitefly with alternate sprays of Neem-based pesticide 1500 ppm at 600 ml/acre and Imidacloprid 17.8 at 70 ml/acre. Manage fungal diseases by treating seeds with Carbendazim at 0.2%, Thiram at 0.3%, or Captan at 0.3%.

## AGROFORESTRY

### Poplar

Irrigate poplar at 7-10-day interval during summers as it requires a lot of water during its growth period from April to October. However, standing water for long period, especially in heavy soils, is harmful for Poplar, therefore proper arrangement for drainage must be ensured during monsoon. Apply urea to plants according to their age: 170 g for 1-year-old, 260 g for 2-year-old, 350 g for 3-year-old, 430 g for 4-year-old, 520 g for 5-year-old, and 610 g for 6-year-old plants. Distribute the total yearly dose evenly, applying one-third in May, one-third in July, and the final third in September. For 1-year-old plants, apply in a 1 m diameter ring; for 2-year-old plants, in a 2 m diameter ring; and for 3-year-old plants, in a 3 m wide strip (1.5 m on each side of the tree row). Adjust the dose by reducing 25% in high nitrogen (N) soils and increasing 25% in low N soils.



Leaf spots in poplar

### Zinc deficiency in Poplar



Zinc deficiency in Poplar

For management of Zinc deficiency in Poplar, apply 100 g, 200 g and 300g Zinc Sulphate Heptahydrate (21%) OR 60g, 120 g and 180 g Zinc Sulphate Monohydrate (33%) for 1<sup>st</sup>, 3<sup>rd</sup>, & 5<sup>th</sup> year of poplar, respectively, in 1 m diameter during 1<sup>st</sup> year, 2 m diameter ring around tree during 3<sup>rd</sup> year and 3 m wide strip during 5<sup>th</sup> year, after harvesting of the Rabi intercrop. Mechanical removal of weeds or intercultural operations with tractor

drawn implements can be followed to keep the field weed free. Avoid the spray of herbicides such as Atrazine, Metribuzin, Diuron, 2,4-D, recommended for control of broadleaved weeds in inter-crops like sugarcane & maize, as these may result in mortality of poplar. Poplar Stem Borer damages the tree by making tunnel in the stem and branches, and hollows out the trees. Prune the infected branches in August-September before entry of larvae into main stem. Plug all the holes on trunk with wet clay, except the lowermost hole. Inject kerosene in this lowermost hole and then plug it with wet clay. Leaf spots in poplar generally appear after onset of monsoons, and result in blight like appearance and early leaf fall, thus affecting the growth of poplar. This can be managed by 3-4 sprays of Bordeaux mixture (0.8%) at 15 days interval. During Kharif season, except Paddy, crops like Maize, Sorghum, Bajra, Cowpea, Moong, Mash, Mentha, Arvi, etc., can be successfully cultivated as intercrops during the first three years of plantation. Avoid pruning of poplar trees during rainy season as this is the

1.0 m wide perforated polythene sheet (50-60 gauge) over 1.0 m wide soil bed with simultaneous uniform seed placement over the soil bed for raising mat type nursery for mechanical paddy transplanters.

## **LIVESTOCK**

### **Shelter Management**

**Ventilation:** Facilitate adequate air circulation in shelters by utilizing exhaust fans, roof ventilators, or open-sided structures.

**Cooling Systems:** Implementing sprinkler or misting systems can alleviate heat stress, while applying whitewash to roofs may enhance sunlight reflection or spreading of straw on roof top also gives relief from heat.

**Drainage:** Ensure adequate drainage around shelters to avoid water accumulation, which may result in mosquito proliferation and associated diseases.

**Washing of shed:** Clean the sheds monthly using a 5% phenyl solution. During the rainy season, the washing frequency should be increased to twice a month. For katcha floors, a layer of 1-2 inches of soil may be excavated and substituted with a layer of clean soil.

**Foot bath:** Make a footbath outside the livestock unit that should carry a mixture of 4-5% Formaldehyde. This helps to avoid the entry of infectious agents in the shed.

### **Feeding Practices**

**Green Fodder:** Employ Kharif crops such as maize, bajra, and sorghum as green fodder. Consider multicut varieties for bajra and sorghum, such as the Napier Bajra hybrid and Pb Sudex Chari No.4, which are both nutritious and abundantly available during this season. Additionally, incorporating leguminous fodders like guara and cowpea with the aforementioned non-leguminous fodders will aid in reducing the reliance on expensive concentrate mixtures. Rate of feeding of fodder should be around 40-50 Kg/animal/day.

**Silage Preparation:** Produce silage from excess green fodder such as maize, bajra, and sorghum to guarantee feed supply during scarcity. Small-scale farmers can utilize polypropylene bags and drums for silage preparation. It is advisable to feed silage post-milking to prevent any smell in the milk.



**Spreading of straw on shed roof**

Rajasthan is predominantly a kharif cropping state due to vast area under rainfed cropping and crops. The diverse crops are grown during Kharif season. During May and June preparations for sowing of different crops like maize, bajra, green gram, cowpea, guar, moth bean are completed by the farmers. The sowing and nursery raising of cotton, groundnut and rice are done under varying situations and climatic conditions. To harness the potential of the crops, an appropriate technology along with management practices are inevitable. Therefore, a proper agro-advisory is need of the day to be issued for the use of the farmers for different agricultural operations. It will help in increasing the productivity and profit from agriculture and empower the farmers while helping in sustainable management of natural resources and applied inputs. The agro-advisories related to different activities and operations in agriculture conducted by the farmers during May-June are discussed in following para:

### **Soil Analysis**

After the harvesting of rabi season crops in March and April and before sowing of kharif crops is the best time for taking soil samples from the farmers' fields for their analysis for physical, chemical and biological properties. The farmers are advised to take soil samples from 5 places of their fields, mix it properly and take them to nearby KVK or district soil testing labs for analysis. The sampling should be done from 0-30 cm soil depth. Attempt should be made get the soil sample analysed before sowing of kharif crops. The nutrient status as evidenced from soil sample analysis will help to decide right and balanced application of fertilizers and avoid extra application. It will also help in enhancing the nutrient use efficiency and saving of cost of cultivation while maximising the profit to farmers.

### **Summer Deep Ploughing and Application of Gypsum**

May and June are the best time for summer deep ploughing and application of Gypsum in the field. The summer deep ploughing is process of solarization of the soil

25-30 days after sowing; apply the full phosphorus dose at sowing. On zinc-deficient soils, apply zinc sulphate at 25 kg/ha (21% Zn) or 15 kg/ha (33% Zn). Alternatively, spray 0.5% zinc sulphate (33% Zn) once at the tillering stage. During prolonged dry spells, skip the nitrogen top-dressing and instead spray a 2% urea solution as a foliar feed.

**Water Management:** If irrigation is available, apply water during critical growth stages: at tillering, flowering, and grain filling. This is especially important under extended dry periods to avoid moisture stress.

**Weed Management:** Two hand weeding (hoeing) at 20 and 25 days after sowing (DAS). This removes competing weeds and helps conserve soil moisture. Otherwise, apply Atrazine 50% WP at 0.5 kg/ha as a pre-emergence herbicide, and follow with one hand-weeding later.

**Pest and Disease Management:** Termites and White Grubs: Apply Fipronil 0.3G (granular insecticide) in the seed furrow at sowing (12 kg/ha) to protect young seedlings. Downy Mildew: Treat seeds with Mancozeb (e.g., Dithane Z-78 or M-45) at 2 kg/ha (as a seed dressing) before planting to prevent seed-borne infection. Smut: Treat seeds with Ceresan (Carboxin) or Thiram at 1-2 g/kg or with fine sulphur (300-mesh) @ 4 g/kg to prevent kernel smut.

## **Maize**

**Time of Sowing:** Last week of May to end of June. In fields, which are prone to damage through water stagnation, sow the crop in end of May to early June, to allow the crop gets firmly established before the rains.

**Seed Treatment:** Seed treatment with Carbendazim 50% WP or Thiram 75% WP @2.0-2.5 g/kg of seed or Captan @2g/kg seed will help against seed-borne and soil-borne fungal diseases. Seed treatment with Imidacloprid 600 FS @5ml/kg seed controls the early shoot fly or termites. Seeds are treated with *Azospirillum* (600 g) and *Phosphobacterium* @200 g/10 kg seed to enhance the availability of nitrogen and phosphorus.

**Improved Varieties:** Pratap QPM Hybrid Maize- 1, HQPM Hybrid -1, HQPM Hybrid -5, Bio-9682, Bio-9544, Pratap Raj Hybrid Maize-1095, Mahi Dhawal, DHM 117, Pratap Hybrid Maize-3, Pratap Maize-9, PEHM-2, Pratap Hybrid-1, DHM-121, IQMH -203, IMHB-1532, DMRHP-1402, LPCH-2, LPCH-3, GM-6, Mahi Kanchan, Madhuri, Pratap Kanchan, Pratap Maize Hybrid -6

**Seed Rate and Spacing:** Sowing of 20-25 kg seed per ha is advised with 60 cm row to row spacing and 25 cm plant to plant spacing. In high rainfall area sowing in ridge - furrow is recommended.

**Nutrient Management:** Apply 10-15 t/ha organic manure/compost to improve soil structure and nutrient availability. Apply 90-120 kg of nitrogen and 40 kg of phosphorus per ha under irrigated conditions. In rainfed conditions, apply 60-90 kg of nitrogen and 35-30 kg of phosphorus per ha. Apply one-third nitrogen and a full dose of phosphorus 10 cm deep in soil at the time of sowing. The remaining half dose of nitrogen 30 days after sowing, and the third dose after the emergence of tasselling and well mixed in soil. Care should be taken to ensure no direct contact of fertilizers with seed at the time of sowing.

**Water Management:** Adequate water-supply is essential throughout the crop season particularly during the pre-tasselling, silking and grain filling stages, if no rain during this stage.

**Weed management:** Spray 1-1.5 kg a.i. atrazine 50 WP (atrazine) per ha on medium to heavy textured soils immediately after sowing using 500 litres of water followed by Tembotrione 42 SC 120 g a.i. per ha in 500 litre of water after 25-30 days after sowing hoeing/interculture at 15 to 30 days after sowing. For the control of broad leaf weed, apply 900 g per ha 2,4-D ester 38 EC in 450 litres water at 30-35 days after sowing.

**Plant Protection:** Fall armyworm: Spray the crop with chlorantraniliprole 18.5 SC @ 200 ml or spinetoram 11.7 SC 250 ml or emamectin benzoate 5 SG 200 g in 500 litres of water per ha, for crop up to 20 days old. Maize stem borer: Apply Phorate 10 G or carbofuran 3% granules @ 7.5 kg per ha in the shoot of the plant after 16 to 30 days after sowing. Alternatively, spray 10% solution of neem leaves/kernels after 10-20 days of germination. Armyworm and grasshoppers: Apply Methyl Parathion 2% power @ 25 kg per ha, summer ploughing in fallow land is advised. Aphids: Apply methyl demeton 25 EC @ 1 liter per ha in 500 litre of water at the time tasseling. Maydis leaf blight: Spray mancozeb or propinab or carbendazim 0.2 percent solution, repeat the spray in 10-15 days.

## Rice

**Time of Nursery Sowing:** The ideal window for paddy nursery sowing in Rajasthan is 20 May to 20 June.

**Seed Rate and Seed Treatment:** 20-25 kg/ha seed is sufficient for transplanting. Treat the seed with Sprint 75 WS (carbendazim + mancozeb) by making slurry of 3 g fungicide formulation in 10-12 mL water for one kg seed (24 g fungicide in 80-100 mL water for 8 kg seed) before sowing.

**Nursery Raising:** Prepare 1.0-1.5 m wide nursery beds. Generally, 400-500 m<sup>2</sup> nursery area is sufficient for transplanting in one ha area. Paddy nursery can be raised in second fortnight of May and 24-25 kg seed is sufficient for transplanting in one ha area. In nursery paddy seed should be covered properly with sand or with mixture of sand and manure, nursery bed should be always moist without standing water.

**Suitable Varieties:** PR-126, PR-106, PR-1121, BK-190, PB-1718, PB-1509, PB-1885. These varieties offer a mix of high yield, stress tolerance, and market preference.

**Transplanting:** Manual transplanting: Transplant the seedlings in lines, 2-3 seedlings per hill at a spacing of 15 x 15 cm, Depth of transplanting should not be more than 2-3 cm. Mechanical transplanting: Done by paddy transplanter in puddled and un-puddled conditions. Raising of mat type nursery is required

**Fertilizer Management:** Medium duration varieties & Hybrids (e.g., PB-1718, PB-1509): 150 kg N, 60 kg P<sub>2</sub>O<sub>5</sub>, 60 kg K<sub>2</sub>O, 25 kg Zinc sulphate per hectare. Early Duration Varieties (e.g., PR-126): 120 kg N, 60 kg P<sub>2</sub>O<sub>5</sub>, 60 kg K<sub>2</sub>O, 25 kg Zinc sulphate per hectare.

**Application Schedule:** Apply the full dose of P, K, Zn, and one-third of N at the time of puddling. The remaining two-thirds of N should be top-dressed in two equal splits at 3 and 6 weeks after transplanting. Incorporate 15 tons of well-decomposed FYM/ha before transplanting.

**Improved Varieties:** MH-1142, MH-421, IPM-410-3 (Shikha), IPM-205-7 (Virat), GM-7, GM-6 & GM-4.

**Fertilizer Management:** Fertilizers and micronutrients should be applied based on the soil test. In case of soil tests are not available, apply 15-20 kg N, 40-50 kg  $P_2O_5$ , 20 kg  $K_2O$ , 20 kg S per ha at the time of sowing as basal application. Zinc sulphate @25 kg/ha should be applied at every 2-3 years, and 0.5 g/kg sodium molybdate.

**Water Management:** Irrigate once or twice if dry spells prevail for 10-15 days especially between flowering and pod formation and grain development stages. Light irrigation is recommended. Critical stages for irrigation are flowering and early grain formation/pod filling stages.

**Weed Management:** The critical weed control period is 15-35 DAS. Hand weeding at 20-25 DAS and 35-40 DAS, or using hand hoes/mechanical weeders, is effective. Apply Pendimethalin 30% EC (1.0-1.25 kg a.i./ha) as pre-emergence within 2 days of sowing for broadleaf and grass weeds, and Imazethapyr 10% SL (75-100 g a.i./ha) as post emergence at 15-20 DAS for 2-4 leaf stage weeds.

**Plant Protection:** Whitefly can be managed by growing barrier crops (maize, sorghum), using 10 sticky traps/acre, removing weed hosts (Parthenium, Croton), and spraying Thiamethoxam 25 WG (100 g/ha) or Imidacloprid 17.8 SL (0.3 ml/litre) at early infestation. Aphids' population can be minimised by spraying with Neem oil 2% as a preventive bio-insecticide or Dimethoate 30 EC @ 1 ml/litre. Jassids/ leafhoppers control by spray with Acetamiprid 20 SP @ 0.2 g/litre or Imidacloprid 17.8 SL @ 0.3 ml/litre at ETL. Thrips managed by avoiding dense sowing and spraying Spinosad 45 SC (0.3 ml/litre) or Fipronil 5 SC (1.5 ml/litre). For pod borer management use of pheromone traps (5/acre for adult moths), spray of Emamectin benzoate 5 SG @ 0.4 g/litre or Spinosad 45 SC @ 0.3 ml/litre at flowering to pod formation stage is recommended.

**Disease Management:** For yellow mosaic virus, control the whitefly vector by spraying Thiamethoxam 25 WG (100 g/ha) or Imidacloprid 17.8 SL (0.3 ml/litre) at early infestation. To reduce the Cercospora leaf spot & Anthracnose incidence, spray of Mancozeb (0.25%) or Carbendazim (0.1%) at first symptom appearance, or treatment of seeds with Thiram/Captan @ 2.5 g/kg is beneficial. To control Root rot and Wilt, crop rotation with non-host crops should be taken and seed treat with Trichoderma viride (4 g/kg seed) or carbendazim is recommended.

## Urd bean

**Time of Sowing:** Urd bean is advised to be sown with the onset of the monsoon, ideally between mid-June and early July, but can be extended until end of July in case of delayed rains.

**Seed Rate & Spacing:** For sole cropping, a seed rate of 15-20 kg/ha is recommended, while for intercropping, use 8-10 kg/ha. The ideal spacing (R x P) is 30-40x10 cm to optimize growth and yield.

**Improved Varieties:** Mukundra Urd-2, Kota Urd-4, Kota Urd-3, KU 96-3 (Azad Urd-3), Pratap Urd-1, Pratap Urd-2.



## CEREAL CROPS

### Rice

Farmers can select the medium duration varieties like Sikkim Dhan 1, CAU-R1, Abhishek, Attey etc. Farmers are suggested to start main field preparation for paddy transplantation through repairing old bunds by maintaining 20-30 cm height in low land situation. Well decomposed FYM should be applied 2-3 days before sowing @10 q/1000 m<sup>2</sup>. or vermicompost @ 3-6 t/ha either alone or in combination 15 days before



**Cultivation of Rice Variety Sikkim Dhan 1**

transplanting. Sow the seeds @35-40 kg/ha for transplanting, 60-100 kg/ha for direct seeding, 5 kg/ha for SRI method. Seed should be treated with Azospirillum @ 20 g/kg seed. Farmers of mid and high altitude are suggested to complete the transplanting by month end of May and in low altitude it should be completed by last week of June. An area of 1000 m<sup>2</sup> nursery area is sufficient for planting in one hectare of land. Uprooting of 25-30 days old rice seedlings with 20 cm x 15 cm spacing and 2-3 seedlings/hill is considered best for timely transplanted crop. Neem based formulation (15ppm) @ 0.5% or 5 ml/l of water two to three times can be used for pest management.

### Maize

Promising varieties are Pahlenlo makkai (local variety) and CP-333 (hybrid variety). Seeds should be inoculated with Azospirillum, Azotobacter etc., and phosphorus solubilizing bacteria (PSB) @ 20 g/kg seed before sowing. A seed rate of 15-20 kg/ha is optimum for maize. Sowing should be done in furrows at a depth of 2-3 cm and

spacing of 50 cm between rows and 30 cm between plants. Mixed compost @ 2.5 t/ha + neem cake @ 0.5 t/ha + vermicompost @ 2.5 t/ha or FYM 2.5 t/ha should be applied. Manual weeding or use of biomass can be done for control of weeds. One Earthing up should be followed after plant attains knee height stage. Next earthing up should be done after 40-45 days of sowing. For management of cut worm, mixture of cow urine + water + neem oil (2 litre + 8 litre + 50



**Cultivation of Maize variety CP-333**

ml) @ 100 ml/plant should be applied. For management of FAW, spray spinosad @ 0.3 ml/litre and Neem oil 4ml/L of water at 2 weeks interval and apply mud slurry in the whorl of affected maize plants soon after observation of FAW incidence in the field.

### **Finger Millet**

For organic condition of Sikkim: VL mandua 352, VL Mandua 379, VL Mandua 347 are suitable. Field preparation should be started at second fortnight of May. Field should be ploughed to prepared smooth seedbed. Provide inward slope with shallow drain to take out excess rain water. Well decomposed FYM @ 10t/ha mixed with 50g of *Trichoderma harzanium* and 200g of *Bacillus cereus* should be incorporated at the time of field preparation. 21-25 days old seedlings should be transplanted with spacing of 30\*10 cm with two seedlings/hill. Foliar spray of neem-based formulation @ 0.5% or 5 ml/lit of water at weekly interval is recommended.

### **Spice Crops**

#### **Cherry Pepper**

The seed of cherry pepper is to be sown in the month of February so as to make the seedlings ready for transplanting in the month of April to May. The seeds should be sown 3-5 cm apart for producing healthy seedlings. Generally, 45-50 days old seedling is ready for transplanting in the main field in raised beds, maintaining an optimum spacing of 35 cm x 40 cm. The beds need to be covered by organic mulch like dried leaves, straw or black poly mulch whichever is available. Nutrients should be properly managed through organic sources with the use of different nutrient rich manures, FYM @ 20 t/ha, neem cake @ 2 t/ha, vermicompost @ 2.5t/ha, and drenching of root area with 20% cow urine at 25 days interval till the harvesting stage.

Light hoeing is required to remove weeds and loosen the top soil for better aeration and root growth. The buds from the first and second nodes should be pinched off to promote vegetative growth, more fruit set and yield. Treatment of soil with *Trichoderma harzianum* culture @ 10g/m<sup>2</sup> one week before sowing manages soil borne diseases. As damping off and wilt is the most common disease, it can be managed by dipping the roots of seedlings in *Trichoderma viride* and *Pseudomonas fluorescens* culture @ 10g/l for about 15 minutes. After the fruit set, placing of 8-12 pheromone traps is necessary to manage fruit fly problem. Regular collection and destruction of dropped fruits also controls fruit fly population.

**Stage-Specific Inoculants:** Biofertilizer pack with *Azospirillum*, *Azotobacter*, *Phosphobacteria*, *Potash bacteria*, *Pseudomonas*, and PPFM for different crop stages; enhances yield by 10-15% and reduces chemical fertilizer use by 25%. **Zinc Solubilizing Bacterial Biofertilizer:** Carrier-based or liquid form; enhances Zn nutrition and yields by 17-20%, adding 250-400 kg/ha. **TNAU Micronutrient Mixtures:** Enriched FYM-based micronutrient mixtures for wetland and rainfed rice; alleviates deficiencies, improves quality, and enhances yield by 11-20%. **Nanosci Lure for YSB:** Nano-pheromone lure for eco-friendly pest monitoring and mass trapping of yellow stem borer. **Stage-Specific Inoculants:** Biofertilizer pack tailored to growth stages; increases yield (10-15%) and reduces chemical fertilizers by 25%. *Bacillus altitudinis* FD 48: Dual-use bioagent (seed+foliar) enhancing root traits and yields under both irrigated and stress conditions. **TNAU Biomineralizer for Stubble:** Microbial product (2 kg/tonne) for in-situ rice straw decomposition to reduce burning and enrich soil.

### **Maize**

**Nutrient & Soil Management:** Neem cake @ 250 kg/ha; Zn citrate (0.5%) + Fe citrate (1%) foliar sprays at 30, 40, and 50 DAS in deficient soils. **FAW Management (Technology Capsule):** Seed treatment: Thiamethoxam 30 FS @ 10 ml/kg; maintain spacing (60 × 25 cm) + rogue rows every 10 rows. **Border cropping:** Cowpea, gingelly; **Intercrop** with blackgram; install pheromone traps (50/ha) and solar traps (1/ha). **Insecticides by stage:** Early whorl (15-20 DAS): Emamectin benzoate 5 SG, Chlorantraniliprole 18.5 SC, Flubendiamide 480 SC, or Azadirachtin. Late whorl (35-40 DAS): Metarhizium anisopliae, Emamectin benzoate, Novaluron, or Spinetoram. **Tasseling/cob formation** (60-65 DAS): Use any insecticide not used earlier. **Biologicals:** Release *Telenomus remus* @ 1.25 lakh/ha.

### **COMMERCIAL CROPS**

#### **Cotton**

**High Density Planting:** Closer spacing (90 × 15 cm) for 29,629 plants/acre using suitable varieties like Suraksha, CO-17, and VPT-2.

**Fertilizer:** Apply 100:50:50 kg NPK/ha; weed control using Pendimethalin 38.7% CS @ 700 ml/acre within 24-48 hours of sowing.

**Growth Management:** Spray NAA @ 40 ppm or Cotton Plus @ 2.5 kg/acre; apply Mepiquat Chloride (0.015%) at square/flowering stage; use sodium chlorate (0.9%) or Drop Ultra as defoliant at 60% boll burst.

**Intercropping:** Paired row planting (45/90 × 30 cm) of Bt Cotton with two rows of cluster bean; application of Mepiquat Chloride and Sodium Chlorate enhances suitability of CO 17 for mechanical harvesting.

#### **Coconut**

**Pest Traps:** Install yellow sticky traps (20/ha or 8/ac), trunk banding, and spray jet water on fronds. **Biocontrol:** Release *Apertochrysa astur* (1000/ha), *Encarsia spp.* (250/ha or 100/ac). **IDM Package:** FYM (50 kg) + neem cake (5 kg) + fertilizers;  $\text{CuSO}_4$  (200 g) +  $\text{MgSO}_4$  (1 kg) alternately; bio-agents (*Trichoderma*, *Bacillus*, *Phosphobacteria*, *Azospirillum*, VAM); TNAU Coconut Tonic (40 ml twice/year);

COCOCON drench @ 2 L/tree every 2 months; crown spray of hexaconazole if needed. *TNAU Coconut Tonic*: Nutrient-rich tonic applied via root feeding (200ml/tree every 6 months); increases photosynthetic efficiency, nut yield, and palm vigour. *IIHR Arka Decomposer*: Microbial mix that rapidly composts coconut residues, enhancing organic carbon and soil health. *Copra Guard*: Polymeric nano-formulation (10 ml/litre) replacing sulphur fumigation to control fungal growth on copra.

## FRUIT CROPS

### Banana

**Insect Management:** Apply emamectin benzoate 5 SG @ 0.4 g/L at 5th, 6th & 7th months via pseudo stem injector. Seedling Bio-hardening: *Bacillus velezensis* + *Brachybacterium paraconglomeratum* (1% consortia) twice fortnightly; soil drenching @ 0.5 L/plant at 2, 4, 6 & 8 months.

### Papaya

**Vector & Disease Management:** 3-row maize border; insect-proof nursery; neem oil spray (3%); humic acid drenching @ 2 ml/L every 2 months; foliar spray of ZnSO<sub>4</sub> (5 g), boron (1 g), and urea (10 g)/L at 4<sup>th</sup> and 7<sup>th</sup> month. Jeevamruth: Drenching (200 L + oilcake extracts) @ 1 L/plant on 3<sup>rd</sup>, 5<sup>th</sup> & 8<sup>th</sup> months; Flonicamid 50 WG @ 75 g a.i./ha on need. Hi-Tech Farming Practices: Raised bed planting with drip+fertigation and foliar micronutrients (Zn, B) improving yield and fruit quality by 25-30%.

## VEGETABLE CROPS

### Chilli

**Seed treatment:** *Bacillus subtilis* @ 10g/kg; use three-row maize barrier; install yellow sticky traps @ 12/ha. Nutrition: Apply and spray a micronutrient mix (Fe, Zn, Cu, Mn, B). Virus & pest management: Rogue infected plants till 45 DAT; apply Imidacloprid 17.8% SL @ 3ml/10L and Pyriproxyfen 10% EC @ 1ml/L alternately. Fungal disease management: Spray Azoxystrobin 18.2% + Difenoconazole 11.4% SC @ 0.1% thrice at 15-day intervals. IPM module for chilli: Includes border cropping with cowpea/lab-lab/coriander, bird perches (20/ha), sticky traps (20/ha), pheromone traps (12/ha), need-based foliar sprays with Azadirachtin 1%, NSKE 5%, 3G extract 5%, five-leaf extract 10%, and *Bacillus subtilis* 0.5%-effective in reducing thrips population and increasing yield and B:C ratio.

### Cassava

**IPM module for mealybug:** Application of Azadirachtin 0.15% (1.5 L/acre) on 60 DAP followed by release of *Anagyrus lopezi* and *Apertochrysa astur*; reduces mealybug population and increases tuber yield. *TNAU Cassava Booster*: A mixture involving cow dung slurry enriched with neem cake, bioagents, and nutrients; enhances starch content, reduces CMD, and increases yield by 20-25%.

**Nutrients for Vegetables:** Bio NPK Formulation: Multi-functional biofertilizer (NPK bacteria) improving nutrient uptake and reducing chemical fertilizer dependency. BIOGROW Liquid (Tomato): Microbial liquid with *Bacillus* and *Pseudomonas* boosting yield (25-30%), lycopene, -carotene, and plant health.



Farmers are advised to follow the sowing rule to take-up sowing of rainfed crops like Cotton, Soybean, Maize, Jowar, Redgram, Greengram etc., only after receipt of cumulative rainfall of 50-60 mm in light soils and 60-75 mm in heavy soils or the soil should be wet up to top 15-20 cm depth after enter of South-West monsoon rains into the state.

## CEREAL CROPS

### Paddy

Under assured irrigation farmers are advised to raise rice nurseries of long duration (140-145 days) varieties from May 20th to June 20th. Under assured irrigation farmers are advised to raise rice nurseries of medium duration (130-135 days) or short duration (120-125 days) varieties from June 20th onwards. Month of June is not optimum for raising paddy nurseries using Telangana Sona (RNR15048) variety. Take the Greengram as catch crop to preceding rice in areas where the release of water into the canals and tanks is delayed. Green manuring with Sesbania/ Crotalaria / Pillipesara / grain legumes, residues of blackgram / greengram and in situ incorporation before flowering. Varieties: cultivation of long duration (BPT-5204, WGL-44, RNR 11718 etc.), medium duration (WGL-14, JGL-11470, JGL-384, RNR-28361, JGL-33124 etc), short duration (RNR-15048, KNM-1638, KPS-6251, WGL-362, KNM-118, JGL-24423, RNR-29325, JGL-28639 etc.) varieties.

Direct seeding of rice with Seed-cum ferti drill wherever suitable. In Telangana state, dry converted wet and wet direct seeded rice are popular, and the sowing can be started from June ending. Treat seeds with carbendazim @ 3 g/ kg (dry seed treatment) or 1 g/L (wet seed treatment). Farmers are advised to apply the Carbofuran 3CG granules @ 800 g per acre of nursery one week before pulling of the nursery. Root dipping: prepare the slurry with 5 packets (1000 g/ha) of Azospirillum and 5 packets (1000 g/ha) of phosphobacteria or 10 packets of (2000 g/ha) of Azophos, inoculant in 40 lit. of water and dip. the root portion of the seedlings for 15-

30 minutes in bacterial suspension and transplant. Make 20 cm alleyways at every 2 meters to facilitate free aeration, ease of farm operations and better pest and disease management. Broadcast butachlor @ 1-1.5 l or pretilachlor @ 500-600 ml or 4 kg granules of bensulfuron methyl (0.6%) + pretilachlor (6.0%) with 20 kg sand per acre. Practice alternate wetting and drying in well irrigated areas to save water, better aeration and minimize incidence of BPH besides getting carbon credits with less emission of greenhouse gases. Recommended dose of fertilizers: 40-48 Kg Nitrogen, 20 Kg Phosphorus 16 Kg Potash per acre during Vanakalam. Spray 0.2% ZnSO<sub>4</sub> 2 to 3 times at 5 days interval to manage Zn deficiency, particularly in alkaline soils.

### **Maize**

Deep summer plough to destroy pests and conserve moisture. Optimum sowing window is from June 15<sup>th</sup> to July 15<sup>th</sup> with medium and long duration varieties (90-100 days and 100-120 days) without any yield reduction. Sow medium duration varieties/hybrids DHM-206, DHM-182, DHM-117, Bisco-855, Bio-9637, Bisco-740, Kaveri Ekka-2288, Kaveri -2020, Kaveri-3696 and short duration hybrids DHM-115, Pioneer-3342, KH-5991, DKC-7074 during 15<sup>th</sup> June to 15<sup>th</sup> July. Sowing can be extended up to July end if monsoon is delayed. Sow on receipt of at-least 50-60 mm rainfall for better germination and complete sowing in a week for reducing the incidence of FAW. Spray NSKE 5% or neem oil 1500 ppm @5ml/l after sowing and at weekly intervals. Dibble seeds with tractor drawn cultivator or seed cum ferti drill on ridges and beds. Intercrop red gram varieties PRG-176, WRG-97 in 2:1 ratio in light soils and in 4:1 ratio in heavy soils to reduce incidence of FAW. Follow IPM for FAW management: summer ploughing, growing single cross hybrids, clean cultivation, intercropping with pulses, balanced fertilization, release of egg parasitoids (*Trichogramma* sp) and spraying Chlorantraniliprole @ 0.4 ml or spinetoram @ 0.5 ml or Emamectin benzoate @ 0.4 g per litre in plant whorls. As post emergence, spraying of Tembotrione @ 115 ml + Atrazine 400 g or Toperamezone @ 40 ml + Atrazine 400 g or mesotrione + atrazine @1400 ml/acre in 200 litre of water will effectively control weeds. After 30-35 days, crop may be inter-cultivated followed by earthing up is needed for better nutrient uptake and to avoid lodging.

### **COMMERCIAL CROPS**

#### **Cotton**

Sow the crop early by June 1st/2nd week after receipt of sufficient good rains at least (60mm). Don't sow cotton crop after July 20th as yields will be reduced drastically under rainfed conditions. Sowing of Cotton on ridge and furrow in medium soil and broad Bed furrow in black soil under rainfed conditions facilitates the crop to moisture conservation and excess water drain out. In a village or area all the cotton cultivating farmers should take up the sowings on community basis and complete within a period of 7 - 10 days. Depending on soil type and rainfall pattern cotton should be sown at a spacing of 120 x 60 cm/ 120 x 45 cm/ 90 x 60 cm/ 90 x 45 cm/ 90 x 30 cm. Deep summer plough to destroy the eggs of insect pests and to remove weed stock. Intercrop with greengram/ blackgram/ soybean/ cluster bean in 1:2 or 1:3 ratio, and red gram 4:1 / 6:1 / 8:1 ratio which also helps to manage sucking pests.

Grow 2-3 rows of maize or sorghum as border crop as barrier to prevent pests. Treat the seed with imidacloprid 70 WS@ 5 g per kg or thiamethoxam 70% WS @ 4 g per kg followed by *Trichoderma viridi* or *Pseudomonas fluorescens* @ 10 g per kg.

Pre-emergence herbicides like pendimethalin (Stamp extra) @ 700 ml/acre should be applied within 24 -48 hours after sowing with the help of hand sprayer for better control of weeds in cotton. Application of post emergence herbicides like quizalofop ethyl @ 400 ml + pyrithiobac sodium @ 250 ml in 200 L of water at 30 DAS in cotton to control grasses and broad-leaved weeds. Inspect the crop at squaring and flowering stage for presence of PBW larvae within flowers. Installation of Pheromone traps @ 8/acre and monitoring ETL at this stage is 10 % damaged flowers. Spray  $MgSO_4$  @ 10 g/l twice at 45 and 75 DAS; Boron @ 1.5 g/l twice at 60 and 90 DAS; 0.5%  $FeSO_4$  with citric acid twice at weekly intervals and  $ZnSO_4$  @ 2 g/l twice at 4-5 days interval at 45 DAS to manage micronutrient deficiencies. During excess rainfall, drain excess water and apply 25 kg urea and 20 kg potash per acre. Spray Planofix (NAA 10 PPM) @ 1 ml per 4.5 liters of water twice at 10-15 days interval to manage flower dropping. Spray 1%  $KNO_3$  or 19:19:19, 20:20:20, 21:21:21. Follow rotation of insecticide sprays against insect pests for high efficiency. Stem application of fipronil 1: 4 at 30 DAS and Flonicamid / Imidacloprid with water (1:20) at 45, 60 DAS to be done which protects the crop from early season sucking pests like jassids and aphids.

## PULSE CROPS

### Pigeon pea

Sow high-yielding wilt-resistant varieties (155-165 days) like WRGE-93, WRGE-97, WRGE-121, and medium-duration varieties like WRG-255 and TDRG-4, TDRG-59 & TDRG-272 can be recommended. Sowing should be initiated only after receipt of 60 mm rainfall. The optimum sowing time should be 15<sup>th</sup> June to 15<sup>th</sup> July at a spacing of 150-180 x 20 cm in black soils and 90-120 x 20 cm in red soils. Seed rate should be 2.5 to 3.0 kg per acre only (plant-to-plant spacing should be mandatory). Treat seeds with thiram or captan @ 3g/kg followed by rhizobium culture @200-400 g/ 3 kgs of seed. The raised bed method or ridge and furrow method should be followed to protect the crop from *Phytophthora* blight disease, which occurs due to excess and stagnation of rainwater. Intercropping in Red gram + Cotton (1:4/ 1:6), Soybean (1:4/ 1:6), Green gram/ Black gram (1:7) should be followed. Treat the Red gram seed with *Trichoderma viride* @ 10 g/ kg seed to protect the crop from soil-borne diseases.

Spray pendimethalin @ 1 - 1.5 L in 200 liters of water within 24 - 48 hours after sowing will effectively control the weeds in Red gram Post-emergence application of quizalofop ethyl @ 400 ml/ acre to control grasses and imazethapyr 10 % @ 300 ml/ acre to control grasses and broad-leaved weeds. Both herbicides in the tank mix are also recommended in case both weeds are observed. If there is submergence due to excess rain, drain excess water at the earliest and spray 1%  $KNO_3$  or 19:19:19, 20:20:20 or 21:21:21. Application of Multi K or 13-0-25 @ 10g/l at the time of flowering should be done. Practice nipping at 45 and 60 DAS, wherein terminal branches are removed to facilitate lateral branches, thus improving yields. To control pod fly spraying of Thiomethaxam 25 wg @ 0.2ml/l, Flubendamide 0.2ml/L with Jaggery 5g/L to be sprayed at the time of early pod formation stage.

liquids & probiotics in drinking water during summer. Utilize broken rice or bajra (up to 30-40%), cotton seed meal (10-12%) and rapeseed meal (5 -8%) in poultry when there is shortage of maize during natural calamities. Keeping in view of disease outbreak follow biosecurity measures and mass vaccinate backyard poultry birds against Newcastle and viral diseases to prevent outbreak during stress periods. For hatching purpose of fertile eggs keep it in fridge at temperature of 4°C. To improve the growth rate 1 per cent protein level (soybean meal) may be added in the feed. Suitable coccidiostat should be added in the feed continuously by consulting the local veterinarian to prevent coccidiosis. 3-5 g of oyster shell/limestone/grit per day/bird shall be given to laying hens to avoid leathery eggs.

**Value Addition:** Convert milk into ghee, paneer, curd, khoa for better returns. Explore meat processing and packaging for hygienic sale. Utilize manure for biogas and compost.

### Fisheries

**Failed Monsoon:** Use bore well or alternative water sources to maintain minimum pond water levels. Reduce stocking density to lower water demand. Adopt partial harvest strategies to minimize losses. Use water-saving practices like reduced water exchange. Avoid new stocking if water availability is uncertain.

**Flood:** Strengthen pond bunds and spillways to prevent overflow and escape of fish. Install protective netting around pond perimeters. Avoid feeding during peak flood to prevent water quality issues. Harvest fish early if flood risk is imminent. Apply disinfectants like Benzalkonium Chloride 80% (BKC) @ 2.5 L/ha and Rock Salt @ 75-125 kg/ha/5ft depth/ha after floodwater recedes to control disease outbreaks.

**Drought:** Reduce biomass load in ponds by early or partial harvesting. Avoid use of organic manure if water level is too low to prevent oxygen depletion. Apply aeration measures (e.g., paddle wheel aerators) to maintain DO levels. Maintain good water quality by regular monitoring and minimum exchange.

**Delayed Monsoon:** Postpone seed stocking until adequate water level is available. Increase application of manures/fertilizers once rains begin to promote plankton growth. Avoid breeding activities during uncertainty to protect brood stock. Maintain brood stock in protected tanks.

**Advanced Monsoon:** Prepare ponds early for stocking, complete liming 200 kg/ha and fertilization in advance. Ensure brood stock is not disturbed during monsoon (15th June to 31st August). Avoid overfeeding as heavy rains can cause feed waste and water quality degradation. Watch for sudden temperature or pH fluctuations due to early rains.

**High Temperature:** Maintain higher water levels (5-6 feet) to buffer temperature fluctuations. Avoid feeding during peak heat hours, feed early morning and late evening. Use shade nets or plant trees around ponds to reduce direct sunlight. Monitor dissolved oxygen (DO) and apply aeration during hot afternoons.

**Low Temperature:** Reduce feeding rate and switch to highly digestible feed. Avoid handling fish during early mornings and nights. Provide windbreaks and increase pond depth to retain heat. Avoid use of antibiotics and strong chemicals during cold periods unless essential.



## CEREAL CROPS

### Paddy

Farmers should begin nursery preparation by late May using recommended varieties like CR Dhan 310, CR Dhan 311, Gomati, or Swarna Masuri (135-150 day duration). Treat seeds 24 hours before sowing with Carbendazim/Captan (2g/kg) or Trichoderma viride (5 g/kg) to prevent soil-borne diseases. Transplant seedlings based on variety duration: 20-25 days for short/medium (?120 days), 25-30 days for medium-long (120-130 days), and 30-40 days for long-duration (130-160 days) varieties. To enhance soil fertility, incorporate green manure crops (dhaincha, cowpea, moong) or Gliricidia leaves during field preparation, along with Azotobacter application (12kg/ha soil or 1.5L/ha seed treatment). For weed control, use post-emergence herbicide Metsulfuron Methyl + Chlorimuron Ethyl (50gm/ha). Farmers currently harvesting boro rice should watch for gundhi bugs (control with Ethofenprox 10EC/Imidacloprid sprays) and blast disease (apply Hexaconazole at flowering), harvesting when 85% grains turn straw-colored.

### Maize

Farmers are advised to cultivate high yielding varieties include RCM 1-1, RCM 1-2, RCM 1-3, Vijay Composite, DA 61-A, HQPM-1, and HQPM-2. Perform three deep ploughings to ensure proper soil aeration and seedbed preparation. Apply atrazine herbicide 2-3 days after sowing to effectively manage weeds for up to 30 days. Proper earthing up is essential to prevent lodging due to high winds or cyclonic conditions, ensuring stable maize growth. Ensure a well-maintained drainage system during the kharif season to avoid waterlogging, as excessive moisture can negatively impact maize yield. To control maize stem borer, apply Dimethoate 30 EC (660 ml/ha) or Carbaryl 50 WP (1.0 kg/ha) for effective management. For Fall Armyworm (FAW) control in kharif maize, use Emamectin Benzoate 5% SG (600-700 g/ha) as a systemic insecticide to protect the crop.

## **SPICE CROPS**

### **Ginger & Turmeric**

Farmers are suggested for planting ginger (Varadha, Nadia) and turmeric (Megha Turmeric 1 or Lakadong) rhizomes in the first week of May for optimal growth. Before planting, dip the rhizomes in *Trichoderma Viride* (4g/L of water for 5 minutes) or *Trichoderma Harzianum* (10ml/L) to prevent soil-borne diseases. Additionally, spraying Neem oil (10000ppm @ 2ml/L) or *Trichoderma* solutions can help prevent soft rot. Apply farmyard manure (FYM) at 25t/ha, and fertilizers with a ratio of N:P:K 75:50:50 kg/ha for better growth and yield. Plant rhizomes in ridges and furrows with a spacing of 40x20 cm. Ensure proper weeding and earthing up. Intercrop with crops like soybean, French bean, finger millet, or maize to improve farm income. Mulch the beds with dry paddy straw or green leaves to conserve soil moisture, control weeds, and prevent soil erosion.

## **LIVESTOCK**

Regularly de-worm livestock as per their body weight and condition, especially during the hot and humid kharif season when worm infestations are more prevalent. Keep the livestock shed clean by using potassium permanganate for regular disinfection, and apply phenol compounds outside the shed to reduce the multiplication of disease-carrying vectors like mosquitoes and flies. Begin planting perennial fodder varieties like Nappier, Congo Signal, and Moringa. Additionally, prepare for the lean season by drying maize stalks for hay or making silage. Vaccinate cattle against FMD, HS, and BQ, goats over three months old against PPR, and pigs against Swine Fever. Apply external ectoparasiticides to prevent ticks, mites, and fleas. Keep the poultry shed dry, apply lime regularly to reduce the incidence of coccidiosis, and ensure proper ventilation to prevent disease outbreaks. Restrict poultry from exposure during rainy or stormy weather to protect from seasonal fever and other environmental stress factors. Use herbal or chemical fly repellents on a daily basis to manage external parasites and prevent fly infestations in the poultry farm.

## **FISHERIES**

It is recommended to focus on carp breeding, hatchery management, and carp fry and fingerling production from April to July, as this is the peak time for these activities. For optimal carp fingerling production, dewater seasonal ponds to remove unwanted fish. Plough the pond bottom, expose it to sunlight for one week, and apply lime at a rate of 400 kg per ha to prepare the pond. After a week, apply cow dung at 10,000 kg per ha. Two weeks later, remove aquatic insects by using soap and kerosene emulsion. The exact doses should be confirmed by consulting a fisheries expert. Once the pond is prepared and insect removal is complete, stock healthy 4-day-old carp spawn at a density of 30 lakh/ha. This should be done 1-2 days after insect removal. Ensure regular feeding of carp fry with a mixture of finely powdered MOC and RB powder in a 1:1 ratio to promote healthy growth. It is advisable to excavate or renovate water harvesting ponds to conserve rainwater and runoff, maximizing water storage during the monsoon season. Maintain pond embankment stability by ensuring side slopes of 2:1 (Horizontal: Vertical) for sandy soil, 1.5:1 for loamy soil, and 1:1 for clayey soil. Additionally, create an empty berm of a width equal to the depth of the pond to prevent siltation during the rainy season. To maintain water quality, check the pH level of the pond after heavy rainfall. If the pH falls below 6.5, apply lime at 50 kg/ha to improve the water's alkalinity.

after transplanting. For Pusa Basmati 1509, apply 135 kg urea/ha in three splits at 3<sup>rd</sup>, 6<sup>th</sup>, and 9<sup>th</sup> weeks after sowing (DSR) or in two equal splits after transplanting. Apply urea when fields are non-flooded; irrigate on the third day after application.

### **Maize**

Select suitable hybrids and composites like Shaktiman-1, K-41, Azad Uttam, Sharad Mani, Azad Kamal, Chandra Mani, Azad Shankar Makka-1 and Azad Shankar Makka-2 or other recommended varieties based on agro-climatic zones. Use certified, disease-free seeds treated with Carbendazim or Captan @ 2 g/kg for fungal protection and Imidacloprid @ 5 ml/kg for insect protection at a seed rate of 20-25 kg/ha.

Adjust sowing time to ensure proper crop establishment before the regular onset of monsoon. Time of sowing from mid-June to early July with spacing of 60 cm×20 cm (approximately 83,000 plants/ha). A population of 60-75 thousand plants per hectare at harvest is required for obtaining the optimum yield.

In irrigated area, it is most desirable to complete the sowings 10-15 days before the onset of the rain. For rain-fed areas, which entirely lack irrigation facilities, it will be desirable to sow the crop as soon as adequate amount of soil moisture has been built up to ensure germination and establishment of proper plant stand. Ensure proper field drainage to prevent water stagnation, especially during germination.

Apply FYM or compost @ 8-10 t/ha during land preparation; also apply 120-135 kg nitrogen, 60 kg phosphorus ( $P_2O_5$ ) and 40 kg potash ( $K_2O$ ) per hectare, with full phosphorus, potash, and one-third nitrogen as basal at sowing, followed by top-dressing of one-third nitrogen at knee-high stage and the remaining at tasseling, along with 25 kg/ha zinc sulphate if deficiency is suspected.

Perform inter-culture and weeding at 20 and 40 days after sowing to maintain a clean field, and apply pre-emergence herbicide Atrazine @ 1.0-1.5 kg/ha within 2 days of sowing to control broadleaf and grassy weeds effectively.

Monitor for insect-pest (stem borer, shoot fly, fall armyworm, and aphids) and diseases, and follow recommended practices as per need. Maintain soil moisture during critical stages-tasseling, silking, and grain filling; avoid water stress and drain excess water during heavy rainfall.

Harvest when husk turns brown and grains become hard and glossy, dry cobs properly before shelling; reduce grain moisture to 13-14% for safe storage.

### **Bajra**

Grow improved long-duration hybrids such as ICMB-155, WCC-75, Narendra-3, Raj 171, Pusa 322, Pusa 23, 86M45 and ICMH-451 based on the length of the growing period and purpose. Sowing should be done from mid-July to mid-August. Sow seeds 4 cm deep using the behind-the-plough method, maintaining a row spacing of 50 cm. Use 4 kg/ha seed rate for all recommended varieties/



**Pearlmillet variety 86M45 (Mathura)**

plant hopper and gandhi bug management. Spray of hexaconazole 5%EC 1000ml or azoxystrobin 18.2%+ difenconazole 11.4% SC 500ml or mancozeb 75WP 1.5-2.0 kg or isoprothiolane 40% EC 750ml/ha for of blast and sheath blight disease management in rice.

### **Maize**

Recommended sankul varieties for Tarai, Bhabhar & Plain area are Pant Sankul Makka-3, Sweta, Bajora Makka-1 & Vivek Sankul-11. Hybrids maize are H M 4, 10, 11, 13, HQPM 1, 4, 5, Pant Sankar Makka-1, 4, 5, 6, Sartaj, P-3522 and Vivek Makka-51. Best sowing time is mid-June in Uttarakhand plains. Popcorn varieties: Pant Popcorn-1 and VL Amber Popcorn, fodder maize variety is African Tall. Sowing time is mid-June to mid-July. Sankul varieties suitable for hills: Vivek Sankul Makka-11, 37, 31, 35 and Pant Sankul Makka-4, hybrids maize are Vivek Sanker-9, 39, 45, 53, 55, 57 and DMH-109. Time of sowing in lower hill (1<sup>st</sup> week June to mid-June), mid hill (May end to mid-June) and higher hill during April end to mid-May is suggested. Optimum seed rate of 18-20 Kg for sankul, 20-22 Kg for hybrids, 12-14 Kg for popcorn, 40-45 for baby corn and 8 kg/ha for sweet corn is recommended. Spacing for maize may be maintained at 60\*20-25 cm in line sowing at a depth of 5 cm. Inter-cropping of cowpea/soybean/ginger in maize would be effective to achieve the higher productivity, profitability and also control the soil erosion in hills.

**Fertilizer Application:** The maize crop in Tarai, Bhabhar & Plain area should be fertilized with NPK @ 120:60:40 kg/ha. Full P and K,  $\frac{1}{4}$  N apply at sowing and rest of N applied in 3 equal splits at 3-4 leaf stage, knee stage and flowering time. In case of zinc deficiency symptoms in previous crop, apply zinc sulphate (21%) 25 kg/ha as basal application.

**Weed Management:** For proper weed control minimum two hand weeding at 20-25 and 40-45 days after sowing is recommended. Application of atrazine 2.5 kg/ha as pre-emergence for broad leaf weeds fb tembotrione 42% SC 120 g/ha post-emergence at 3-4 leaf stage (15-20 DAS) for grassy and broad leaf weeds or alachlor 2.0 kg/ ha as pre-emergence for both grassy and BLW is recommended. To control the sedges, application of halosulfuron methyl 70WG 67.5g a.i./ha as post-emergence at 3-4 leaf stage (15-20 DAS) is recommended. For control of broad leaf weeds in maize crop application of 2, 4 -D dimethyl amine salt 58% SL 0.5 kg/ha or 2, 4-D ethyl ester 38% EC 0.9 kg/ha as post-emergence at 15-20 DAS.

**Plant Protection:** Banded Leaf and Sheath Blight disease in maize can be effectively managed through an integrated approach. Soil application of Trichoderma (Pant Bioagent-1) at the rate of 1 kg multiplied in 100 kg of vermicompost (VC) or farmyard manure (FYM), applied 10-15 days prior to sowing, has shown good results. Additionally, seed treatment with Trichoderma (Pant Bioagent-1) at 20 g



**FAW infestation in Maize**

honey (50 part) formed into patties. Old and dark combs should be discarded, and the inside of the hive should not be painted. Open the hive only when necessary and remove highly infested combs, which should be destroyed by burning. To sterilize combs and equipment, dip them in soap solution containing 7% formalin for 24 hours, wash with water, dry, and reuse. Diseases such as European Foulbrood (EFB), caused by *Melissococcus pluton*, can be identified when larvae turn yellow, then coffee brown, dying in uncapped cells as larvae, not pupae. EFB can be treated with oxytetracycline (Terramycin), but not during nectar flow, and robbing of infected hives should be prevented.

Wax moths like the Greater Wax Moth (*Galleria mellonella*) and Lesser Wax Moth (*Achroia grisella*) tunnel through combs, leaving webs and feces, primarily attacking weak colonies. Infested combs should be removed, and regular inspection will help maintain strong colonies that naturally suppress wax moths. As wax moths dislike light, exposing equipment to light or freezing empty frames will deter them. Ants can be controlled by placing hives on stands with legs in earthen cups filled with water. The ectoparasitic mite *Varroa jacobsoni* is highly destructive and requires strict colony hygiene, regular examination, and the use of Formic Acid to manage infestations.

**Care during Migration:** During migration, successful beekeeping demands timely and well-managed movement of colonies, particularly from areas lacking flora to those with abundant floral resources. Before migration, survey the target area to ensure flora availability and extract honey from the colonies. Close the entrance gates of colonies in the evening after all worker bees are inside, and pack the colonies internally and externally to prevent jerking. When loading onto vehicles, ensure the entrance faces the front side of the vehicle. Migration should start late in the evening, ensuring colonies reach the destination within 10-12 hours and entrance gates are opened upon arrival in the morning. For long-distance moves, colonies should be halted at appropriate locations during the day, the entrance gates opened, and the process repeated. Avoid jerking during transport to prevent colony stress and losses.

**Bee Forage Plants:** At this time weeds, grasses, and many kharif crops like coriander, maize and oilseeds flower and bees start collecting pollen. Comparatively, nectar is not freely available. Plants which are good source of nectar are tamarind, neem, eucalyptus. Plants which are good source of pollen are sorghum, sweet potato, maize, millets like ragi, roses, castor, pomegranate. Plants which are good source of both pollen and nectar are banana, peach, citrus, guava, apple, Sunflower, berries, safflower, pear and plum.

## MUSHROOM PRODUCTION PRACTICES DURING RAINY SEASON

During the rainy season in Uttarakhand, mushroom production requires careful management to ensure healthy growth and prevent contamination in mushroom units. Mushrooms thrive in a humid environment (80-90%), but excessive humidity during the rainy season can become a challenge; hence, proper ventilation is crucial to prevent moisture buildup. Maintaining a consistent temperature within the recommended range for the specific mushroom species, typically between 15-30°C, is essential, while the mushroom house should be well-insulated to avoid temperature fluctuations that may hinder growth or cause diseases. Dehumidifiers

avoid heatstroke, animal should be kept in shady areas and if possible, arrangement for fans and cooler should be made. Animal should be given fresh and clean water ad libitum especially during summer season. The feed given should be balanced consisting of appropriate quantity of green, dry fodder and concentrate along with mineral mixture, common salt. In higher altitude region, where the temperature is low throughout the year the animal whether cattle, goats, sheep, poultry should be given jaggery water in order to maintain optimum energy level. During summer and monsoon season the parasitic infestation rises especially in plain areas of the state. For control of ectoparasites various ecto-parasitocidal/acaricides are applied topically such as amitraz, cypermethrin, deltamethrin @ 2-3 ml/L of water and for control of endoparasites various anti-helminthic drugs such as Fenbendazole, Albendazole, oxclozanide, levamisole etc. are used at appropriate dose and dosage. To combat any adverse condition such as outbreak of any disease the farmer should be in close contact with nearby Veterinary hospital, dispensary.

### **Cow and Buffalo**

Prior to monsoon season cattle should be vaccinated against economically important diseases such as Foot and mouth disease, Hemorrhagic disease, brucellosis etc. Now a days vaccination is also done in cross breed cattle against theleriosis (haemoprotozoan diseases) in disease endemic areas. The feed should be nutritious consisting of green fodder such as sorghum, maize, napier etc. ( $\frac{2}{3}$ ), dry fodder such as wheat/paddy straw ( $\frac{1}{3}$ ) and adequate amount of concentrate (1 kg for maintenance and 1kg for 2.5-3 L of milk). Regular feeding of mineral mixture @ 50 mg/day/cattle and common salt @ 30 g/day/animal should be done for lactating/dry/pregnant/heifers. Deworming should be done at regular interval preferably in every 3 months particularly prior and end of monsoon season as prevalence of endoparasites particularly liver fluke are highest during this season. Topical application of acaricides is done to control ectoparasite such as ticks, lice, mites etc. However special precaution should be taken during application of acaricides as these are highly toxic and if ingested could be fatal. During lean period when there is scarcity of quality fodder feeding of hay, silage should be practiced. In consultation with veterinarian animal health camp and awareness programs should be organized at village level.

### **Sheep and Goats**

Prior to monsoon season the sheep and goats should be vaccinated against PPR, Hemorrhagic septicemia, Foot and mouth disease and Enterotoxemia. Deworming of sheep and goat in every 2-3 months should be done with emphasizes on rotational grazing. Dipping of goats and sheep is done in medicated water to control ectoparasites however, certain precautionary measures should be taken before dipping the animal. Store the forage in dry and moist free condition to prevent any fungal growth which could be detrimental for the health of sheep and goats. For optimum production, productivity and growth mineral mixture should be fed on regular basis @ 15-20 g/day/animal along with common salt. Trimming of hoofs should be done in order to prevent lameness, foot rot and other such conditions. To enhance the conception rate and probability of twinning flushing ration i.e., over and

above the maintenance ration be given prior to breeding season. The floor of the shed should be raised and to prevent moisture straw, wood husk etc. should be used as bedding material and be replaced at regular interval.

### **Poultry**

The shed of the poultry farm should be free from any leakages and the floor should be elevated. Feeding cost constitute 70% of total cost incurred in poultry farming so special focus should be given that feed is moisture-free with antifungal and anticoccidials are added to feeds as any contamination such as aflatoxins, aspergillosis, coccidiosis could lead to disease outbreak causing mortality and thus a great loss to farmers. Fumigation with potassium permanganate and formalin (1:2) is done prior to new lot in order to destroy any microbes present in farm. The water given should be fresh, clean and replaced at regular interval and supplemented with multivitamin, liver tonics, calcium supplement, probiotics in order to maintain growth of the poultry birds. Biosecurity measures include footbath at entrance, restricted entrance by visitors, use of foggers, fans during moisture/summer months. Vaccination of birds against important diseases such as IBD, Ranikhet, Pox disease.

### **Swine**

The Swine shed should be durable and leak proof to withstand heavy rains and wind. To prevent water logging and fungal growth the shed should be proper drainage and ventilation. Regular deworming and supplementation of mineral mixture optimizes the growth and productivity of swine. Swine should be vaccinated against classical swine fever. Dipping should be avoided during monsoon. Good hygiene and sanitation are important to combat any disease outbreak.

## **FODDER AND FEEDING MANAGEMENT**

Livestock feed is generally classified according to the amount of nutrient present. They are generally classified as roughage and concentrate. Roughages are bulky feed containing large amount of crude fiber (>18%) and low Total digestible nutrients (<60% TDN). While concentrate have less fiber (<18%) and high TDN (>60%). Some of the commonly cultivated fodder crops in Uttarakhand during kharif season are Maize, sorghum, pearl millet, cow pea, guar, hybrid napier, perennial forages, lucerne etc. During scarce period silage, hay feeding should be fed to animals. Feeding management is an important aspect, the feeding should be done 2-3 times a day however fresh clean water should be available 24 hrs. The fodder should be chopped as it avoids wastage and increases the palatability of the feed. The ratio of green fodder ( $\frac{2}{3}$ ) and dry fodder ( $\frac{1}{3}$ ) should be maintained. Feeding of concentrate is based on the stage of animal i.e., dry animal are given maintenance ration, dairy cattle are fed concentrate based on milk production, pre-starter, starter, finisher ration is given in poultry etc. Along with dry, green fodder, concentrate regular feeding of mineral mixture, common salt should be given. Animal should be given maximum time for rumination so that all the nutrient available in feed of animal are fully utilized. Azolla, an aquatic fern is also now a days as a source of green fodder @ 1.5-2 kg in cattle, swine @ 300-500 g in goats and 20-30 g is poultry.

with brine water solution followed by treatment with Mancozeb + Carbendazim or T. viridae. Community nursery, tier nursery and alternative methods such as DSR through drum seeder are very good options to combat with flood or drought like situation, as per applicability.

Use pre-emergence herbicide like Butachlor 5 G (20-25 kg/ha), Pretilachlor (0.75-1.0 kg a.i./ ha) or post-emergence herbicide like Bispyribac-Sodium 10% SC (250 ml/ha) or triafamone (20%) plus ethoxysulfuron (10%) at 15-20 DAT.

Soil application of Zinc through 40-50 kg of zinc sulphate heptahydrate (21%) or 20-30 kg zinc sulphate monohydrate (33%) per ha and boron through solubor 20% at puddling is advocated, if the previous crop had shown zinc and/or boron deficiency symptom.

The crop yield is mostly affected by pests like yellow stem borer, brown plant hopper & leaf folder and diseases like blast, sheath blight, bacterial leaf blight and false smut. Root dipping of the seedlings in *Pseudomonas fluorescens* (mixed in water @ 10 g/L) for 30 min, before transplantation, will give protection against stem rot. Apply cartap hydrochloride 4 G @ 25 kg/ ha at the time of transplanting or chlorantraniliprole 0.4 G @ 10 kg/ha after 18 days of transplanting to manage stem borers and other lepidopteron insects in the main field. One can also use pheromone traps @ 35/ ha.

Follow skip row planting (skip two rows after every 10 rows of planting) to prevent brown plant hoppers severity. Spray Dinotefuran 20 SG @ 0.25 g/L or buprofezin 25 SC @ 1.5 ml/L as prophylactic measure. Apply Emamectin Benzoate 0.4 g/L to control leaf folder. Use Tricyclazole 0.6 g/L to control blast, Validamycin 2.5 ml/L or cow-based biopesticide (4 L/15 L water) to control sheath blight and Streptomycin sulphate + Tetracycline combination 300 g + copper oxychloride 1.25 kg/ha. If necessary, repeat 15 days later. If crop faces any short- or long-term dry spell or faces flood like situation contingent measures shall have to be adopted depending upon the growth stages. Contact KVK of your district or block Agriculture office for necessary contingent measure to be taken.

## Maize

Short duration hybrid maize varieties are suitable for cultivation during kharif season. Some preferred varieties are: Early duration (78-85 days) varieties like Dekan 109 and Prakash; medium duration (85-95 days) varieties like Dekan 107, Ranjit and Ganga Safed; and late varieties (95-125 days) like PS-740, D course- 9081, 900 M Gold, P-3522, P-3396.

Crop should be planted preferably in upland/medium upland and coarse textured soil having good drainage facility as maize is very sensitive to moisture stress and excess moisture condition.

Weeds can be controlled manually by two hoeing at about 15 to 30 days after sowing or by spraying of Atrataf/ Atragold/ Atrazine @ 2.0 kg/ha on medium to heavy textured soils and 1.25 kg/ha in light soils as pre-emergence or within 10 DAS using 500 litres of water. Weeds can also be controlled with post emergence application of Laudis 420 SC (tembotrione) 262.5 ml/ha in 375 L of water at 20 DAS.

Crop should be planted in rows along wind direction. Earthing up to be done 20-25 days after sowing. Treat seeds with Mancozeb + Carbendazim (2 g/kg) or T. viride (10 g/kg) using 10-12 ml water per kg seed to prevent foliar and sheath diseases.

Fall Army Worm is the prevalent pest hampers maize yield. Installation of pheromone trap @ 5 per ac at 30 DAS followed by spraying of NSKE @ 5 ml/L water at 35-40 DAS or spraying of novaluron 5.25% + Emamectin benzoate 0.9% SC @ 40 g a.i./ha at 45 DAS or spraying of Btvarhurstaki 17600 IU/mg @ 2.5 ml/L of water at 60 DAS or spraying of spinetoram 11.7 SC @ 35 g a.i./ha at 70 DAS and spraying of chlorantraniliprole 9.5% + lambda cyhalothrin 4.6% ZC @ 85 g a.i./ha at 85 DAS to be applied depending upon the stages of appearance.

## **COMMERCIAL CROPS**

### **Jute**

This is the main cash crop of summer season of the state of West Bengal which faces severe challenges due to scarcity of water in retting ponds. Jute semilooper, stem weevil, yellow mite and root knot nematode are major pests and stem rot and wilt are major diseases of jute. Spray Emamectin benzoate and Indoxacarb or alternatively B. thuringiensis and B. bassiana for controlling jute semilooper. Cypermethrin 0.03% or quinalphos @ 2 ml/L are used to control the Stem weevil. Dicofol 0.04% or Fenazaquin 0.02% are effective in managing the yellow mite. Spraying of Carbendazim @ 2 g/L of water or Copper oxychloride @ 4 g/L of water and Mancozeb @ 5 g/L of water is recommended for management of stem rot and validamycin 3% EC @ 2 ml/L water and Streptocyclin @ 1 ml/L water may be used against bacterial wilt. Covering jute bundle with polythene and use of CRIJAF Sona/NINFET Sathi powder during retting at pond reduces retting period and improve fibre quality. A solution of DAP and liquid molasses (7 kg DAP + 2 kg molasses in 10 L water followed by addition of 90 lit water) can also be used to improve fibre quality. During this process use of mud, banana plants or water hyacinth must be avoided.

## **PULSE CROPS**

### **Black gram**

Up or medium upland with well drainage facility is suitable for the crop. Improved varieties like PU-31/ PU10/VBN-8/WBU-109 etc. to be collected from reliable source for high yield. Seed should be treated with fungicides Mancozeb or Carbendazim @ 2 g/kg seed and after seven days of chemical treatment inoculation of seed with Rhizobium @ 1.5 kg / 30 kg of seed (for one ha) followed by mixing of 5 g T. viridae/kg seed is recommended for better performance.

Apply pendimethalin @ 2 ml/L pre-emergence herbicide followed by one manual weeding at 20-25 days after sowing will manage weed population. For better performance of the crop foliar spraying of micronutrient mixture @ g/lit water twice i.e. at 21 DAS and at pre-flowering stage and DAP @ 2 g/L water during flowering stage is advocated.

YMV is the major disease and for controlling the disease infected plants must be rogued, installation of yellow sticky traps @ 15/acre need based spraying of acetamiprid 20 SP @ 250g/ha is recommended to control the disease. For pod borer or leaf eating caterpillar, spray Cartap 1 g/L or acephate 1 g/L twice at 10 days interval.

## OILSEED CROPS

### Sesame

Use improved varieties like Suprava, Unnat Rama and Savitri. Seed dressing with Azotobacter and PSB followed by soil application of *T. viridae* + *P. fluorescence* (with 50 kg FYM) @ 1.5 kg to be followed. For better crop performance integrated basal application of Zn- 5 kg + sulphur- 25 kg + boron- 1 kg + RD of NPK per ha followed by 2 foliar sprays of 0.25% boric acid at 40 and 60 after sowing is advocated. Crop yield is adversely affected due to stem and collar rot. Foliar spray of *T. viridae* + *P. fluorescence* (oil-based) @ 5 ml/L each at flower initiation stage followed by foliar spray of difenoconazole 25 EC @ 0.5 ml/L of water depending upon the field condition. Spray imidachloprid 17.8 SL @ 1 ml/L of water to control phyllody of sesame.

## VEGETABLE CROPS

### Okra and Brinjal

Select improved varieties of okra like Pusa Makhmoli, Azad Kranti, Arka Anamika, Arka Abhay etc. Some improved varieties of brinjal are Pusa Purple Long, Sweta, Pusa Bhoirab, Punjab barshati, Punjab Sadabahar etc. (long varieties) and Pusa Uttam, Pusa Purple round, Punjab Neelam etc. (oval varieties). Planting of grafted seedlings (VNR-212) raised in pro-trays can be a good option. Seeds must be treated with Mancozeb + Carbendazim @ 2 g or *T. viridae* @ 10 g by dissolving in 10-12 ml water per kg seed. Use 45-60 mesh net covers to prevent viral vectors like whiteflies.

Select upland or medium up-land with good drainage facility for planting of seedling. Planting should be done in broad raised bed and furrow to avoid or minimize damage of planted seedling during heavy rain. Ensure mulching to reduce nutrient loss and conserve moisture. To improve fruit setting and to minimize flower or immature fruit drop spray soluble B @ 1.5 g/L and NAA 4.5% W/W (Planofix) @ 0.2 ml/L, alternately at 10-12 days interval.

Fruit and shoot borer can be controlled by installation of pheromone trap @ 15 nos./acre or leucilure @ 15 nos./acre followed by spraying of tetraniliprole @ 0.5 ml/lit or broflanilide @ 0.3 ml/L. Spraying of Flonicamid @ 0.5 g/L or Afidopyropen @ 1 ml/L to control aphid and jassid is recommended. To control whitefly installation of 60 yellow sticky traps/ha and spraying of Pyrifluquinazon @ 1 g/L or Dimpropridaz @ 0.5 ml/L at 15-day intervals advocated or alternatively moringa root extract (10%) + Afidopyropen/Flupyradifurone in 10:1 ratio is advocated. For managing red spider mite, Spiromesifen @ 1 ml/L, Etoxazole @ 1.5 ml/L or diafenthion @ 1 ml/L are to be sprayed at weekly interval. Barrier cropping with maize may be suitable option for sucking pest management.

For controlling YMV disease in okra infected plants must be rogued, installation of yellow sticky traps @ 15 nos./acre and need based alternate spraying of Acephate (1.5 g/L) + neem oil and Imidachloprid (0.5 ml/l) + neem oil or admire @ 2 g/litre + neem oil and Hostathion @ 1 ml/L + neem oil or spraying of acetamiprid 20 SP @ 250 g/ha is recommended to control the disease. For managing nematode problem in brinjal, use VEPAM (15 days before sowing) and drench the soil with Fytolan (2.5 g/L

of water) after 7 days of sowing to control. To overcome bacterial wilt use grafted brinjal seedlings. Apply bleaching powder @ 15-18 kg/ha to soil before 1 month of planting to reduce the chance of bacterial wilt. Apply organic amendments like neem cake (5 q/ha) to reduce wilt incidence. Consider pre-colonized *Trichoderma harzianum* or *Pseudomonas fluorescens* in soil applications for biocontrol.

### **Chilli**

Select improved varieties like Andhra Jyoti, Punjab Lal, and Arka Lohit, Arka Tejasvi etc. Seeds must be treated with Mancozeb + Carbendazim @ 2 g or *T. viridae* @ 10 g by dissolving in 10-12 ml water per kg seed.

Select upland or medium up-land with good drainage facility for planting of seedling. Planting should be done in broad raised bed and furrow to avoid or minimize damage of planted seedling during heavy rain. Ensure mulching to reduce nutrient loss and conserve moisture.

To improve fruit setting and to minimize flower or immature fruit drop spray soluble B @ 1.5 g/L and NAA 4.5% W/W (Planofix) @ 0.2 ml/L alternately at 10-12 days interval. Chilli leaf curl complex is the main threat and for controlling leaf curl complex a holistic approach is required. Scientific management of nursery bed through covering with 45 nylon mesh; carbofuran 3 G @ 2.5 kg/bigha during final land preparation and sowing of maize around the main field as border crop at 15 days prior to transplanting of chilli seedlings; spraying of castor oil @ 3 ml/L of water at 20 DAT; spraying Ofchlorfenapyr 10 SC @ 75 g a.i./ha at 35 DAT; spraying of Pymetrozine 50 WG + moringa root extract 10% w/v as tank mix @ 200 g a.i./ha + 5 ml/L of water at 55 DAT and spraying of Spirotetramat 11.01% + Imidacloprid 11.01% SC @ 55 g a.i./ha at 70-75 DAT.

To control whitefly in the main field installation of 60 yellow sticky traps/ha and spraying Pyrifluquinazon @ 1 g/L or Dimpropridaz @ 0.5 ml/L at 15-day intervals is advocated or alternatively moringa root extract (10%) + Afidopyropen/Flupyradifurone in 10:1 ratio can be used. Spraying of Flonicamid @ 0.5 g/L or Afidopyropen @ 1 ml/litre to control aphid and jassid is recommended.

Barrier cropping with maize may be suitable option for sucking pest management. Marigold can be grown as trap crop for shoot and fruit borer, along with pheromone traps and Heli NPV. For controlling mites spraying of Diafenthiuron @ 12.5 g/15 L of water is or Etoxazole @ 1.5 ml/L is recommended.

### **Cucumber and Pumpkin**

Select improved varieties for cultivation. Some improved varieties of cucumber are Japanese Long Green, Poinsette, Pusa Barkha, KPCH 2, Pusa Barkha, Pusa Uday, Pusa Barkha, Pusa Long Green, Swarna Sheetal etc. Some improved varieties of pumpkin- Arka Surjamukhi, Arka Chandan, Solan Badami can be grown.

Seeds must be treated before sowing in nursery bed with Mancozeb + Carbendazim 2 g or *Trichoderma viridae* 10 g by dissolving in 10-12 ml water per kg seed. Select upland or medium up-land with good drainage facility. Seeds should be done in broad raised bed to avoid or minimize damage of planted seedling during heavy rain.

Comfortable and ventilated housing should be there and practice regular deworming after every quarter. Provide daily molasses-based anti-heat drink (molasses, wheat bran, salt, water) or jaggery-gram flour beverage to reduce heat stress. Check feed for fungal contamination. Start cultivating J-1006 maize post-flood for fodder. Use enriched straw (with molasses, urea, minerals), tree leaves, aquatic weeds, and silage for nutritional improvement. Frequent cleaning and sanitation of milk utensils and equipment's should be practised with suitable detergent. Maintain hygiene during feeding and milking; wash hands, teats, and feed utensils. Conduct timely vaccinations against FMD, HS and BQ, and Deworm animals with proper anthelmintics. Contact Veterinarian posted at the nearby Block under State Govt.

### **Sheep, Goat and Pig**

The floor of the animal shed should be clean and dry to prevent foot rot and other diseases. In case of foot rot of goat and sheep, wash affected feet (foot bath) with 10% ZnSO<sub>4</sub> or CuSO<sub>4</sub> daily until lesions subside. In severe case, provide antibiotic Enrofloxacin @ 7 mg/kg b wt. Construct elevated low-cost shelters for goats and pigs in flood-prone areas with proper ventilation. Provide plenty of clean water along with supplementation mineral mixture @ 20-30 g/day in extreme hot season to combat with the heat stress. Provide daily molasses-based drink (molasses, wheat bran, salt, water) or jaggery-gram flour beverage to reduce heat stress. Practice regular deworming once in pre-monsoon and post-monsoon each. Give enterotoxaemia vaccination @ 2 ml s/c to pregnant Ewe/Doe before one month of expected date of lambing/kidding. 25-50 g molasses per day may be provided to each pregnant Ewe/ Doe to prevent pregnancy Toxaemia.

### **Poultry**

Adopt hardy breeds like Vanaraja or Hit CARI for better heat tolerance. Provide electrolyte-enriched water frequently and maintain a balanced diet. Follow vaccination schedules to prevent disease outbreaks. Poultry birds may be fed with green leaves along with grinded maize/rice. Poultry sheds should be properly cleaned and disinfected with 60-70 g of KMnO<sub>4</sub> mixed with 120-150 ml formaline solution per 10 cubic feet space. Place the birds in the sheds after 24 hours of fumigation. Disinfect the premises of poultry houses with 1% hypochloride + bleaching powder (7 g/L of water). The saw dust should be dried completely before using as bedding material. Apply hydrated lime powder @ 1-2 kg per 10 sq. ft area in the shed before birds are placed in the shed. All the equipments in the shed should be disinfected using hot water.

### **Fisheries**

Strengthen/repair pond dykes and avoid deciduous plants like bamboo. Prefer planting banana, coconut, arecanut, cucurbits or solanaceous crops. Remove pond bottom mud to prevent gas buildup and ensure proper de-weeding while retaining some Eichhornia for shade and evaporation control.

Ensure 1-1.5 m water depth and use polythene lining or dyke plantation to reduce seepage. Lime dry pond before stocking (500-1000 kg/ha) and monitor water pH (7.5-8.5), DO (>5 mg/L), and salinity as per species requirement. Ponds which

## Field Crops

India has one of the largest agricultural sectors worldwide, which is the primary source of livelihood for ~55% of India's population. India has the world's largest cultivated land under wheat, rice, and cotton. India is the largest producer of milk, pulses, and spices in the world. It is the second-largest producer of fruits, vegetables, tea, farmed fish, cotton, sugarcane, wheat, rice, cotton, and sugar. The agriculture sector in India holds the record for second-largest agricultural land in the world, generating employment for about half of the country's population. Thus, the farming community comprise an integral part of this sector to provide food security with a means of sustenance.

During the last 9 years (2014-2023), a total of 2279 varieties of field crops have been released in India under the aegis of ICAR, which include 1092 of cereals, 343 of oilseeds, 343 of pulses, 137 of forages crops, 279 of fibrecrops, 68 of sugarcane and 17 of potential (underutilized) crops. These varieties include 1888 climate resilient varieties (83%) tolerant to one or more biotic and/or abiotic stresses. Out of these 1888 varieties tolerant to one or more biotic and/or abiotic stresses, 407 varieties have been developed specially for extreme climate including flood/ water submergence/ water logging tolerance (73), drought/moisture stress/ water stress tolerance (220), salinity/ alkalinity/ sodic soils tolerance (52), heat stress/ high temperature tolerance (49), cold/ frost/ winter chilling tolerance (13) using the precision phenotyping tools. In horticultural crops also, 22 climate resilient varieties have been developed. This includes varieties of potato (5), tomato (2), spinach (1), radish (1), cassava (4), sweet potato (2), greater yam (1), white yam (1), coconut (3) and taro (2).

Further, to tackle the widespread problem of malnutrition in globally, concerted efforts have been made to develop varieties rich in various essentially required micro-nutrients (Protein, Fe, Zn, Vit. A, Vit. C, oleic acid, anthocynin, lysine, tryptophan, removal of anti-nutritional factors like erucic acid, glucosinolates, KTI) for proper growth and development of the body and as a result 124 field crops and horticultural crops varieties have been released and notified during past nine years [Rice (10), Wheat (41), Maize (20), Pearl millet (9), finger millet (4), little millet (1), Lentil (2), Mustard (8), Linseed (1), Soybean (7), groundnut (3) and 18 Horticultural crops]. Concerted efforts for seed production have been made to take the biofortified varieties in general cultivation, which have resulted in coverage of more than 10 million hectares during 2022-23. Biofortified varieties have more than 50% share in total wheat breeder seed indents.

### 1. Implementation of GAP

The ministry of Agriculture and Farmer's Welfare and Processed Food Products Export Development Authority (APEDA), Govt. of India implements the GAP. APEDA adopts the approach of developing product specific clusters in different agro- climatic zones of the country to help in dealing with various supply side issues through Good Agriculture Practices addressing issues such as, soil nutrients management, higher productivity, adoption of market-oriented varieties etc.

|                                                                                                                                                           |                                                                                                      |                                                                       |                                                                                                                                                                                                               |                                                                                                                                   |                                                                                                                                                                                                                              |                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                           |                                                                                                      |                                                                       | Multicut:<br>70:30:30 kg<br>N: P2O5:<br>K2O/ha as<br><br>basal and<br>50<br>kg N/ha as<br>top dressing                                                                                                        |                                                                                                                                   | hoeing<br>through<br>weed-<br>cum-<br><br>mulcher<br>at<br>3-4 week<br>crop stage.                                                                                                                                           | at 40-45<br>DAS and<br>subseque<br>nt cut<br>at<br>30 days<br>interval                                                                                                       |
| <b>Pearl millet</b><br>Single cut:<br><br>Raj Bajra<br><br>Chari-2,<br>CO-8,<br>APFB-2<br><br>Dual:<br>AVKB-19<br>Multi cut:<br>Giant Bajra,<br>Proagro-1 | Summer:<br>March to<br>mid April.<br><br>Monsoon<br>season:<br>first<br><br>fortnight<br>of July     | 25 cm row<br>to row<br>using<br>a seed<br>rate<br>of 10-12<br>kg/ha   | FYM: 10<br>t/ha<br><br>50:30:30 kg<br><br>N: P2O5:<br>K2O/ha as<br>basal and<br>30<br>kg N/ha as<br>top<br>dressing. In<br>rainfed 20-<br>30 kg N/ha<br>coinciding<br>with rain at<br><br>30-35 days<br>stage | Rainy<br>season<br><br>crop: 1-2<br><br>irrigations<br>Summer<br>sown crop:<br><br>4-5<br>irrigations                             | Pre<br>emergence<br><br>applicatio<br>n of<br>atrazine @<br>0.5 kg<br><br>a.i./ha in<br>450 litres<br>of water is<br>effective.<br>One<br>hoeing<br>through<br><br>weed-<br>cum-<br>mulcher<br>at<br>3-4 week<br>crop stage. | Single<br>cut: 55-<br>60 DAS<br><br>(initiatio<br>n of<br>flowerin<br>g)<br>Multicut:<br>first cut<br>at 40-45<br>DAS and<br>subseque<br>nt cut<br>at<br>30 days<br>interval |
| <b>Maize</b><br>African<br><br>Tall, Vijay,<br>Moti,<br>Jawahar<br><br>Composite,<br>VL-54,<br>APFM-8,<br>J-1006                                          | Summer:<br>February<br><br>to March.<br>Monsoon<br>season:<br><br>Beginning<br>of rains<br>June-July | 30-40 cm<br>row to<br>row,<br>using a seed<br>rate of 40-<br>50 kg/ha | FYM: 12-15<br>t/ha,<br><br>80-100 kg N<br>+ 40 kg<br>P2O5/ha.<br>15-<br>20 kg<br>ZnSO4 ai/ha                                                                                                                  | Rainy<br>season<br><br>crop: 1-2<br>irrigations<br>Summer<br><br>sown crop:<br>5-6<br>irrigations<br>at 10-12<br>days<br>interval | Applicatio<br>n of<br><br>atrazine @<br>0.75-1.0<br>kg a.i./ha<br><br>in 450<br>litres of<br>water is<br>effective.<br>One<br>hoeing<br>through<br>weed-<br>cum-<br>mulcher<br>at<br>3-4 week                                | Silk<br>stage<br><br>(60-75<br>DAS) for<br>fodder<br><br>purpose<br>which<br>continue<br>up to<br>milk<br>stage                                                              |